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Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis

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Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis

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ABSTRACT

Objectives: (1) To summarize the literature on the impact of pediatric weight management interventions on health outcomes in preschool age children with overweight or obesity. (2) To evaluate the completeness of intervention description and real-world applicability using validated tools.

Design: Systematic review and meta-analysis.

Data sources: Medline, EMBASE, CINAHL, Cochrane Library and PsychInfo were searched between 10 March 2015 through 31 January 2020.

Eligibility criteria: Randomized controlled trials (RCTs) addressing weight management in preschool children (two to six years) with overweight or obesity.

Data extraction and synthesis: Two reviewers independently extracted key information from each study and assessed risk of bias. Random-effects meta-analysis was performed where there was evidence for homogeneous effects. The certainty of evidence was assessed by the Grading of Recommendations Assessment, Development and Evaluation.

Results: Of the 14018 studies retrieved, four trials (937 participants) met the inclusion criteria. These interventions utilized motivational interviewing (MI) or education related to health behaviour approaches and were 6 to 12 months in duration. All studies contained some risk of bias. No difference was found in the intervention groups compared to controls for BMI z score, though heterogeneity was high (mean difference (MD) -0.14, 95% CI -0.29 to 0.01; 3 trials, 754 participants; p=0.070; I^2 79%). In a subgroup analysis, those receiving educational interventions had a reduction in BMI z score compared to controls (MD -0.27, 95% CI -0.41 to -0.12; 2 trials, 242 participants; p<0.001; I^2 40%); no differences were noted with MI. The certainty of the evidence was considered low to moderate. The trials were reported in sufficient detail and were considered pragmatic.

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Conclusions: Educational interventions delivered to the parents of young children with obesity result in decline in BMI z score. There was insufficient evidence to examine other health outcomes.

PROSPERO registration number: CRD42020166843.

Strengths and limitations of this study

- This review updates our understanding of the effectiveness of weight management interventions on health outcomes of preschool aged children over the past five years.
- This review highlights a knowledge gap for the best practice for pediatric weight management in preschool children.
- This review evaluates the description of interventions and their real-world applicability by using the TIDieR checklist and RITES tool, respectfully.
- This review investigated children ages two to six years old with overweight or obesity and not those with a medical condition known to affect weight status or syndromic causes for obesity.

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INTRODUCTION

The prevalence of overweight and obesity in children has been increasing globally.[1] There is estimated to be 124 million children and adolescents with obesity worldwide.[2] Nationally representative data from 144 countries suggests that 43 million children under five years of age have obesity.[3] Children with obesity carry an increased risk for cardiometabolic disease and poor psychosocial health.[4-7] Childhood obesity often persists into adulthood and is associated with increased risk of morbidity throughout life.[8-10] These alarming rates and health risks underscore a need for effective interventions to treat obesity in early childhood and to prevent continued obesity related health issues across the lifespan.

Family-based behaviour modification remains the foundation of pediatric weight management programs.[11] In a 2016 Cochrane systematic review, multicomponent interventions tested with randomized controlled trials (RCTs) in preschool children with obesity aged two to six years old were effective in reducing body mass index (BMI) z score, a common metric to classify obesity in children.[12] Interventions typically consisted of lifestyle behaviour modification, including dietary and physical activity advice, along with parenting skills. Four trials (n=210 participants) were included in that analysis, and a mean change of -0.3 units (95% CI -0.4 to -0.2) in BMI z score was reported. However, the quality of evidence was deemed to be low and a high risk of bias was identified. Since this time, there has been increased interest in this field given the increasing prevalence of pediatric obesity.[13] This systematic review will update the effectiveness of weight management interventions on preschool age children since the Cochrane review in 2016.[12]

Although systematic reviews can be invaluable in facilitating clinical decision-making by providing a critical evaluation of the evidence on a specific topic or condition,[14] to do so the intervention must be adequately described. Further, the pragmatism of the trial will inform the potential for 'real-world' implementation. The Template for Intervention Description and Replication (TIDieR) checklist was created to ensure interventions are described in sufficient

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detail to allow for replication.[15] The Rating of Included Trials on the Efficacy-Effectiveness Spectrum (RITES) tool is used to assess the real-world applicability and pragmatism of trials.[16]

The objectives of this systematic review were: (1) to assess the impact of pediatric weight management interventions on health outcomes in preschool children aged two to six with overweight or obesity; and (2) to evaluate the description of these interventions using the TIDieR checklist and their real-world applicability using the RITES tool. This knowledge will help inform future clinical practice guidelines and the planning of health programming for the treatment of young children with obesity.

METHODS

This review was conducted and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses.[17] A PRISMA study flow diagram was completed to present a summary of searching, screening, and inclusion of studies in this systematic review.[17] The review protocol for this study was registered with PROSPERO (CRD42020166843).

Patient and public involvement statement

No individual participant data were involved in the study; therefore, patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Eligibility

Study designs eligible for inclusion were RCTs with the aim of weight management in preschool children with overweight or obesity. Any type of intervention of at least six months duration and delivered in a clinical setting was included. The interventions could be delivered by a single professional or be multidisciplinary or multicomponent. Multicomponent interventions typically address health behaviours including diet and physical activity. Participation of family members was not mandatory. Interventions for prevention of obesity, studies with observational designs and studies with self-reported height and weight were excluded. Participants included children with overweight or obesity who were between the ages of two and six years at time of enrolment into a weight management intervention. Studies of children with obesity who had a medical

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condition known to affect weight status or syndromic causes for obesity were excluded. These included genetic syndromes such as Prader-Willi Syndrome and Bardet-Biedl Syndrome, for example. The control group in eligible RCTs was defined as no intervention or usual or standard care.

The primary outcomes were change in BMI or change in BMI z score or BMI percentile. BMI changes with age during normal growth,[6] and so age and sex-specific BMI z scores, based on a normative standard were included.[18] Additional secondary health outcomes were included in the search terms: body weight, waist circumference, waist-to-height ratio, skin fold thickness, body fat percentage measured with bioelectric impedance (BIA) or dual-energy x-ray absorptiometry, lipid profile (including LDL, HDL, triglycerides, and total cholesterol), blood pressure, glycemia (fasting plasma glucose, glucose level at 2 hours in a standard oral glucose tolerance test, Hemoglobin A1C) and quality of life. Adverse events were defined as any adverse outcome occurring during or after the intervention, including events not necessarily caused by the intervention.

Search of studies

We conducted a comprehensive literature search in the following electronic databases: Ovid Medline, Embase, CINAHL, Cochrane Library, PsychInfo, WHO International Clinical Trials Registry Platform, Web of Science, and Clinicaltrials.gov. These databases were searched from March 10 2015 (the end date of the 2016 Cochrane review)[12] to Jan 31 2020. The search strategy was developed in consultation with a research librarian using the following major themes: ‘preschool’, ‘overweight or obesity’, and ‘intervention’. Key terms within the search strategy were aligned to medical subject headings (MeSH) in Ovid Medline and were expanded to include more descriptive terms. The search strategy for each database is presented in Appendix 1. Reference lists of included publications and reviews and of other systematic reviews or meta-analyses were manually searched to identify other studies that might not have been identified through the database searches.

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Study selection

Titles, abstracts, and full texts were reviewed independently for eligibility by two groups of two authors (SN and CD, PM and RG) using a pre-tested, standardized screening form. Publications selected for inclusion were reviewed by an additional author and expert in the field and compared against the inclusion criteria. Only papers published in English were included. Every attempt was made to identify duplicate publications or multiple reports/companion pieces based on the same study and, if that occurred, the publication with the longest follow-up period for the primary or secondary outcomes was included.

Data extraction

Key information from each included study was extracted in duplicate using an adapted screening form and data extraction template. If data were missing, attempts were made to obtain the data from the corresponding author noted in the publication. Any disagreement during screening or extraction was resolved by discussion or in consultation with a third author.

Risk of bias and certainty of evidence

Risk of bias of each of the included studies was assessed using the Cochrane Risk of Bias tool.[14] The overall certainty of the evidence for each outcome was judged using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. The certainty of the evidence was assessed across the domains of risk of bias, precision, consistency and publication bias (internal validity) as well as directness (external validity). The quality of evidence was rated for each outcome (high, moderate, low or very low). Certainty was judged as high (further search is very unlikely to change the confidence in the effect estimate), moderate (further research is likely to have an important impact on the confidence in the effect estimate and may change the estimate), low (further research is very likely to have an important impact on the confidence of the effect estimate and is likely to change the estimate), or very low (very uncertain about the estimate of the effect). A checklist was created and used to help ensure consistency and reproducibility of the GRADE assessments.[14]

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The TIDieR checklist was completed to ascertain if the intervention was reported in sufficient detail to allow for reproduction of the study.[15] The RITES tool was completed to evaluate the pragmatism of the intervention, which uses a 5-point Likert scale in four domains: (1) Participant characteristics, (2) Trial setting, (3) Flexibility of the intervention, and (4) Clinical relevance.[16] Two reviewers independently classified each trial on the Likert scale. Disagreements were resolved with a discussion to achieve consensus. All items were summed for each study to provide an efficacy/effectiveness score. The score is a continuum from maximum efficacy (score of 4) on one end to maximum effectiveness (or most pragmatic) at the other (score of 20). A low total score (i.e. 4) suggests a strong emphasis on efficacy and a high total score (i.e. 20) is consistent with a strong emphasis on effectiveness or a more pragmatic trial.

Statistical analysis

Statistical analyses were performed using Review Manager5 (RevMan 2013) software package (Cochrane Training, Cochrane). The risk ratio (RR) or odds ratio (OR) was calculated for binary data. For continuous data measured on the same scale, the mean difference (MD) was calculated and for continuous data measured on different scales, the standardized mean difference (SMD) was calculated. Overall effects were reported and classified as small, medium, or large based on Cohen's d effect sizes.[19] We used a random-effects model taking into account the interstudy variation which provided a more conservative effect than a fixed model. All effect sizes were reported with 95% CIs. The statistic I^2 was calculated to investigate the variation in effect size due to heterogeneity across studies. 0% to 40% corresponded with low heterogeneity, 30% to 60% corresponded with moderate heterogeneity, 50% to 90% corresponded with substantial heterogeneity, and >75% corresponded with considerable heterogeneity.[20] For three-arm trials (two interventions with a control group), the control group was split for meta-analyses to compare interventions to control.

Synthesis of results

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Review Manager5 was used to combine and calculate each outcome (RevMan 2013). If there was good evidence for homogeneous effects (i.e. tests of heterogeneity were not significant) across trials, a random effects meta-analysis was performed. If there was substantial heterogeneity, a qualitative synthesis was conducted. We anticipated intervention modality and length of intervention would introduce heterogeneity and planned to conduct subgroup analyses with available data.

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

RESULTS

Search results

The database searches resulted in a combined total of 14018 studies after removal of duplicates (Figure 1). After title and abstract screening, 441 studies remained and underwent full text screening. After full text review and removal of studies that did not meet the inclusion criteria, 4 studies remained. A manual search of the articles in the reference lists from the included studies did not generate any additional relevant studies. Therefore, a total of four RCTs were included in this review.

Summary of included studies

The four included studies, published between 2017 and 2019, were conducted either in the USA (n=3)[21-23] or in Sweden (n=1).[24] Of the 930 participants, 537 were randomized to an intervention and 393 to a comparator group. Three studies were multi-center trials[21; 22; 24] while one study was conducted at a single center.[23] Three of the studies were delivered in a primary care setting,[23; 21; 24] while one study was delivered in community-centered programs and primary care clinics.[22] The interventions included motivational interviewing (MI) for parents[23] or families,[21] or education on the principles of good nutrition and activity.[23; 22; 24] (Table 1). Interventions ranged from six[23] to 12 months[21; 22; 24] in duration and one study included a 12-month follow-up period.[21]

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Interventions varied in intensity, from 3 hrs total contact time (MI)[21] to 27 hrs total contact time (education).[22]

Effects of interventions

The BMI outcome was BMI z score for 3 studies and %BMIp95 was reported for two trials. BMI was reported in three trials, but as secondary outcomes, and change in BMI from baseline to post intervention was not reported in two of these trials. No difference in BMI z score between intervention and control groups was identified at the end of intervention (mean difference (MD) -0.14, 95% CI -0.29 to 0.01; 3 trials, 754 participants; p=0.070; I^2 79%) (Figure 2). Similarly, no between group difference in %BMIp95 was identified (MD -1.60%, 95% CI -4.74 to 1.55; 2 trials, 299 participants; p=0.320; I^2 73%) (Figure 3). Of note, the heterogeneity in both of these analyses exceeded 70%. There was no difference in change in BMI between intervention (1.11 ± 1.99 kg/m^2) and usual care (1.22 ± 1.82 kg/m^2).[21]

Subgroup analyses

Our subgroup analysis, based on intervention type, education vs MI, resulted in lower heterogeneity. The educational interventions resulted in a greater decline in BMI z score in treatment group compared to controls (MD -0.27, 95% CI -0.41 to -0.12; 2 trials, 242 participants; p<0.001; I^2 40%) (Figure 4), but no differences were found when the outcome was measured as %BMIp95 (MD -2.80%, 95% CI -6.72 to 1.13; 2 trials, 228 participants; p=0.050; I^2 73%) (Figure 5). In contrast, no significant impact of the MI based interventions on BMI z score (MD 0.01, 95% CI -0.08 to 0.11; 2 trials, 512 participants; p=0.770; I^2 24%) (Figure 6) was identified.

Secondary outcomes

Two of four trials included secondary outcomes. Butte 2017 included weight, blood pressure and health-related quality of life. They identified a significant time effect for weight post-intervention in both the intervention (3.84kg ± 0.23; p<0.05) and control groups (3.26kg ± 0.25; p<0.05). There were no time or group-by-time interactions for changes in blood pressure or

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health-related quality of life (PedsQL total score).[22] Ek 2019 found no significant difference in waist circumference in the Booster intervention (MD -1.09cm; 95% CI -3.35 to 1.17), but found significant increases in waist circumference in the No booster group (MD 2.12cm; 95% CI 0.23 to 4.01; p=0.03) and standard treatment (MD 4.06cm; 95% CI 2.64 to 5.47; p<0.001).[24]

Adverse events

Two trials reported no adverse events,[23; 24] while adverse events from the other two trials were not reported.

Risk of bias

All four trials reported adequate sequence generation and allocation concealment and used a form of computer random generator for randomization. As noted in the risk of bias assessment presented in Figure 7, all trials did not blind participants or healthcare providers to the intervention or control groups resulting in a high risk for performance bias for study outcomes. Two of four trials did not mention blinding for data collectors or outcome assessors; these trials were rated as uncertain risk for detection bias.[21; 22] Stark 2018 blinded study staff collecting outcome measures to participant assignment; this trial was judged as low risk for detection bias.[23] Ek 2019 stated that data were collected by child healthcare professionals participating in the study and was therefore rated high risk for detection bias.[24] The drop-out rates were unbalanced across studies: only 38% of intervention participants in the study by Rifas-Shiman 2017 completed more than 1 visit.[21] Of the remaining three trials, drop-out rates for the intervention groups ranged from 11%[24] to 36%.[22] All trials reported at least 20% drop-out in either an intervention or control group in their trials; consequently, all trials were rated as high risk for attrition bias. We judged three of four trials to have low risk of reporting bias.[21-23] We did not draw funnel plots or perform Egger's test to assess reporting bias due to limited number of trials per outcome (n=4).

Certainty of the evidence

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The overall quality of the evidence was considered low to moderate as a result of the high risk for performance bias and attrition bias (Table 2).

Template for Intervention Description and Replication (TIDieR)

TIDieR checklists for each RCT are available in Appendix 2. In 3 of the studies,[21; 23; 24] there was limited information provided regarding standard of care or usual treatment. Ek 2019 described standard of care based on an action plan specific to Stockholm County, but did not include more detailed information of what this entailed.[24] The article by Butte did provide some information regarding their standard treatment control in addition to a link for the provided materials; participants were provided activity booklets to work on physical activity and nutrition targets in a self-directed manner, but the extent of doing so was unreported.[22]

The interventions from each of the 4 studies were described in varying but sufficient detail. Of note, the materials required to conduct the intervention were not provided or were provided in insufficient detail to reproduce in two of the studies.[22; 24] Two of the included interventions were based on previously reported interventions. Stark's RCT was a three pillar RCT: LAUNCH, was a previously described intensive obesity intervention program and Butte et al report on the MEND program. Although it was well described, the intervention required extensive training in order to effectively administer and information regarding the training was not included. The second intervention group, MI, also required highly trained providers to facilitate the groups participating in the interview, again without information pertinent to the training. Materials and training details for both the LAUNCH protocol as well as the motivational interviews were not provided, which makes the reproduction of these interventions difficult, and required the reviewers to search for previously described protocols. Butte et al had previously published a protocol referencing their study design in adequate detail.

Rating of Included Trials on the Efficacy-Effectiveness Spectrum (RITES)

The RITES tool data for all four trials are presented in Appendix 3. All of the four included trials scored between 12 to 15, indicating more pragmatic trials. Three of the trials[22; 24; 23] had more

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explanatory trial settings as they required extensive training for implementation of the intervention and/or ongoing supervision to ensure the precise implementation. All of the included studies were found to be highly effective in terms of their clinical relevance. This was a result of all studies having change in BMI z score or BMI percentile and performing intention to treat analysis. However, all four trials excluded participants with underlying conditions or comorbidities, resulting in greater emphasis on efficacy for participant characteristics.

DISCUSSION

Summary of evidence

Since the previous review published in 2016,[12] four RCTs examining obesity treatment interventions in young children were published. Three of the four RCTs utilized an educational approach for intervention and these resulted in lower BMI z score at end of intervention (6-12 months duration). The change was however modest, and the trials were found to have a high risk of bias. The trials did lack some detail in description of the control based on the TIDieR Checklist, but all were noted to be relatively pragmatic, suggesting they could be applied in real-world settings.

In the previous Cochrane review, multicomponent interventions were found to be more successful at reducing BMI z score compared to controls.[12] When grouped together, the trials included in the current systematic review had high heterogeneity, and were not effective in reducing BMI z score or %BMIp95. In the subgroup analysis, the three educational interventions, all of which were described as having multiple components, had efficacy in reducing BMI z score. MI, either focused on the parents or on the family, was not effective in any of the included studies for the treatment of overweight and obesity in preschool aged children. Two of the interventions included MI over the telephone if parents missed a session. The implications of providing an intervention virtually are largely unknown and may have affected the overall impact of the intervention, which is consistent with the previous Cochrane systematic review.[12]

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In addition to the intervention type, educational vs MI, it is important to note the educational programs had greater contact time with the families. A previous systematic review concluded that at least 26 hours of multicomponent intervention over a minimum of 6 months improved BMI z score.[25] Of the four trials reported in this review, only the MEND educational intervention by Butte et al achieved at least 26 total hours of intervention.[22] On the contrary, MI interventions were limited to 3 or 7.5hrs of total contact time.[21; 23] Given these differences, one cannot suggest with confidence that it was the intervention type (i.e. education vs MI) and not the 'intensity' of the intervention that led to differences in effectiveness.

Overall, the four trials included in this study described their interventions in sufficient detail in the methods sections, or via reference to a previously published protocol paper. However, the standard care provided to the control group was not often described in sufficient detail to allow replication. Trials need to be fully transparent when describing the details of usual or standard care control groups, to truly understand the effectiveness of interventions in the real-world setting. This is an important area for further exploration in future studies and reviews.

To the best of our knowledge, this was the first systematic review in children with obesity to utilize the RITES tool. Generally, the four trials were more pragmatic than efficacious on the effectiveness-efficacy scale. Effective or pragmatic trials are similar to routine clinical practices and indicate a more 'real-world' setting. The fact that all trials were conducted in the primary care setting, or a combination of primary care and the community, supports a pragmatic approach; however, the level of training and resources required to implement the interventions in real-world scenarios may pose a challenge. Additionally, most trials recruited participants following stringent criteria, excluding those with comorbidities, and resulting in less pragmatism.[26] This limits the generalizability of the findings beyond preschool age children with overweight or obesity only.

Limitations

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The search strategy for this systematic review was influenced by the previously published Cochrane review.[12] Bias was introduced as the searches performed and trials included in this review were restricted to English. Another limitation was the methodological heterogeneity between trials precluded interpretation of an overall effect of interventions on improving BMI z score. However, our subgroup analysis based on intervention modality resulted in lower heterogeneity, and we calculated a greater decline in BMI z score for those undergoing educational interventions. Nonetheless, the certainty of the evidence for primary outcomes ranged from low to moderate. The main methodological concern from review authors was the substantial attrition bias in all trials (>20%). Unfortunately this is common in pediatric weight management.[27] Most trials declared that the participants and healthcare providers were not blinded to their intervention or control group, leading to concerns of performance bias. However, weight management interventions that involve services from healthcare providers make it difficult to blind personnel. Because of this, we did not downgrade for blinding when examining overall quality of evidence for the primary outcomes.

Future directions

There continues to be limited evidence surrounding the best practice for pediatric weight management in young children. Furthermore, efficacy has been almost entirely based on anthropometric health outcomes alone. Secondary outcomes are inconsistently reported and further evidence of the impact of these intervention programs on secondary outcomes including measures of cardiometabolic health and health related quality of life will be important given their relevance to clinical practice. Future interventions should also consistently consider potential adverse health outcomes.

Fundamentally, emphasis on a continued search for novel pathways to intervene upon or new approaches to current paradigms must be undertaken given the modest outcomes to date. Furthermore, although modest changes in BMI z score were noted, the persistence of the

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impact of these interventions over time remains uncertain and is an important area for further research.

Conclusions

There continues to be limited evidence regarding the treatment of obesity in children of preschool age. In both this updated systematic review and the previous Cochrane review, there was limited evidence of the benefit of MI as a treatment option. There is some evidence that education programs involving parents from the start of the intervention appear to assist with BMI z score reduction, but further research is required to improve on the care of young children with obesity.

Contributors

SN, PGM, RG, and KMM contributed to the conceptualization of the study. SN, PGM, RG, and CD led the data searches, data extraction, and quality appraisal. PGM was responsible for data analysis with guidance from LM. SN, PGM, and RG drafted the initial manuscript. All authors contributed to the interpretation of findings, critically reviewed and provided feedback on multiple iterations of the manuscript, and approved the final version of the manuscript for submission.

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Competing interests

KMM is an advisory board member for Novo Nordisk and a member of the Data Safety Monitoring Board for Novartis.

Ethics statement

This meta-analysis did not involve individual participant data; therefore, Institutional Review Board approval was not required for this study.

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Data availability statement

Data are available upon reasonable request from the corresponding author (kmorrison@mcmaster.ca).

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Table 1. Characteristics of included studies

Source	Setting	Intervention(s) and comparator(s)	Sample size	Sex (F, (%))	Age (mean (SD))	BMI measure (baseline)	Intervention (duration)	Outcomes
Rifas-Shiman 2017	Primary care clinics	I: High five for kids (motivational interviewing)	253	121 (47.8)	4.8 (1.2)	BMI z score: 1.88 (0.69) BMI: 19.2 (2.6)	12 mo. intervention: 4 (25 min) in-person visits + 2 (15 min) phone calls 12 mo. maintenance period: 2 in-person intervention visits Total contact = 8 sessions (3hrs)	Primary: BMI z score (CD Secondary: Television viewing, intake of fast foo and sugar sweetened beverages
		C: Usual care	192	94 (49.0)	5.2 (1.1)	BMI z score: 1.82 (0.56) BMI: 19.1 (2.0)	Baseline and annual well child care visits	
Butte 2017	Community-centered programs and Primary care clinics	I: Mind, Exercise, Nutrition… Do it (MEND)	100	51 (51.0)	2 – 5*	BMI %ile: 113.0 (15.2) BMI: 20.5 (0.3)	3 mo. intensive phase: 9 (90 min) weekly sessions 9 mo. transition phase: (90 min) sessions/mo. Total contact = 18 sessions (27hrs)	Primary: %BMIp95 (CDC) Secondary: Body composition, blood pressu Peds QL and Strengths a Difficulties measured at baseline, 3 and 12 month
		C: 12-month clinic-based program	60	29 (49.0)	2 – 5*	BMI %ile: 110.2 (13.7) BMI: 20.0 (0.3)	12 mo. clinic based program Total contact = 8hrs	
Stark 2018	Primary care clinics	I1: Learning about Activity and Understanding Nutrition for Child Health (LAUNCH)	47	25 (53.2)	4.6 (1.0)	BMI z score: 2.41 (0.53) BMI %ile: 98.6 (1.23)	3 mo. intensive phase: weekly (90/60 min) sessions 3 mo. maintenance phase: biweekly (90/60 min) sessions Total contact = 18 sessions (23hrs)	
		I2: Motivational interviewing	50	29 (58.0)	4.6 (0.9)	BMI z score: 2.41 (0.56) BMI %ile: 98.5 (1.3)	3 mo. intensive phase: weekly sessions 3 mo. maintenance phase: biweekly sessions Total contact = 18 sessions (7.5hrs)	Primary: BMI z score (CD Secondary: %BMIp95, (C parental BMI
		C: Standard care	54	32 (59.3)	4.6 (0.9)	BMI z score: 2.48 (0.70) BMI %ile: 98.6 (1.3)	Informed caregivers of child's weight status, but no treatment given	
Ek 2019	Primary care clinics	I1: No booster	43	23 (53.2)	5.2 (0.9)	BMI z score: 3.1 (0.70) BMI: 21.9 (2.3)	12 mo. intervention: 10 (90 min) sessions Total contact = 10 sessions (15hrs)	Primary: BMI z score (IOT Secondary: Child BMI, wa circumference, metabolic health, lifestyle patterns,

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	I2: Booster	44	19 (43.2)	5.2 (0.8)	BMI z score: 3.0 (0.50) BMI: 21.4 (1.5)	12 mo. intervention: 10 (90 min) sessions + 7 (30 min) phone calls Total contact = 17 sessions (18.5hrs)	obesity-related child behaviors, lifestyle specifi self-efficacy
	C: Standard care	87	56 (64.4)	5.3 (0.7)	BMI z score: 2.9 (0.60) BMI: 21.3 (1.7)	≥4 individual visits of ~30 minutes over 12 months Total contact = min. 4 sessions (2hrs)	

* = average age not reported in study

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Table 2. Quality of evidence

No. of studies	Outcome	Design	Risk of bias	Precision	Consistency	Directness	Other considerations	No. of patients	Effect	Quality
3	BMI z score	RCT	Serious Limitations[1]	No serious imprecision	No serious consistency	No uncertainty	None	329	-0.14 (-0.29, 0.01)	Moderate ⊕⊕⊕⊖
2	BMI %95	RCT	Serious Limitations[1]	Serious[2]	Serious consistency[3]	No uncertainty	None	110	-1.60 (-4.74, 1.55)	Low ⊕⊕⊖⊖

1. Studies rated down for attrition bias with over 20% loss to follow-up
2. Rated down due to wide confidence intervals that overlap no effect
3. High I^2, even after sub-group analysis was performed

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Figure 1. PRISMA Flow Diagram

Figure 2. Forest plot showing the mean difference in **BMI z score** comparing intervention versus control

Figure 3. Forest plot showing the mean difference in **%BMIp95** comparing intervention versus control

Figure 4. Forest plot showing the mean difference in **BMI z score** comparing educational intervention versus standard care

Figure 5. Forest plot showing the mean difference in **%BMIp95** comparing educational intervention versus standard care

Figure 6. Forest plot showing the mean difference in **BMI z score** comparing motivational interviewing versus standard care

Figure 7. Methodological quality summary: authors' judgments for each methodological quality item for each included study

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Figure 1. PRISMA Flow Diagram

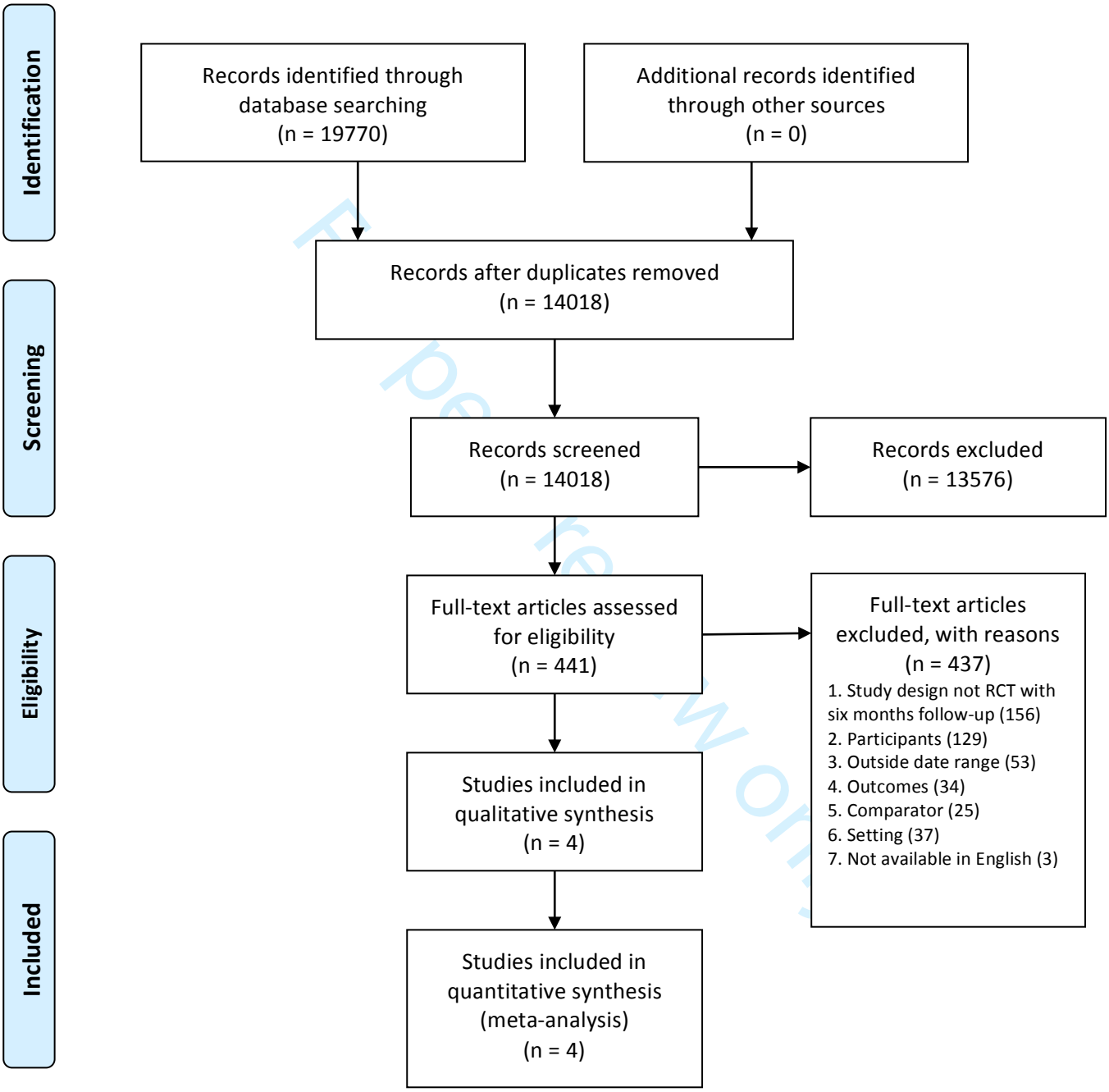
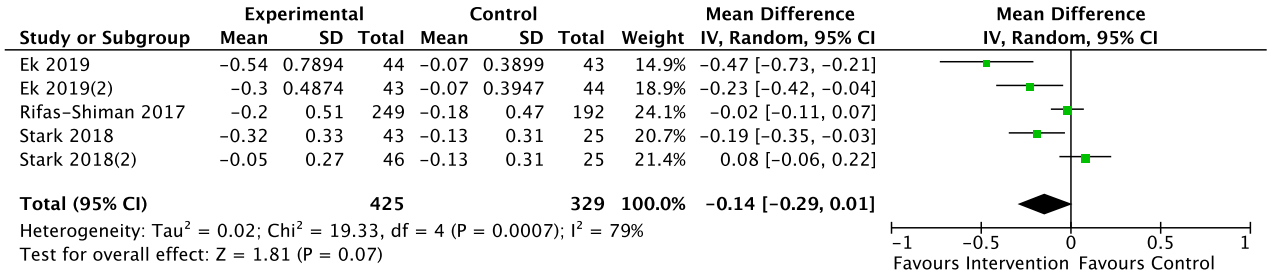


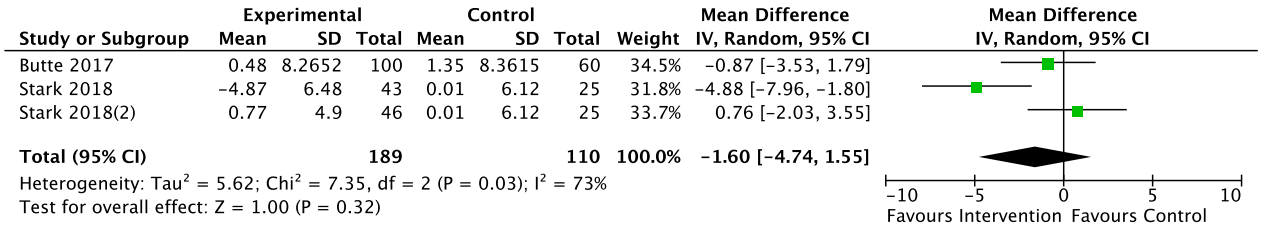
Figure 2. Forest plot showing the mean difference in **BMI z score** comparing intervention versus control



Note: Ek 2019 = Booster intervention; Ek 2019(2) = No booster intervention
Stark 2018 = LAUNCH intervention; Stark 2018(2) = Motivational interviewing (parent-only) intervention

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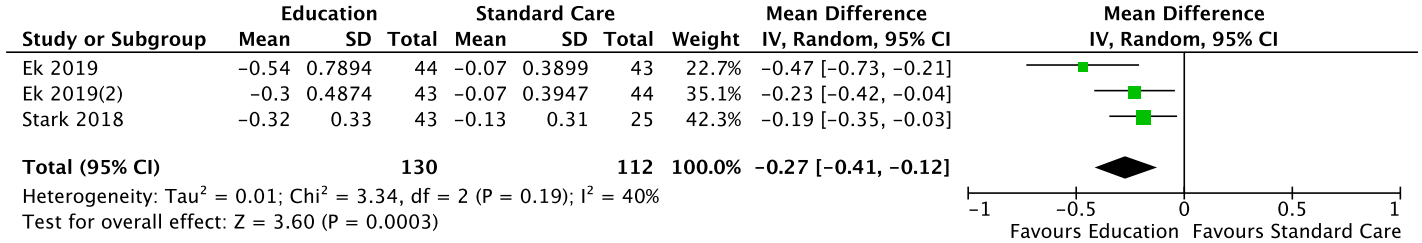
Figure 3. Forest plot showing the mean difference in **%BMIp95** comparing intervention versus control



Note: Stark 2018 = LAUNCH intervention; Stark 2018(2) = Motivational interviewing (parent-only) intervention

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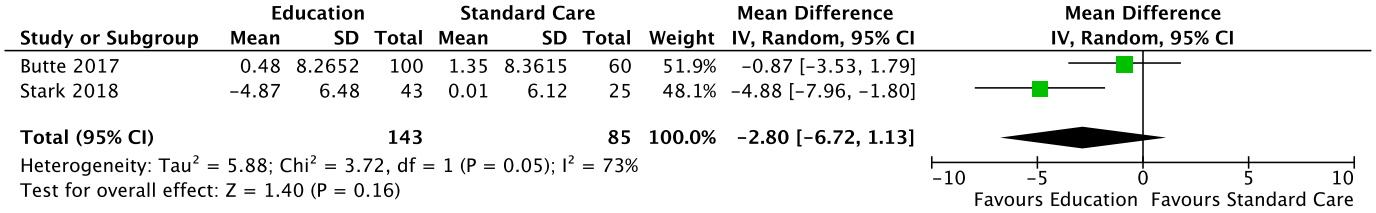
Figure 4. Forest plot showing the mean difference in **BMI z score** comparing educational intervention versus standard care



Note: Ek 2019 = Booster intervention; Ek 2019(2) = No booster intervention; Stark 2018 = LAUNCH intervention

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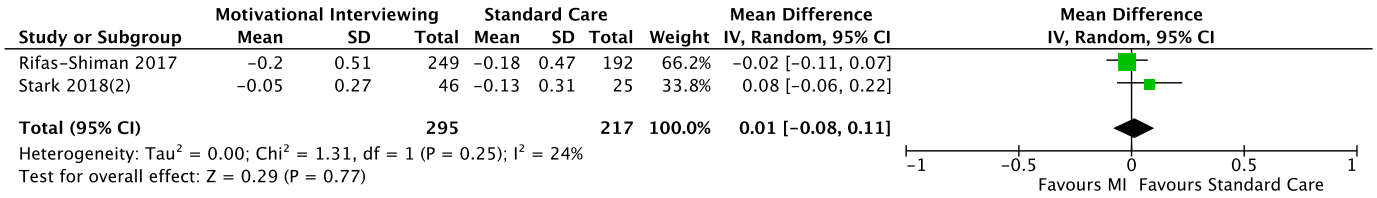
Figure 5. Forest plot showing the mean difference in **%BMIp95** comparing educational intervention versus standard care



Note: Stark 2018 = LAUNCH intervention

For peer review only

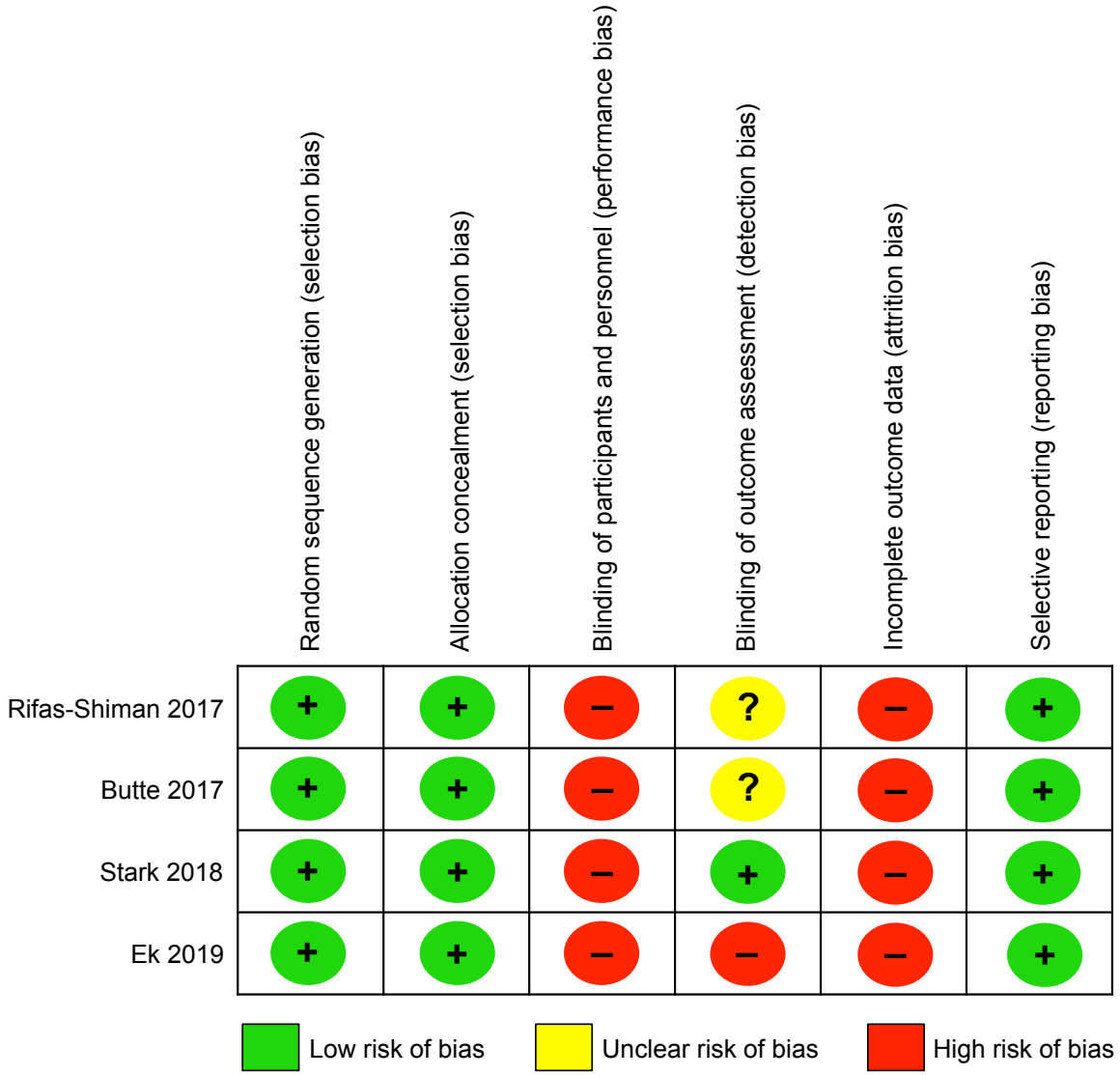
Figure 6. Forest plot showing the mean difference in **BMI z score** comparing motivational interviewing versus standard care



Note: Stark 2018(2) = Motivational interviewing (parent-only) intervention

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Figure 7. Methodological quality summary: authors' judgments for each methodological quality item for each included study



Medline

Part I: Obesity

1. Obesity/
2. Obesity, Morbid/
3. Obesity, Abdominal/
4. Pediatric Obesity/
5. Overweight/
6. Weight Loss/
7. Body Mass Index/
8. (adipos* or obes*).tw.
9. (overweight* or over weight*).tw.
10. (weight adj1 (reduc* or los* or control* or manage*)).tw.
11. or/1-10

Part II: Intervention

12. exp Behavior Therapy/
13. exp Counseling/
14. Family Therapy/
15. Social Support/
16. Program Evaluation/
17. exp Exercise/
18. exp Exercise Therapy/
19. exp Physical Education/
20. exp Exercise Movement Techniques/
21. Motor Activity/
22. exp Diet/
23. exp Diet Therapy/
24. Patient Education as Topic/
25. Health Education/
26. exp Health Behavior/
27. exp Health Promotion/
28. Life style/
29. ((obesity adj3 intervention) or program or programme or camp?).tw
30. (lifestyle or life style).tw.
31. exercis*.tw.
32. (physic* adj (activ* or fit*)).tw.
33. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
34. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*).tw
35. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
36. ((group or family or cognit* or behav*) adj therap*).tw.
37. counsel?ing.tw.
38. educat*.tw.
39. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
40. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw

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41. or/12-40

Part III: Part I + Part II and additional MeSH/subheading combination

42. 11 and 41
43. Obesity/ or Obesity, Morbid/ or Overweight/ or Weight Loss/
44. diet therapy.fs. or prevention & control.fs. or rehabilitation.fs. or therapy.fs. or psychology.fs
45. 43 and 44
46. 42 or 45

Part IV: Population [adapted from Leclercq 2013]

47. exp Child/
48. Pediatrics/
49. minors.tw.
50. (boy*).tw.
51. girl*.tw.
52. toddler?.tw
53. (kid or kids).tw.
54. (child or childs or children* or childhood* or childcar*)tw
55. (preschool* or pre-school* or toddler* or kindergar* or nursery school*).tw
56. juvenil*.tw.
57. youth*.tw.
58. (underage* or under age*).tw.
59. p?ediatric*.tw.
60. or/47-59

Part V: Part III AND IV and additional MeSH/subheading combination

61. 46 and 60
62. Pediatric Obesity/
63. diet therapy.fs. or prevention & control.fs. or rehabilitation.fs. or therapy.fs. or psychology.fs
64. 62 and 63
65. 61 or 64

Part VI: Study filter [Cochrane Handbook 2008 RCT filter - sensitivity and precision maximizing version]

66. randomized controlled trial.pt.
67. controlled clinical trial.pt.
68. randomi?ed.ab.
69. placebo.ab.
70. clinical trials as topic/
71. randomly.ab.
72. trial.ti.
73. or/66-72
74. exp animals/ not humans/
75. 73 not 74

Part VII: Part V + Part VI

76. 65 and 75

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Embase

Part I: Obesity

1. obesity/
2. morbid obesity/
3. abdominal obesity/
4. childhood obesity/
5. weight reduction/
6. weight control/
7. body mass index/
8. (adipos* or obes*).tw.
9. (overweight* or over weight*).tw.
10. (weight adj1 (reduc* or los* or control* or manage*)).tw.
11. or/1-10

Part II: Intervention

12. behavior therapy/
13. cognitive therapy/
14. exp counseling/
15. family therapy/
16. social support/
17. exp program evaluation/
18. exp exercise/
19. exp physical education/
20. exp physical activity/
21. exp motor activity/
22. training/
23. exp diet/
24. exp diet therapy/
25. nutritional health/
26. child nutrition/
27. feeding behavior/
28. patient education/
29. health promotion/
30. health literacy/
31. nutrition education/
32. health education/
33. lifestyle/
34. lifestyle modification/
35. ((obesity adj3 intervention) or program or programme or camp?).tw
36. (lifestyle or life style).tw.
37. exercis*.tw.
38. (physic* adj (activ* or fit*)).tw.
39. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
40. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*).tw
41. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
42. ((group or family or cognit* or behav*) adj therap*).tw.

For peer review only

43. counsel?ing.tw.
44. educat*.tw.
45. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
46. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw
47. or/12-45

Part III: Part I + Part II and additional MeSH/subheading combination

48. 11 and 47
49. obesity/ or morbid obesity/
50. pc.fs or rh.fs or th.fs. [prevention.fs. or rehabilitation.fs. or therapy.fs.]
51. 49 and 50
52. 48 or 51

Part IV: Population [adapted from Leclercq 2013]

53. juvenile/
54. child/
55. toddler/
56. preschool child/
57. pediatrics/
58. minors.tw.
59. boy*.tw.
60. girl*.tw.
61. toddler?.tw.
62. (kid or kids).tw.
63. (child or childs or children* or childhood*).tw
64. juvenil*.tw.
65. youth*.tw.
66. (underage* or under age*).tw.
67. p?ediatric*.tw.
68. or/53-67

Part V: Part III AND IV and additional MeSH/subheading combination

69. 52 and 68
70. childhood obesity/
71. pc.fs or rh.fs or th.fs. [prevention.fs. or rehabilitation.fs. or therapy.fs.]
72. 70 and 71
73. 69 or 72

Part VI: Search Filters (Cochrane Handbook)

74. 'crossover procedure':de
75. 'double-blind procedure':de
76. 'randomized controlled trial':de
77. 'single-blind procedure':de
78. (random* OR factorial* OR crossover*
79. cross NEXT/1 over*
80. placebo*
81. doubl* NEAR/1 blind*
82. singl* NEAR/1 blind*
83. assign*

For peer review only

84. allocat*
85. (volunteer*):de,ab,ti
86. or/74-85

Part VII: Part V + Part VI

87. 73 and 86

PsychINFO

Part I: Obesity

1. exp Overweight
2. exp Body Mass Index/
3. (adipos* or obes*).tw.
4. (overweight* or over weight*).tw.
5. or/1-4

Part II: Intervention

6. Weight Control/
7. Weight Loss/
8. Aerobic Exercise/
9. Diets/
10. exp Exercise/
11. Movement Therapy/
12. Dance Therapy/
13. exp Physical Activity/
14. Physical Fitness/
15. Health Behavior/
16. Health Promotion/
17. Health Knowledge/
18. Health Literacy/
19. Health Education/
20. Client Education/
21. Lifestyle/
22. Physical Education/
23. exp Program Evaluation/
24. Educational Programs/
25. Educational Therapy/
26. exp Program Development/
27. Counseling/
28. Group Counseling/
29. Family Therapy/
30. Support Groups/
31. Social Support/
32. exp Behavior Modification/
33. Cognitive Behavior Therapy/
34. Cognitive Therapy/
35. ((obesity adj3 intervention) or program or programme or camp?).tw

For peer review only

36. (lifestyle or life style).tw.
37. exercis*.tw.
38. (physic* adj (activ* or fit*)).tw.
39. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
40. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*)
41. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
42. ((group or family or cognit* or behav*) adj therap*).tw.
43. counsel?ing.tw.
44. educat*.tw.
45. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
46. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw
47. or/6-46

Part III: Part I + Part II

48. 5 and 47

Part IV: Population [adapted from Leclercq 2013]

49. minors.tw.
50. (boys*).tw.
51. girl*.tw.
52. toddler?.tw.
53. (kid or kids).tw.
54. (child or childs or children* or childhood*)tw
55. juvenil*.tw.
56. youth*.tw.
57. (underage* or under age*).tw.
58. p?ediatric*.tw.
59. or/49-58

Part V: Part III AND IV and additional MeSH/subheading combination

60. 48 and 59

Part VI: Study filter [Eady 2008filter - BS version]

61. control*.tw. OR random*.tw. OR exp Treatment/

Part VII: Part V + Part VI

62. 60 and 61

CINAHL

Part I: Obesity

1. MH ”Obesity+“
2. MH "Body Mass Index"
3. TX (adipos* or obes*)
4. TX (overweight* or ”over weight*“)
5. S1 OR S2 OR S3 OR S4

Part II: Intervention

For peer review only

6. MH ”Weight Loss“
7. MH ”Behavior Modification+“
8. MH ”Counseling“
9. MH ”Family Therapy“
10. MH ”Support, Psychosocial“
11. MH ”Support Groups“
12. MH ”Program Evaluation“
13. MH ”Program Implementation“
14. MH ”Exercise+“
15. MH ”Sports+“
16. MH ”Therapeutic Exercise+“
17. MH ”Physical Fitness“
18. MH ”Physical Education and Training+“
19. MH ”Health Education+“
20. MH ”Diet+“
21. MH ”Diet Therapy+“
22. MH ”Health Behavior “
23. MH ”Eating Behavior“
24. MH ”Health Promotion“
25. MH ”Life style changes“
26. MH ”Life style“
27. TX (weight N1 (reduc* or los* or control* or manage*))
28. TX ((obesity N3 intervention) OR program OR programme OR camp#)
29. TX (lifestyle or ”life style“)
30. TX exercis*
31. TX (physic* N1 (activ* or fit*))
32. TX (walk* or jog* or swim* or weight lift* or danc* or aerobics)
33. TX ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) N3 train*)
34. TX (behavio#ral or behavio#r modification or psychoth* or psychosocial)
35. TX ((group or family or cognit* or behav*) N1 therap*)
36. TX counsel#ing
37. TX educat*
38. TX ((parent# or family) N1 (based or focused or directed or centered or only or led))
39. TX (diet* or ”healthy nutrition“ or (nutrition* N1 (knowledge or educat* or therap* or program* or intervention*)))
40. S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR 37 OR 38

Part III: Part I + Part II and additional MeSH/subheading combination

41. S5 AND S40
42. MH ”Obesity+/DH/ED/PC/PF/RH/TH“) [diet therapy or education or prevention & control or psychosocial factors or rehabilitation or therapy]
43. S40 OR S41

Part IV: Population [based on Leclercq 2013]

For peer review only

44. MH ”Child+“
45. MH ”Pediatrics“
46. TX minors
47. TX (boy*)
48. TX girl*
49. TX toddler#
50. TX (kid OR kids)
51. TX (child OR childs OR children* OR childhood*)
52. TX juvenil*
53. TX youth*
54. TX (underage* or under age*)
55. TX (paediatric* OR pediatric*)
56. S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54 OR S55

Part V: Part III AND IV and additional MeSH/subheading combination

57. S43 AND S56
58. MH ”Pediatric Obesity/DH/ED/PC/PF/RH/TH“) [diet therapy or education or prevention & control or psychosocial factors or rehabilitation or therapy]
59. S57 OR S58

Part VI: Study filter [Wong 2006bfilter - SDSSGS version]

60. MH ”treatment outcomes+“ OR MH ”experimental studies+“ or random*

Part VII: Part V + Part VI

61. S58AND S59

Cochrane Library

Part I: Obesity

1. Obesity.mp
2. Obesity, morbid.mp
3. Obesity, abdominal.mp
4. Pediatric obesity.mp
5. Overweight.mp
6. Weight loss.mp
7. Body mass index.mp
8. (adipos$ or obes$).ti, ab
9. (overweight$ or (over ADJ1 weight$)).ti,ab
10. (weight ADJ1 (reduc$ or los$ or control$ or manage$)).ti,ab
11. {or #1-#10}

Part II: Intervention

12. Behavior therapy.mp
13. Counseling.mp

[mh ”Behavior Therapy“]

13. [mh ”Counseling“]
14. [mh ^”Family Therapy“]
15. [mh ^”Social Support“]

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16. [mh ^"Program Evaluation"]
17. [mh "Exercise"]
18. [mh "Exercise Therapy"]
19. [mh "Physical Education and Training"]
20. [mh "Exercise Movement Techniques"]
21. [mh ^"Motor Activity"]
22. [mh Diet]
23. [mh "Diet Therapy"]
24. [mh ^"Patient Education"]
25. [mh ^"Health Education"]
26. [mh "Health Behavior"]
27. [mh "Health Promotion"]
28. [mh ^"Life style"]
29. (("obesity" near/4 "intervention") or "program" or "programme" or "camp" or "camps"):ti,ab
31. exercis*:ti,ab
32. (physic* next (activ* or fit*)):ti,ab
33. (walk* or jog* or swim* or ("weight" next lift*) or danc* or "aerobics"):ti,ab
34. ((physic* or strength* or resist* or "circuit" or "weight" or aerob* or "cross" or "endurance" or structur*) near/4 train*):ti,ab
35. ("behavioral" or "behavioural" or (("behavior" or "behaviour") next "modification") or psychoth* or "psychosocial"):ti,ab
36. (("group" or "family" or cognit* or behav*) next therap*):ti,ab
37. (counseling or counselling):ti,ab
38. educat*:ti,ab
39. (("parent" or "parents" or "family") next ("based" or "focused" or "directed" or "centered" or "only" or "led")):ti,ab
40. diet* or "healthy nutrition" or (nutrition* next ("knowledge" or educat* or therap* or program* or intervention*))):ti,ab
41. {or #12-#40}

Part III: Part I + Part II and additional MeSH/subheading combination

42. #11 and #40
43. [mh Ôbesity] or [mh ^"Obesity, Morbid"] or [mh Ôverweight]
44. [mh /DH,PC,RH,TH,PX][diet therapy or prevention & control or rehabilitation or therapy or psychology]
45. #43 and #44
46. #42 or #45

Part IV: Population [adapted from Leclercq 2013]

47. [mh Child]
48. [mh ^Pediatrics]
49. "minors":ti,ab
50. "boys*":ti,ab
51. girl*:ti,ab
52. ("kid" or "kids"):ti,ab
53. ("toddler" or "toddlers"):ti,ab
54. ("child" or "childs" or children* or childhood*):ti,ab

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55. juvenil*:ti,ab
56. youth*:ti,ab
57. (underage* or (”under“ next age*)):ti,ab
58. (paediatric* or pediatric*):ti,ab
59. {or #47-#58}

Part V: Part III AND IV and additional MeSH/subheading combination

60. #46 and #59
61. [mh ^”Pediatric Obesity“]
62. [mh /DH,PC,RH,TH,PX]
63. #61 and #62
64. #60 or #63

Web of Science:

Obesity

TS =(obesity)
TS=(obesity, morbid)
TS=(obesity, abdominal)
TS=(pediatric obesity)
TS=(overweight)
TS=(weight loss)
TS=(body mass index)
#7 OR #6 OR #5 OR #4 OR #3 OR #2 OR #1

Intervention:

TS=(behavior therapy)
TS=(counseling)
TS=(family therapy)
TS=(social support)
TS=(program evaluation)
TS=(exercise)
TS=(exercise therapy)
TS=(physical education and training)
TS=(exercise movement techniques)
TS=(motor activity)
TS=(diet)
TS=(diet therapy)
TS=(patient education)
TS=(Health Education)
TS=(Health Behavior)
TS=(Health Promotion)
TS=(Life style)
#9 OR #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21or #22 or #23 or #24 or #25

Part III: Part I + Part II

1. #8 AND #26

Population

TS=(Child*)

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TS=(Pediatric*)
TS=(Preschool child*)
TS=(Toddler*)
TS=(Minor*)
TS=(Youth*)
TS=(Kid*)
#28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34

Study filters

TS= (clinical trial*)
TS= (research design)
TS= (controlled trial*)
TS= (single blind*)
TS= (double blind*)
TS= (follow-up stud*)

Clinicaltrials.gov

Conditions: obese OR overweight OR obesity
Study Type: Interventional Studies
Age Group: Child (birth-17)

WHO Clinical Trial Registry

Activated "search for clinical trials in children"
Title: obes* OR overweight*
OR
Condition: obes* OR overweight*
Recruitment: All

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Article Title	Efficacy of a community- versus primary care-centered program for childhood obesity: TX CORD RCT	
Authors	Butte, N., Hoelscher, D., Barlow, S., Pont, S., Durand, C., Vandewater, E., Liu, Y., Adolph, A., Pérez, A., Wilson, T., Gonzalez, A., Puyau, M., Sharma, S., Byrd-Willams, C., Oluyomi, A., Huang, T., Finkelstein, E., Sacher, P. & Kelder, S.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	TX CORD: Texas Childhood Obesity Demonstration project **Intervention:** MEND2-5 (Mind, Exercise, Nutrition…Do it!) program **Comparison**: Next Steps program	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	The TX CORD project was designed to address childhood obesity through an integrated, systems-oriented model that incorporated primary and secondary obesity interventions. Within this population-level, systems-based intervention was a 12-month randomized controlled trial targeting low-income, ethnically diverse children with overweight or obesity and their families. It is hypothesized that the 12-month, community-centered program would significantly reduce BMI compared to the health care-centered program.	1585 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Intervention :** Conducted at the YMCA within 5 miles of the participants. Education materials surrounding general nutrition, PA and behaviours were needed (not included in paper/protocol). Based on CATCH program. All programs were delivered in English or Spanish. A digital scale and stadiometer for weight and height measurements. Transition program includes the book *A Year of Being Well* (role models stories of low-income families using CDC obesity prevention behaviours), cooking classes	74-76 (2)

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	based on *The Happy Kitchen/La Coicina Alegre* and MED World online/print material. **Control:** Electronic medical record at the primary care physician's office. Counselling materials for the primary care provider. Activity booklet for parents and children. Material was available in both English and Spanish, and available through the AAP website (https://shop.aap.org/next-steps-a-practitioners-guide-for-themed-follow-up-visits-for-their-patients-to-achieve-a-heal/).	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention:** A multicomponent, healthy lifestyle program including behavioural, nutrition and physical activity sessions. Included 10 weekly 90-minute sessions relating to general nutrition, physical activity and behaviours for children including: - 30 minutes of guided active play - 15 minutes of healthy snack time evidence-based, exposure-based technique to promote acceptance and increased intake of fruit and vegetables - 45 minutes where children participate in creative play activities while parents attend an interactive education and skill development session based on group-based parent training principles. The transition program includes monthly 90-minutes post-program sessions that children and parents attend together. Sessions include a brief review of the MEND concepts, role model stories and cooking lessons. **Control:** A clinic-based program that optimizes health care procedures to assess BMI and provides brief counselling materials to support family/child self-improvements in healthy lifestyles. Electronic health record (BMI screening, alerts for BMI ≥ 85th centile, specific diagnostic codes, lab tests and referrals common in obesity management) changes to support childhood obesity visits. Counselling materials were provided to the primary care providers. Next Steps activity booklets for parents and children to work on nutrition and physical activity targets in a self-directed manner. Families were encouraged to seek follow up clinic visits to address	74-75 (2) 75 (2), 1586 (1)

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	childhood obesity if identified as being obese. The brief counselling materials include information on facilitating clinical intervention and supporting educational materials for families.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention**: All sessions were delivered by bilingual allied health professionals (dieticians and health educations) and were assisted by community health workers. Community health workers were trained in MEND, motivational interviewing, nutrition and physical activity programming. **Control:** Program was delivered by primary care physicians in their offices, and self-directed learning by the families/children. No specific training, however counselling material was provided to the primary care physicians Trained research staff measured weight and height (?unclear training)	75 (2) 1586 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Intervention:** The intervention was delivered at the YMCA face-to-face. ? It was not stated whether this was individual sessions or group sessions. Measurements were performed at the Children's Nutrition Research Centre in Houston or YMCAs/offices in Austin. **Control:** Delivered face-to-face in primary care physician's office, as well as workbooks provided to complete individually at home.	1585 (1) 1585 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Intervention:** conducted at YMCAs in Austin and Houston. Measurements were performed at the Children's Nutritional Research Centre in Houston or YMCAs/offices in Austin. ?No information regarding how participants accessed this **Control**: primary care clinics in Texas	1585 (1)

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When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Intervention:** The intensive phase consisted of nine weekly 90-minute sessions. The transition phase offered 90-minutes session including MEND reviews (nine months in total). In total, MEND2-5 offered 27 hours of contact. **Intervention:** A clinic-based program – families were provided with self-directed education material as well as encouraged to seek follow-up clinic visits to address childhood obesity. An estimated 8-hours of contact was provided.	1585 (1), 75 (2) 1585-1586 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	"extensively adapted for the US population" (from British study) - ?did not report how it was adapted "pilot tested program elements and used qualitative data to further tailor the program to our population" TX CORD culturally adapted Next Steps material for low-income, urban, black and Latino children in both English and Spanish versions.	1585 (1) 76 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Fidelity to the session content and participant engagement was assessed using instructor checklists. Compliance in the intervention arm was based on attendance (the number of sessions attended out of the number offered). Retention was based on completion of measurements at 3 and 12 months.	1587 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	Retention rates were 64% for the intervention group compared to 87% for the comparison at 3 months. The retention rate at 12 months was 54% for the intervention group compared to 82% for the comparison group.	1587-1588 (1)

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	During the Intensive Phase, 86% of the sessions were delivered completely as planned. During the Transition Phase, 75% of the sessions were delivered completely as planned. Mean attendance was 46% during the Intensive Phase and 16% during the Transition Phase. Fidelity was not assessed for the comparison group.	1590 (1)

** **Authors -** use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

(1) Butte, NF, Hoelscher, DM, Barlow, S, et al. Efficacy of a community- versus primary care-centered program for childhood obesity: TX CORD RCT. Pediatric Obesity 2017; 25:1584-1593

(2) Hoelscher DM, Butte NF, Barlow S, et al. Incorporating primary and secondary prevention approaches to address childhood obesity prevention and treatment in a low-income, ethnically diverse population: study design and demographic data from the Texas Childhood Obesity Research Demonstration (TX CORD) study. Child Obes 2015;11:71-91.

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Article Title	A parent treatment program for preschoolers with obesity: A randomized controlled trial	
Authors	Ek, A., Chamberlain, K., Sorjonen, K., Hammar, U., Malek, Mahnoush., Sandvik, P., Somaraki, M., Nyman, J., Lindberg, L., Nordin, K., Ejderhamn, J., Fisher, P., Chamberlain, P., Marcus, C., & Nowicka, P.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	More or Less Study: 1) Parent-only group treatment **with** follow-up booster sessions 2) Parent only group treatment **without** booster sessions 3) Standard treatment	2 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Given the significant role of parents in a young child's life, the treatment program created was grounded in evidence-based parenting practices to evaluate whether a focus on parenting would be especially suitable for this age group as suggested by previous papers. It is hypothesized that the parent-only treatment would be more effective compared to standard treatment.	2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Intervention** - Based on the KEEP (Keeping Foster and Kin Parents Supported and Trained) program – a positive parenting program. The More or Less program was developed in collaboration with the KEEP program developers with conceptual and cultural adaptations designed for Swedish population. Involved home assignments (not provided as part of paper/protocol) In collaboration with the KEEP developers, the authors translated and adapted the parent group manual outlining the content of each session and parent handouts (not included in the protocol/paper).	2 (1) 6 (2)

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	Control -receive treatment as usual in an outpatient pediatric unit. A questionnaire was developed to monitor the healthcare providers involved (as it varied), and the number and focus of the visits. The questionnaire was not included as part of the protocol or paper.	6 (2)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention -** ML program consisted of 10 weekly 90-minute sessions with parents only, with childcare provided. Each session centered around a parenting practice as well as a lifestyle component aimed to provide parents with knowledge/skills to support a healthy home environment. The session was finished with a home practice assignment related to the focus of the day. If parents missed a session, material was sent home and key parts were reviewed over the phone by 1 of the group leaders. **More or Less with Booster** – Group leaders called participants every 4-6 weeks for a 30-minute phone call (max 7 calls). The parents were referred back to the discussions had during the program and to the manual for additional support **Control** – based on the action plan for childhood obesity in Stockholm County (not described in the paper). Received at least 4 treatment visits, typically the first with the physician, and subsequent with the pediatric nurse. The visits centered around lifestyle advice. Reasons for treatment are discussed including weight gain and future risk factors. Treatments varied between pediatric units – a questionnaire was included regarding questions about the professions of the involved healthcare providers, number and focus of visits.	3 (1) 6-8 (2) 2(1) 6 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention** – Led by 5 registered dieticians – they received training by the KEEP developers, who also provided weekly supervision on videotaped sessions (translated from Swedish to English) for program certification and fidelity. The initial training was a 5-day experiential training including information on program's theory and practice in the delivery of group sessions. Each session of the ML program was thoroughly discussed, and each trainee role played facilitating several key sessions.	3 (1) 7 (2)

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	The training continued through external supervision from the KEEP developer group. **Control** – first and last visit were with a pediatrician, and the remaining visits were with a pediatric nurse (minimum of 4 visits). Some children were referred to dieticians and/or PTs.	3 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**More or Less without Boosters**– face-to-face group sessions **More or Less with booster** – face-to-face group sessions followed by seven 30-minute phone calls every 4-6 weeks (9 months in total) Neither the group leaders or parents were informed about which families were allocated to the boosters until after the group sessions were finished in order to avoid differential treatment. **Control** – delivered in person during routine visits to pediatrician	3 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Intervention -** Meetings were held in health care facilities. Both parents were invited to participate and childcare was provided. **Control** – Delivered at 14 outpatient pediatric clinics	3 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**No Booster** – 10 weekly 90-minute sessions **Booster** – 10 weekly 90-minute sessions with 30-minute phone calls (max 7) every 4-6 weeks for the following 9 months. **Control** – Were offered ≥ 4 individual visits of around 30 minutes over 12 months. Treatment setup varied between clinics – each clinic completed a protocol at every visit documenting details (*not available in paper*)	3 (1) 6 (2)
Tailoring		

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If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention** – The parent group manual outlining each session and parent handouts was translated and adapted from the original KEEP developers' version. **Control** – treatment set up varied between practitioners and location. Some children were referred to dieticians and/or PTs. Number of appointments with pediatrician/pediatric nurse also varied.	6 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	A 6-month follow-up was added post baseline to better evaluate the treatment program	2 (1)
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Intervention** - Training of the ML group leaders was conducted by the KEEP developers. 5-day experiential training session. The training continued through external supervision from the KEEP group – all parent groups were videotaped (and translated to English). KEEP consultants viewed the recordings, rated them for fidelity and identified areas for reinforcement and feedback Group leaders completed a session review with questions regarding what went well and challenges experienced. Also completed weekly forms on parent attendance and engagement ratings – these measures informed the consultation process **Control -** Because the treatments vary between pediatric units, a questionnaire including questions about the profession of healthcare providers involved as well as the number and focus of visits	7 (2) 6 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**Intervention**– Not reported in the final paper. **Control** – Each clinic filled out a protocol at every visit documenting what treatment the family had attended and on how many occasions.	3 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

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(1) Ek A, Lewis Chamberlain K, Sorjonen K, et al. A Parent Treatment Program for Preschoolers With Obesity: A Randomized Controlled Trial. Pediatrics. 2019;144(2): e20183457

(2) Ek A., Lewis Chamberlain K., Ejderhamn J., et al. The more or less study: A randomized controlled trial testing different approaches to treat obesity in preschoolers. Public Health. 2015. 15:735: DOI 10.1186/s12889-015-1912-1

For peer review only

Article Title	Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study (1)	
Authors	Sheryl L. Rifas-Shiman, Elsie M. Taveras, Steven L. Gortmaker, Katherine H. Hohman, Christine M. Horan, Ken P. Kleinman, Kathleen Mitchell, Sarah Price, Lisa A. Prosser and Matthew W. Gillman	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	High Five for Kids primary care-based pre-schooler obesity intervention	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Targeting preschool-age children may be beneficial because children of this age visit their primary care clinicians frequently for routine medical care and parents have more control over their children's health-related behaviours, and these behaviours may be more malleable. The purpose of the study was to assess the effectiveness of the primary care-based obesity intervention over a 2-year period. The overarching model for this intervention was the Chronic Care Model which posits that changes in primary care to produce functional patient outcomes requires changes for all members of the practice team.	24 (2) 2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where	**Intervention:** Posters were developed for the waiting rooms that encouraged the five target behaviours of the High Five. An image of this poster is available at: https://pubmed.ncbi.nlm.nih.gov/21464376/. Educational modules were developed to target fast-food and sugar-sweetened beverage consumption, as well as television viewing (https://www.researchgate.net/publication/261773967_Health_Behavior_Change_A_Guide_For_Practitioners). A list of local resources for physical activity was also available but not provided in the study write-up, as was a website with recipes and educational materials. Small physical incentives (water bottles, snack containers, books) were provided by nurse practitioners to participants, as was an	2-3 (2)

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the materials can be accessed (e.g. online appendix, URL).	electronic tool to monitor television-viewing time (more information on this tool, including where to get it, was not provided in the study). The electronic medical record was enhanced to help physicians make decisions, track patients, follow-up, schedule and bill.	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention:** Intervention was based on the Chronic Care Model (3), which requires change from all members of the health care team. Participants (including families) received motivational interviewing by nurse practitioners and educational modules targeting certain behavioural modifications. Specifically, the interviewing and modules targeted reducing sugar-sweetened beverage intake, reducing television viewing time, increasing fruit and vegetable consumption, decreasing fast-food consumption and encouraging more time in active play. **Usual care:** Participants got the standard of care offered by their pediatric practice. This included the baseline and annual well-child care visits as well as follow ups with the paediatrician or subspecialist (ex. Dietician).	2 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention:** Pediatric nurse practitioners were trained for their role as the central intervening clinician. They also received training in motivational interviewing. This training included enhancing motivation for change, enhancing self-efficacy, how to help recognize inconsistencies between desired and actual behaviours, and how to help reduce this dissonance. Medical assistants were trained on how to conduct research-standard anthropometric measurements (for collecting height and weight data). Medical receptionists were trained to schedule the initial and follow-up visits with the nurse practitioners in accordance with the study protocol. Primary care pediatricians were trained in obesity management and to use brief, focused negotiation skills. **Usual care:** Participants got the standard of care offered by their pediatric practice. This included the baseline and annual well-child care visits as well as follow-ups with the pediatrician or subspecialist (ex. Dietician).	2 (2)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone)	**Intervention:** In the first year, four visits were face-to-face and two were by telephone. In the second year, both visits were face-to-face. Visits were with individual families. **Usual care:** At the discretion of the participants regular pediatrician.	2 (2)

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of the intervention and whether it was provided individually or in a group.		
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Done at 10 pediatric primary care practices within the Harvard Vanguard Medical Associates, which is a multi-site group practice in Massachusetts. Half of the practices provided usual care and half provided the intervention care.	2 (2)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Intervention:** Four in person (each 25 minutes) visits with the clinician over the first year, as well as two phone calls (15 minutes each). Two in-person visits in the maintenance year (length not described). **Usual care:** Frequency of usual care visits was determined by the routine care procedures for each particular pediatric practice.	2 (2)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention:** During their visits, the families could choose to work on one or more of the behavioural targets (previously listed).	4 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	
How Well		

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Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Intervention:** Adherence was assessed by completion of the various visits/phone calls, during data analysis. No strategies were used to improve fidelity.	5 (2), 3 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**Intervention:** In the first year (the intervention period), only 56% of participants in the intervention completed at least 2 of the 6 intervention activities (in person or phone visits). During the second year (the maintenance period), only 17% went to one visit and only 21% of participants went to both visits. Overall adherence was thus low. **Usual care:** adherence was not measured for usual care participants.	5 (2), 3 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Rifas-Shiman, S. L., Taveras, E. M., Gortmaker, S. L., Hohman, K. H., Horan, C. M., Kleinman, K. P., ... & Gillman, M. W. (2017). Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study. *Pediatric obesity*, *12*(3), e24-e27.

2. Taveras, E. M., Gortmaker, S. L., Hohman, K. H., Horan, C. M., Kleinman, K. P., Mitchell, K., ... & Gillman, M. W. (2011). Randomized controlled trial to improve primary care to prevent and manage childhood obesity: the High Five for Kids study. *Archives of pediatrics & adolescent medicine*, *165*(8), 714-722.

3. Wagner, E. H. (1998). Chronic disease management: what will it take to improve care for chronic illness?. *Effective clinical practice*, *1*(1).

For peer review only

Article Title	Clinic and Home-Based Behavioral Intervention for Obesity in Preschoolers: A Randomized Trial (1)	
Authors	Lori J. Stark, PhD, Stephanie Spear Filigno, PhD, Christopher Bolling, MD, Megan B. Ratcliff, PhD, MPH, Jessica C. Kichler, PhD, Shannon M. Robson, PhD, MPH, RD, Stacey L. Simon, PhD, Mary Beth McCullough, PhD, Lisa M. Clifford, PhD, Cathleen Odar Stough, PhD, Cynthia Zion, BS, and Richard F. Ittenbach, PhD	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Comparing LAUNCH (Learning about Activity and Understanding Nutrition for Child Health), motivational interviewing and standard care as in intervention for pre-school obesity	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Obesity in preschool years increases the risk of being overweight and obese in later childhood through to adulthood. Focusing on children with obesity is important as they are at higher risk for mortality compared with overweight peers. This study hypothesizes that a skills-based behavioural family clinic in combination with a home-based intervention would reduce BMI z score compared with motivational interviewing and standard care in preschool age children with obesity.	1,2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**LAUNCH**: Caregivers were given an iPod Touch so that they could use the "DietTracker" application to track the child's food and beverage intake. Pedometers were also provided to the children and their caregivers, to help caregivers monitor physical activity levels. Caregivers were also given a two-week supply of fresh vegetables at each of the clinic sessions to help expose children to new vegetables. **Motivational Interviewing:** Caregivers were given a pack of informational handouts from the Let's Go 5-2-1-0 program (https://www.mainehealth.org/Lets-	13, 16 (2), 135 (3)

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	Go/Childrens-Program/Parents). A motivational interviewing (MI) manual was used for MI review by interviewers and study staff (Moyers et al., *Revised global scales: Motivational interviewing treatment integrity 3.1.1*). **Standard Care:** N/A	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**LAUNCH**: Sessions alternated between clinic and home-based sessions. During the clinic sessions, the caregiver(s) and child were in different groups. Caregiver groups received education and problem-solving skills on a number of diet and physical activity, as well as strategies to manage the behaviour of their child. A list of topics by session is available in Table 1 (2). In following the teachings of the sessions, families aimed to decrease the child's screen time, increase physical activity (for which they had daily step goals for both the parents and the child), stay within the prescribed daily calorie goals for the child as well as choose healthier foods, and to record child's food and beverage intake. The behavioural strategies that they learned to help achieve these goals included praising their child's healthy dietary and physical activity choices, modelling healthy behaviours, managing tantrums by ignoring and using time-outs, contingency management, as well as how to best set up the food environment to encourage healthy choices (ex. Not having unhealthy food available). More information about the teaching sessions and strategies learned is available here (3). The child group included education about healthy eating and opportunities for physical exercise. Sessions also included a light nutritious meal provided to caregivers and children, which served as exposure to fruits and vegetables for the children. During home visits, families got a chance to practice and receive feedback on the skills they had learned in clinic, as well as to problem solve any barriers they were facing in implementing those strategies in the home setting. **Motivational Interviewing:** MI was a caregiver-only intervention. During the first	4-5 (1), 13-15 (2), 135 (3)

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	session, caregivers were asked about their concerns for their child's diet, weight, or physical activity habits and then given a chance to set a goal that was meaningful to them. The subsequent sessions (either by phone or in person) involved a discussion of the previous goals, how their implementation was going, enhancing motivation, working on establishing the desired behaviours and choosing new goals to work towards. **Standard Care:** No intervention was provided outside of the routine pediatric care the participant was already receiving, other than to be notified of their weight.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**LAUNCH**: The LAUNCH interventionists were therapists trained on the protocol in pilot studies (more information on training was not provided). A new LAUNCH interventionist was brought on midway through the study. She had extensive previous training in pediatric behavioural weight management and was trained on the LAUNCH protocol by shadowing and co-leading sessions with one of the previous interventionists. Home visits were completed by postdoctoral fellows, who were trained by reviewing the LAUNCH manual (not provided), listening to audiotapes of home sessions, watching videotapes of child group sessions, administering role-played sessions, and shadowing a second year fellow during home visits for one cycle before being assigned their own visits. The group-sessions in the clinic were done by a licensed clinical psychologist. Fidelity coders had PhDs and expertise in obesity treatment. **Motivational Interviewing:** Interviewers were either pediatricians or PhD level psychologists. The first session was done by the pediatrician and the following sessions were done by the psychologists. Interviewers were selected for their previous MI training and experience; each interviewer had approximately 10 years of MI experience and had participated in previous structured MI training. **Standard Care:** Standard care was provided by the participant's regular pediatrician.	13, 16 (2)
How		

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Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**LAUNCH**: All sessions were done in-person. The clinic sessions were done in groups, with one group for the parents and one group for the children. Home-based sessions were done with individual families. **Motivational Interviewing:** There were 4 in-person sessions (sessions 2, 12, 16) and 14 sessions done over the phone. All sessions were done with individual caregivers. **Standard Care:** Participants in the standard care group received only a notification of their child's weight, provided by phone call and during the initial (in-person) assessment.	13 (2)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**LAUNCH**: The home sessions were done in the participants' homes, which were all within 50 miles of the treatment centre. The clinic sessions were done at a medical centre (the treatment centre). **Motivational Interviewing:** The initial assessment interview was done at the medical centre, but subsequent sessions were done at the families' homes or over the telephone. **Standard Care:** At the discretion of the participants' regular pediatricians.	117 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**LAUNCH**: Intervention sessions alternated between clinic (10 sessions) and home sessions (8 sessions), for a total of 18 sessions. For the first three months there were weekly sessions, and for the following three months sessions were every second week. The clinic-based sessions were each 90 minutes and the home visits were 60 minutes. Sessions were scheduled for the late afternoons/early evenings to minimize scheduling conflicts/barriers. **Motivational Interviewing:** The 18 sessions were delivered in alignment with the LAUNCH session schedule. Phone sessions varied in length depending on caregiver preference, but were generally between 15-35 minutes long. Home visits were all	13, 15 (2)

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	scheduled for 60 minutes. **Standard Care:** Length and frequency of appointments was determined by the participant's normal pediatrician, in line with their practice policies.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**LAUNCH**: Home visits provided the opportunity to tailor the program for individual children/families. If there was a barrier to using a clinic-taught strategy in the home, the home visit provided an opportunity to problem-solve. It was also an opportunity for targeted feedback, since therapist was able to witness the taught behavioural strategies being used in vivo. Caregivers also received weekly written feedback on their child's food and beverage consumption, including feedback on their progress towards their goals. Feedback was also provided on activity levels, based on the number of steps recorded on the pedometers. **Motivational Interviewing:** Caregivers were asked to select a specific nutrition or physical activity goal from the Let's Go 5-2-1-0 (5 fruits or vegetables a day, limit recreational screen time to 2 hours or less, get 1 or more hours of physical activity a day, limit or eliminate sugary drinks) and AAP recommendations (https://pediatrics.aappublications.org/content/120/Supplement_4/S164). This was then used as the focus of the motivational interviewing sessions. **Standard Care:** Care was personalized as appropriate by the pediatrician providing care.	13 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	**LAUNCH**: N/A **Motivational Interviewing:** N/A **Standard Care:** N/A	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve	All LAUNCH and MI sessions were videotaped and/or audiotaped. **LAUNCH**: Fidelity coders reviewed the first session of the first cycle of LAUNCH and	16 (2)

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fidelity, describe them.	provided feedback to the interventionist. One quarter of all LAUNCH sessions (both clinic and home-based) were compared to a LAUNCH checklist by a PhD level consultant with experience in weight modification treatment. If fidelity was found to drop below 90%, the therapist was provided with feedback on the components that needed improvement. **Motivational Interviewing:** For the first three cycles of interviewing, the two interviewers met together to review audiotapes of each other's MI interviews and provide feedback on MI fidelity. After the third interviewer joined the study, her audiotapes were reviewed and given regular feedback by the other two interviewers. MI fidelity coders also reviewed audiotapes regularly throughout the study and provided feedback to each interviewer on a quarterly basis. 25% of MI sessions were compared to a MI checklist ((Moyers et al., *Revised global scales: Motivational interviewing treatment integrity 3.1.1*)) to assess fidelity. If fidelity was found to drop below 90%, the therapist was provided with feedback on the components that needed improvement. **Standard Care:** N/A	
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**LAUNCH**: Fidelity was assessed as planned – see previous box. **Motivational Interviewing:** Fidelity was assessed as planned – see previous box. **Standard Care:** N/A	16 (2)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark, L. J., Filigno, S. S., Bolling, C., Ratcliff, M. B., Kichler, J. C., Robson, S. M., ... & Zion, C. (2018). Clinic and home-based behavioral intervention for obesity in preschoolers: a randomized trial. *The Journal of pediatrics*, *192*, 115-121.

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2. Stark, L. J., Filigno, S. S., Bolling, C., Ratcliff, M. B., Kichler, J. C., Robson, S. L., ... & Zion, C. (2017). Learning about Activity and Understanding Nutrition for Child Health (LAUNCH): Rationale, design, and implementation of a randomized clinical trial of a family-based pediatric weight management program for preschoolers. *Contemporary clinical trials*, *52*, 10-19.

3. Stark, L. J., Spear, S., Boles, R., Kuhl, E., Ratcliff, M., Scharf, C., ... & Rausch, J. (2011). A pilot randomized controlled trial of a clinic and home-based behavioral intervention to decrease obesity in preschoolers. *Obesity*, *19*(1), 134-141.

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Participant Characteristics	" ☐ "	" ☒ "	" ☐ "	" ☐ "	" ☐ "
Trial Setting	" ☐ "	" ☒ "	" ☐ "	" ☐ "	" ☐ "
Flexibility of Intervention	" ☐ "	" ☐ "	" ☒ "	" ☐ "	" ☐ "
Clinical relevance of experimental and comparison intervention(s)	" ☐ "	" ☐ "	" ☐ "	" ☐ "	" ☒ "

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Flexibility of Intervention	" ☐ "	" ☐ "	" ☐ "	" ☒ "	" ☐ "
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Participant Characteristics	" ☐ "	" ☐ "	" ☒ "	" ☐ "	" ☐ "
Trial Setting	" ☐ "	" ☐ "	" ☒ "	" ☐ "	" ☐ "
Flexibility of Intervention	" ☐ "	" ☒ "	" ☐ "	" ☐ "	" ☐ "
Clinical relevance of experimental and comparison intervention(s)	" ☐ "	" ☐ "	" ☐ "	" ☒ "	" ☐ "

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Flexibility of Intervention	" ☐ "	" ☐ "	" ☒ "	" ☐ "	" ☐ "
Clinical relevance of experimental and comparison intervention(s)	" ☐ "	" ☐ "	" ☐ "	" ☒ "	" ☐ "

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P.1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P.2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P.4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P.5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P.5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P.6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P.6-7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P.7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P.6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P.6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P.7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P.8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P.8-9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P.8-9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P.8-9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P.8-9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P.8-9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P.7-9
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P.7-9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P.9 Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	P.9 Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P.11 Figure 7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P.9-11 Figures 2-6
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P.11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P.9-11 Figures 2-6
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P.9-11 Figures 2-6
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P.11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P.13-15
	23b	Discuss any limitations of the evidence included in the review.	P.14-15
	23c	Discuss any limitations of the review processes used.	P.14-15
	23d	Discuss implications of the results for practice, policy, and future research.	P.15-16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P.3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P.3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P.16
Competing interests	26	Declare any competing interests of review authors.	P.16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis

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Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis

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ABSTRACT

Objectives: (1) To summarize the literature on the impact of pediatric weight management interventions on health outcomes in preschool age children with overweight or obesity. (2) To evaluate the completeness of intervention description and real-world applicability using validated tools.

Design: Systematic review and meta-analysis.

Data sources: Medline, EMBASE, CINAHL, Cochrane Library and PsychInfo were searched between 10 March 2015 through 21 November 2021.

Eligibility criteria: Randomized controlled trials (RCTs) addressing weight management in preschool children (two to six years) with overweight or obesity.

Data extraction and synthesis: Two reviewers independently extracted key information from each study and assessed risk of bias. Random-effects meta-analysis was performed where there was evidence for homogeneous effects. The certainty of evidence was assessed by the Grading of Recommendations Assessment, Development and Evaluation.

Results: Of the 16908 studies retrieved, nine trials (1687 participants) met the inclusion criteria. These interventions utilized motivational interviewing (MI) or multicomponent educational interventions related to health behaviour approaches and were 6 to 12 months in duration. All studies contained some risk of bias. A difference was found in the intervention groups compared to controls for BMI z score (mean difference (MD) -0.10, 95% CI -0.12 to -0.09; 8 trials, 1491 participants; $p<0.001$; I^2 68%), though there was substantial heterogeneity. In a subgroup analysis, those receiving multicomponent educational interventions had a reduction in BMI z score compared to controls (MD -0.11, 95% CI -0.12 to -0.09; 6 trials, 504 participants; $p<0.001$; I^2 68%); no differences were noted with MI. The certainty of the evidence was considered low to moderate. The trials were reported in sufficient detail and were considered pragmatic.

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Conclusions: Multicomponent educational interventions delivered to the parents of young children with obesity result in decline in BMI z score. The results should be interpreted cautiously as the quality of the evidence ranged from low to moderate.

PROSPERO registration number: CRD42020166843.

Strengths and limitations of this study

- This review updates our understanding of the effectiveness of weight management interventions on health outcomes of preschool aged children over the past six years.
- This review highlights a knowledge gap for the best practice for pediatric weight management in preschool children.
- This review evaluates the description of interventions and their real-world applicability by using the TIDieR checklist and RITES tool, respectfully.
- This review investigated children ages two to six years old with overweight or obesity and not those with a medical condition known to affect weight status or syndromic causes for obesity.

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INTRODUCTION

The prevalence of overweight and obesity in children has been increasing globally.[1] There is estimated to be 124 million children and adolescents with obesity worldwide.[2] Nationally representative data from 144 countries suggests that 43 million children under five years of age have obesity.[3] Children with obesity carry an increased risk for cardiometabolic disease and poor psychosocial health.[4-7] Childhood obesity often persists into adulthood and is associated with increased risk of morbidity throughout life.[8-10] These alarming rates and health risks underscore a need for effective interventions to treat obesity in early childhood and to prevent continued obesity related health issues across the lifespan.

Family-based behaviour modification remains the foundation of pediatric weight management programs.[11] In a 2016 Cochrane systematic review, multicomponent interventions tested with randomized controlled trials (RCTs) in preschool children with obesity aged two to six years old were effective in reducing body mass index (BMI) z score, a common metric to classify obesity in children.[12] Interventions typically consisted of lifestyle behaviour modification, including dietary and physical activity advice, along with parenting skills. Four trials (n=210 participants) were included in that analysis, and a mean change of -0.3 units (95% CI -0.4 to -0.2) in BMI z score was reported. However, the quality of evidence was deemed to be low and a high risk of bias was identified. Since then, there has been increased interest in this field given the increasing prevalence of pediatric obesity.[13] This systematic review will update the effectiveness of weight management interventions on preschool age children since the Cochrane review in 2016.[12]

Although systematic reviews can be invaluable in facilitating clinical decision-making by providing a critical evaluation of the evidence on a specific topic or condition,[14] to do so the intervention must be adequately described. Further, the pragmatism of the trial will inform the potential for 'real-world' implementation. The Template for Intervention Description and Replication (TIDieR) checklist was created to ensure interventions are described in sufficient

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detail to allow for replication.[15] The Rating of Included Trials on the Efficacy-Effectiveness Spectrum (RITES) tool is used to assess the real-world applicability and pragmatism of trials.[16]

The objectives of this systematic review were: (1) to assess the impact of pediatric weight management interventions on health outcomes in preschool children aged two to six with overweight or obesity; and (2) to evaluate the description of these interventions using the TIDieR checklist and their real-world applicability using the RITES tool. This knowledge will help inform future clinical practice guidelines and the planning of health programming for the treatment of young children with obesity.

METHODS

This review was conducted and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses.[17] A PRISMA study flow diagram was completed to present a summary of searching, screening, and inclusion of studies in this systematic review.[17] The review protocol for this study was registered with PROSPERO (CRD42020166843).

Patient and public involvement statement

No individual participant data were involved in the study; therefore, patients or the public were not involved in the design, conduct, reporting, or dissemination plans of our research.

Eligibility

Study designs eligible for inclusion were RCTs with the aim of weight management in preschool children with overweight or obesity. Any type of intervention of at least six months duration and delivered in a clinical setting was included. The interventions could be delivered by a single professional or be multidisciplinary or multicomponent. Multicomponent interventions typically address health behaviours including diet and physical activity. Participation of family members was not mandatory. Interventions for prevention of obesity, studies with observational designs and studies with self-reported height and weight were excluded. Participants included children with overweight or obesity who were between the ages of two and six years at time of enrolment into a weight management intervention. Studies of children with obesity who had a medical

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condition known to affect weight status or syndromic causes for obesity were excluded (e.g., genetic syndromes such as Prader-Willi Syndrome and Bardet-Biedl Syndrome). The control group in eligible RCTs was defined as no intervention or usual or standard care.

The primary outcomes were change in BMI or change in BMI z score or BMI percentile. BMI changes with age during normal growth,[6] and so age and sex-specific BMI z scores, based on a normative standard were included.[18] Additional secondary health outcomes were included in the search terms: body weight, waist circumference, waist-to-height ratio, skin fold thickness, body fat percentage measured with bioelectric impedance (BIA) or dual-energy x-ray absorptiometry, lipid profile (including LDL, HDL, triglycerides, and total cholesterol), blood pressure, glycemia (fasting plasma glucose, glucose level at 2 hours in a standard oral glucose tolerance test, Hemoglobin A1C) and quality of life. Adverse events were defined as any adverse outcome occurring during or after the intervention, including events not necessarily caused by the intervention.

Search of studies

We conducted a comprehensive literature search in the following electronic databases: Ovid Medline, Embase, CINAHL, Cochrane Library, PsychInfo, WHO International Clinical Trials Registry Platform, Web of Science, and Clinicaltrials.gov. These databases were first searched from March 10 2015 (the end date of the 2016 Cochrane review)[12] to January 31, 2020 and this search was updated on November 21, 2021. The search strategy was developed in consultation with a research librarian using the following major themes: ‘preschool’, ‘overweight or obesity’, and ‘intervention’. Key terms within the search strategy were aligned to medical subject headings (MeSH) in Ovid Medline and were expanded to include more descriptive terms. The search strategy for each database is presented in Appendix 1. Reference lists of included publications and reviews and of other systematic reviews or meta-analyses were manually searched to identify other studies that might not have been identified through the database

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searches. The database searches resulted in a combined total of 16908 studies after removal of duplicates (Figure 1).

Study selection

Titles, abstracts, and full texts were reviewed independently for eligibility by two groups of two authors (SN and CD, PM and RG) using a pre-tested, standardized screening form. Publications selected for inclusion were reviewed by an additional author and expert in the field and compared against the inclusion criteria. Only papers published in English were included. Every attempt was made to identify duplicate publications or multiple reports/companion pieces based on the same study and, if that occurred, the publication with the longest follow-up period for the primary outcome was included.

Data extraction

Key information from each included study was extracted in duplicate using a screening form and data extraction template. If data were missing, attempts were made to obtain the data from the corresponding author noted in the publication. Any disagreement during screening or extraction was resolved by discussion or in consultation with a third author.

Risk of bias and certainty of evidence

Risk of bias of each of the included studies was assessed using the Cochrane Risk of Bias tool.[14] The overall certainty of the evidence for each outcome was judged using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. The certainty of the evidence was assessed across the domains of risk of bias, precision, consistency and publication bias (internal validity) as well as directness (external validity). The quality of evidence was rated for each outcome (high, moderate, low or very low). Certainty was judged as high (further search is very unlikely to change the confidence in the effect estimate), moderate (further research is likely to have an important impact on the confidence in the effect estimate and may change the estimate), low (further research is very likely to have an important impact on the confidence of the effect estimate and is likely to change the estimate), or very low

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(very uncertain about the estimate of the effect). A checklist was created and used to help ensure consistency and reproducibility of the GRADE assessments.[14]

The TIDieR checklist was completed to ascertain if the intervention was reported in sufficient detail to allow for reproduction of the study.[15] The RITES tool was completed to evaluate the pragmatism of the intervention; this tool uses a 5-point Likert scale in four domains: (1) Participant characteristics, (2) Trial setting, (3) Flexibility of the intervention, and (4) Clinical relevance.[16] Two reviewers independently classified each trial on the Likert scale. Disagreements were resolved with a discussion to achieve consensus. All items were summed for each study to provide an efficacy/effectiveness score. The score is a continuum from maximum efficacy (score of 4) on one end to maximum effectiveness (or most pragmatic) at the other (score of 20). A low total score (i.e. 4) suggests a strong emphasis on efficacy and a high total score (i.e. 20) is consistent with a strong emphasis on effectiveness or a more pragmatic trial.

Statistical analysis

Statistical analyses were performed using Review Manager5 (RevMan 2013) software package (Cochrane Training, Cochrane). For continuous data measured on the same scale, the mean difference (MD) was calculated. Overall effects were reported and classified as small, medium, or large based on Cohen's d effect sizes.[19] We used a random-effects model taking into account the interstudy variation which provided a more conservative effect than a fixed model. All effect sizes were reported with 95% CIs. The statistic I^2 was calculated to investigate the variation in effect size due to heterogeneity across studies. 0% to 40% corresponded with low heterogeneity, 30% to 60% corresponded with moderate heterogeneity, 50% to 90% corresponded with substantial heterogeneity, and >75% corresponded with considerable heterogeneity.[20] For three-arm trials (two interventions with a control group), the control group was split for meta-analyses to compare interventions to control (Cochrane Handbook 5.1 Ch:16.5.4).[21]

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Synthesis of results

Review Manager5 was used to combine and calculate each outcome (RevMan 2013). If there was good evidence for homogeneous effects (i.e. tests of heterogeneity were not significant) across trials, a random effects meta-analysis was performed. If there was substantial heterogeneity, subgroup analyses based on sex or differences in BMI z score (e.g., overweight vs. obesity) were to be performed (PROSPERO CRD42020166843). We anticipated intervention modality (e.g., motivational interviewing or multicomponent educational interventions) would introduce heterogeneity and planned to conduct exploratory subgroup analyses with available data. If substantial heterogeneity remained, a qualitative synthesis was conducted.

RESULTS

Search results

After title and abstract screening, 456 studies remained and underwent full text screening. After full text review and removal of studies that did not meet the inclusion criteria and considering articles and ongoing trials from the previous Cochrane review, 9 studies remained. A manual search of the articles in the reference lists from the included studies did not generate any additional relevant studies. Therefore, a total of 9 RCTs were included in this review.

Current study versus previous Cochrane systematic review

The current study is an updated systematic review and meta-analysis of the previous study by Colquitt et al.[12] The previous study included seven RCTs and identified four ongoing trials that might be relevant in future meta-analyses. Upon review and consideration of these previous studies, four of the seven previously analyzed studies were also included in the current meta-analysis.[22-25] Three studies from the previous review were excluded: Bocca 2012 was a 4-month intervention (<6 months);[26] Lanigan 2013 included an age range of 1-5 years old (<2 years old);[27] Taveras 2011[28] included the same participants as Rifas-Shiman 2017.[29] Consistent with our inclusion criteria, we selected the study with the longest follow-up period[29] and excluded

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Taveras 2011.[28] Of the four ongoing trials identified in the previous study,[12] one status was unknown (NCT00916318), another was terminated due to no funding to launch RCT (NCT01698606), and two were unidentifiable in the literature (NCT02292602; Reifsnider 2012).

Summary of included studies

The nine included studies, published between 2009 and 2021, were conducted either in the USA (n=6),[29-31; 22; 24; 25] in Sweden (n=2),[32; 33] or in Iran (n=1).[23] Of the 1687 participants, 937 were randomized to an intervention and 750 to a comparator group. Five studies were multi-center trials,[29; 30; 32; 22; 33] while four studies were conducted at a single center.[31; 24; 25; 23] Five studies were delivered in a primary care setting,[31; 29; 32; 33; 30] one study was delivered in community-centered programs and primary care clinics,[30] while four studies were implemented in the outpatient care setting under the guidance of healthcare professionals.[22; 24; 25; 23] The interventions included motivational interviewing (MI) for parents[31] or families,[29; 33] dietary interventions with an educational component[23] or education on the principles of good nutrition and activity.[31; 30; 32; 24; 25] (Table 1). Interventions ranged from six[31] to 12 months[29; 30; 32] in duration; three studies included a 12-month follow-up period.[24; 29; 25] and one study had a three year follow-up.[23] Interventions varied in intensity, from less than 1 hour contact time[33] to 3 hrs total contact time (MI)[29] to upwards of 27 hrs total contact time (education).[30]

Effects of interventions

The BMI outcome was BMI z score for eight studies and %BMIp95 was reported for four trials. A difference in BMI z score between intervention and control groups was identified at the end of intervention (mean difference (MD) –0.10, 95% CI -0.12 to -0.09; 8 trials, 1491 participants; p=<0.001; I^2 68%) (Figure 2). No between group difference in %BMIp95 was identified (MD -1.34, 95% CI -3.17 to 0.49; 4 trials, 349 participants; p=0.15; I^2 79%) (Figure 3). Of note, the heterogeneity in both analyses exceeded 50% indicating substantial heterogeneity. Two RCTs included BMI as a primary outcome. There was a small improvement in BMI between

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intervention and usual care (MD -0.17, 95% CI -0.33 to -0.01; 2 trials, 931 participants; p=0.03, I^2 0%) (Figure 4).

Subgroup analyses

A subgroup analysis was completed based on intervention modality due to substantial heterogeneity. This analysis resulted in a lower I^2 for MI, however remained unchanged for multicomponent educational interventions. The educational interventions resulted in a greater decline in BMI z score in the treatment group compared to controls (MD -0.11, 95% CI -0.12 to -0.09; 6 trials, 504 participants; p<0.001; I^2 68%) (Figure 5). However, similar to the primary analysis, there was no difference found when the outcome was measured as %BMIp95 (MD -1.73, 95% CI -3.85 to 0.39; 4 trials; 278 participants; p=0.11; I^2 83%) (Figure 6). There was no significant impact of MI based interventions on BMI z score (MD -0.02, 95% CI -0.11 to 0.06; 3 trials, 987 participants; p=0.59; I^2 49%) (Figure 7). The heterogeneity remained above 50% despite the subgroup analyses, which is consistent with substantial heterogeneity.

Secondary outcomes

Six of the nine trials included secondary outcomes. Unfortunately, not all data were able to be interpreted in the context of a meta-analysis due to missing results (i.e.,, no mean standard change reported).[32; 30] Kelishadi 2009 included many secondary outcomes including waist circumference, body fat, lipid profile and blood pressure, however this was the only RCT to include these outcomes.[23] Two of the trials[24; 25] included pediatric quality of life, and noted an improvement in quality of life (MD 6.32, 95% CI 0.91 to 11.73; 2 trials, 50 participants; p= 0.02; I^2 0%) (Figure 8). Weight (kg) was also reported in two trials,[22; 25] with a clinically significant improvement (MD -1.27, 95% CI -1.93 to -0.61; 2 trials, 129 participants; p<0.001; I^2 0%) (Figure 9).

Adverse events

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Two trials reported no adverse events,[31; 32] , while Derwig 2021 reported no adverse events aside from worsening weight status of several participants.[33] Adverse events from the other trials were not reported.

Risk of bias

All trials reported adequate sequence generation and allocation concealment and used a form of computer random generator for randomization. As noted in the risk of bias assessment presented in Figure 10, all but one trial[22] did not blind participants or healthcare providers to the intervention or control groups resulting in a high risk for performance bias for study outcomes. Three of the trials did not mention blinding for data collectors or outcome assessors; these trials were rated as uncertain risk for detection bias.[29; 30; 33] Four of the RCTs blinded study staff collecting outcome measures to participant assignment and therefore were judged as low risk for detection bias.[22; 23; 25; 31] Ek 2019 stated that data were collected by child healthcare professionals participating in the study and was therefore rated high risk for detection bias.[32] The drop-out rates were unbalanced across studies: only 38% of intervention participants in the study by Rifas-Shiman 2017 completed more than 1 visit.[29] Of the remaining trials, drop-out rates for the intervention groups ranged from 2.5%[23] to 36%.[30] Only four RCTs reported less than a 20% drop out rate, making these studies low risk for attrition bias.[22-24; 33] The remaining five RCTs reported at least a 20% drop out rate resulting in a high risk for attrition bias. We judged three of four trials to have low risk of reporting bias.[29-31] We did not draw funnel plots or perform Egger's test to assess reporting bias due to limited number of trials per outcome (n=8; BMI z score).

Certainty of the evidence

The overall quality of the evidence was considered low to moderate as a result of the high risk for performance bias and attrition bias (Table 2). We did not downgrade for blinding of participants when examining overall quality of evidence for the primary outcomes.

Template for Intervention Description and Replication (TIDieR)

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Most of the studies included adequate information regarding the intervention (Appendix 2). In three of the studies,[30; 32; 22] the intervention was well described, although some of the materials required to conduct the intervention were not provided or were not described in sufficient detail to reproduce the intervention. Three of the included trials were variations on the same intervention: Stark has published three RCTs using their LAUNCH intervention.[24; 25; 31] While the papers did include well-described interventions, there is extensive training required to effectively administer the intervention, and training information was not included. The lack of information regarding the training or education for those implementing the intervention was common throughout many of the included studies.[30; 24; 25; 31; 23; 22] In some cases, the reviewers were required to search for previously described protocols to gain an understanding of the materials and training details.[24; 25; 31; 30] Derwig 2021 did require training to implement their intervention, however they did provide an additional paper describing the training and materials in detail.[33]

While Tidier is not typically used for describing the standard of care or usual treatment, in the case of preschool aged obesity treatment, many of the studies included interventions beyond the current standard of care. As such, this review also includes the TIDiER checklist for the control intervention. Both Kelshadi 2009 and Quattrin 2012 control groups received the same education sessions as the intervention groups, which was significantly more than the standard of care. Butte 2017 and Stark also included additional interventions for their control groups, over and above standard of care. The standard of care was not always adequately described for all the included studies. Ek 2019 described standard of care based on a specific action plan but did not include more detailed information of what this entailed. Butte 2017 described their standard treatment control with additional material, although the extent to which the material was utilized was unreported.

Rating of Included Trials on the Efficacy-Effectiveness Spectrum (RITES)

The RITES tool data for all nine trials are presented in Appendix 3. All of the included trials scored between 12 to 16, indicating more pragmatic trials. Five of the trials[30; 32; 31; 24; 25] had more

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explanatory trial settings as they required extensive training for implementation of the intervention and/or ongoing supervision to ensure the precise implementation. All the included studies were found to be highly effective in terms of their clinical relevance. However, all the trials, aside from Derwig 2021,[33] excluded participants with underlying conditions or comorbidities, resulting in greater emphasis on efficacy for participant characteristics pertaining to overweight or obesity.

DISCUSSION

Summary of evidence

Since the previous review published in 2016, five RCTs examining obesity treatment interventions in young children have been published. This review included RCTs from the original Cochrane study that were consistent with this review's inclusion criteria. This resulted in an additional four studies being included. Seven of the nine RCTs included an educational component as part of the intervention, however, one of those had a primarily dietary intervention.[23] The other two interventions included family guidance (family-based therapy) and motivational interviewing.

In the previous Cochrane review, multicomponent interventions were found to be more successful at reducing BMI z score compared to controls.[12] When grouped together, the trials from the Cochrane review along with the updated search in this current review had high heterogeneity with an overall small effect in reducing BMI z score. Most of the trials' interventions were well described using the TIDiER checklist, although some did not include the materials or staff training required for the intervention. The control group did lack some detail in their description. All of the trials were noted to be relatively pragmatic, suggesting that they could be applied in real-world settings.

An exploratory subgroup analysis considering intervention modality (multicomponent educational interventions vs motivational interviewing) was completed due to substantial heterogeneity in the primary analysis. The heterogeneity remained unchanged with the

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subgroup analysis. The six RCTs with a multicomponent educational intervention utilizing BMI z-score as their primary outcome showed efficacy in reducing BMI z-score. MI, either focused on the parents, child or the family did not show consistent effectiveness for the treatment of overweight and obesity in preschool aged children. Two of the interventions included MI over the telephone if parents missed a session. The implications of providing an intervention virtually are largely unknown and may have affected the overall impact of the intervention.

It is important to note that the multicomponent interventions had greater contact time with the families compared with motivational interviewing or family guidance. A previous systematic review concluded that at least 26 hours of multicomponent intervention over a minimum of 6 months improved BMI z score.[34] Of the included RCTs, only three reported at least 26 hours of intervention time.[30; 24; 25] Kelishadi 2009 did not document total intervention contact time. On the contrary, motivational interventions were limited to 3 to 7.5 hours of total contact time, while family guidance had very limited contact time at 1 hour.[33] Given these differences, one cannot suggest with confidence that it was the intervention type (i.e. education vs MI) and not the 'intensity' of the intervention that led to differences in effectiveness.

Overall, the trials included in this study described their interventions in sufficient detail in the methods sections, or via reference to a published protocol paper. However, the standard care provided to the control group was not often described in sufficient detail to allow replication. Trials need to be fully transparent when describing the details of usual or standard care control groups, to truly understand the effectiveness of interventions in the real-world setting. This is an important area for further exploration in future studies and reviews.

To the best of our knowledge, this was the first systematic review in children with obesity to utilize the RITES tool. Generally, the trials were more pragmatic than efficacious on the effectiveness-efficacy scale. Effective or pragmatic trials are similar to routine clinical practices and indicate a more 'real-world' setting. The fact that all trials were conducted in the primary care setting, or a combination of primary care and the community, supports a pragmatic

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approach; however, the level of training and resources required to implement the interventions in real-world scenarios may pose a challenge. Additionally, most trials recruited participants following stringent criteria, excluding those with comorbidities, and resulting in less pragmatism.[35] This limits the generalizability of the findings beyond preschool age children with overweight or obesity only.

Limitations

The search strategy for this systematic review was influenced by the previously published Cochrane review.[12] Bias was introduced as the searches performed and trials included in this review were restricted to English. Splitting the control group in three arm trials limits the overall variability within trials; however, it is a recommended approach in the Cochrane Handbook to analyze such multicomponent interventions.[36] Further, the methodological heterogeneity between trials weakens our confidence in the conclusion regarding an overall effect of interventions on improving BMI z score. The subgroup analysis based on intervention modality did not result in a change in heterogeneity, which remained >50% and consistent with substantial heterogeneity. As such, this results in a downgrade of the evidence due to imprecision. Although there was a greater decline in BMI z score for those undergoing educational interventions, the certainty of the evidence for primary outcomes ranged from low to moderate. The main methodological concern from review authors was the substantial attrition bias in most trials (>20%). Unfortunately this is common in pediatric weight management.[37] Most trials declared that the participants and healthcare providers were not blinded to their intervention or control group, leading to concerns of performance bias. However, weight management interventions involving services from healthcare providers make it difficult to blind personnel. Due to this, we did not downgrade for blinding when examining overall quality of evidence for the primary outcomes.

Future directions

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There continues to be limited evidence surrounding the best practice for pediatric weight management in young children. Furthermore, efficacy has been almost entirely based on anthropometric health outcomes alone. Other health outcomes are inconsistently reported and further evidence of the impact of these intervention programs on these outcomes including measures of cardiometabolic health and health related quality of life, will be important given their relevance to clinical practice. Future interventions should also consistently consider potential adverse health outcomes.

Fundamentally, emphasis on a continued search for novel pathways to intervene upon or new approaches to current paradigms must be undertaken given the modest outcomes to date, while also considering the relevance and potential impact of observational evidence. Furthermore, although modest changes in BMI z score were noted, the persistence of the impact of these interventions over time remains uncertain and is an important area for further research.

Conclusions

There continues to be limited evidence regarding the treatment of obesity in children of preschool age. In this updated systematic review, there was limited evidence of the benefit of MI as a treatment option, though the interventions were quite brief. There is some evidence education-based programs involving parents from the start of the intervention appear to assist with BMI z score reduction, but further research is required to improve on the care of young children with obesity.

Contributors

SN, PGM, RG, LM and KMM contributed to the conceptualization of the study. SN, PGM, RG, and CD led the data searches, data extraction, and quality appraisal. PGM was responsible for data analysis with guidance from LM. SN, PGM, and RG drafted the initial manuscript. All authors contributed to the interpretation of findings, critically reviewed and provided feedback on multiple iterations of the manuscript and approved the final version of the manuscript for submission.

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Competing interests

KMM is an advisory board member for Novo Nordisk and a member of a Data Safety Monitoring Board for Novartis.

Ethics statement

This meta-analysis did not involve individual participant data; therefore, Institutional Review Board approval was not required for this study.

Data availability statement

Data are available upon reasonable request from the corresponding author (kmorrison@mcmaster.ca).

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Table 1. Characteristics of included studies

Source	Setting	Intervention / comparator	Sample size	Sex (F, (%))	Age (mean (SD))	BMI measure (baseline)	Intervention (duration)	Outcomes
Kelishadi 2009	Outpatient	I1: Calcium rich diet	40	NR	5.4 (0.2)	z score: 2.4 (0.01) BMI: 22.1 (0.9)	6 mo. with 3-year follow-up >800mg calcium/day Total contact = 6 education sessions	Primary: BMI z score (CDC) Secondary: Waist circumference, % body fat, dietary intake, physical activity, glycemic control, blood pressure, lipid profile
		I2: Energy-restricted group	40	NR	5.5 (0.7)	z score: 2.3 (0.04) BMI: 22.7 (0.8)	6 mo. with 3-year follow-up energy content restricted Total contact = 6 education sessions	
		C: Control	40	NR	5.7 (0.3)	z score: 2.4 (0.01) BMI: 22.4 (0.5)	No dietary recommendation	
Stark 2011	Outpatient	I: LAUNCH	8	2 (25.0)	4.4 (0.9)	BMI %ile: 99.0 (0.9)	6 mo. with 12 mo. follow-up Phase 1: 12 weekly sessions (90 min) Phase 2: 12 biweekly sessions, alternating between group and home Total contact = ≥30 hrs	Primary: BMI z score, BMI %ile (CDC) Secondary: Dietary intake, home food environment, children's PA, parenting styles, Peds QL
		C: Control	10	4 (40.0)	3.9 (1.1)	BMI %ile: 97.7 (2.5)	45 min: diet and PA recommendations	
Quattrin 2012	Outpatient	I: Family-based weight control	46	31 (67.4)	4.6 (1.1)	z score: 2.2 (0.8) BMI %: 32.4 (22.4)	12 mo. 60 min sessions/mo. (13 total). Control + behavioural modification Total contact = 13hrs (+8 phone calls)	Primary: BMI z score (CDC) Secondary: BMI percent
		C: Information control	50	33 (66.0)	4.4 (1.1)	z score: 2.1 (0.7) BMI %: 29.8 (17.1)	60 min sessions/mo. (13 total). Dietary and PA education Total contact = 13hrs	
Stark 2014	Outpatient	I1: LAUNCH – home + clinic	10	8 (80)	4.7 (1.3)	z score: 2.1 (0.2)	6 mo. with 12 mo. follow-up Phase 1: 12 sessions/week (90 min) Phase 2: 12 weeks group clinic + home Total contact = ≥30 hrs	Primary: BMI z score (CDC) Secondary: Dietary intake, home food environment, children's PA, parenting styles
		I2: LAUNCH - clinic	11	7 (64)	4.2 (1.1)	z score: 2.5 (0.8)	Phase 1: Biweekly sessions months 1-3 Phase 2: Monthly clinic visits for months 4-6 for 10 total sessions Total contact = 10 hrs	
		C: Control	12	8 (67)	4.8 (0.7)	z score: 2.4 (0.4)	45 min: diet and PA recommendations	
Rifas-Shiman 2017	Primary care clinics	I: High five for kids (motivational interviewing)	253	121 (47.8)	4.8 (1.2)	z score: 1.9 (0.7) BMI: 19.2 (2.6)	12 mo. 4 (25 min) in-person visits + 2 (15 min) phone calls 12 mo. maintenance: 2 in-person visits Total contact = 8 sessions (3hrs)	Primary: BMI z score (CDC) Secondary: Television viewing, intake of fast food

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		C: Usual care	192	94 (49.0)	5.2 (1.1)	z score: 1.8 (0.6) BMI: 19.1 (2.0)	Baseline + annual well child care visits	and sugar sweetened beverages
Butte 2017	Primary care clinics + Community programs	I: MEND	100	51 (51.0)	2–5*	BMI %ile: 113.0 (15.2) BMI: 20.5 (0.3)	3 mo. intense: 9 (90 min) sessions/week 9 mo. transition: (90 min) sessions/mo. Total contact = 18 sessions (27hrs)	Primary: %BMIp95 (CDC) Secondary: Body composition, blood pressure, Peds QL and Strengths and Difficulties
		C: 12-month clinic-based program	60	29 (49.0)	2–5*	BMI %ile: 110.2 (13.7) BMI: 20.0 (0.3)	12 mo. clinic based program Total contact = 8hrs	
Stark 2018	Primary care clinics	I1: LAUNCH	47	25 (53.2)	4.6 (1.0)	z score: 2.4 (0.5) BMI %ile: 98.6 (1.2)	3 mo. intense: 90/60 min sessions/week 3 mo. maintenance: biweekly sessions Total contact = 18 sessions (23hrs)	Primary: BMI z score (CDC) Secondary: %BMIp95, (CDC) parental BMI
		I2: Motivational interviewing	50	29 (58.0)	4.6 (0.9)	z score: 2.4 (0.6) BMI %ile: 98.5 (1.3)	3 mo. intense: weekly sessions 3 mo. maintenance: biweekly sessions Total contact = 18 sessions (7.5hrs)	
		C: Control	54	32 (59.3)	4.6 (0.9)	z score: 2.5 (0.7) BMI %ile: 98.6 (1.3)	Informed caregivers of child's weight	
Ek 2019	Primary care clinics	I1: No booster	43	23 (53.2)	5.2 (0.9)	z score: 3.1 (0.7) BMI: 21.9 (2.3)	12 mo. 10 (90 min) sessions Total contact = 10 sessions (15hrs)	Primary: BMI z score (IOTF) Secondary: Child BMI, waist circumference, metabolic health, lifestyle patterns, obesity-related child behaviors, lifestyle specific self-efficacy
		I2: Booster	44	19 (43.2)	5.2 (0.8)	z score: 3.0 (0.5) BMI: 21.4 (1.5)	12 mo. 10 (90 min) sessions + Booster (7 (30 min) phone calls) Total contact = 17 sessions (18.5hrs)	
		C: Standard care	87	56 (64.4)	5.3 (0.7)	z score: 2.9 (0.6) BMI: 21.3 (1.7)	≥4 visits of 30 minutes over 12 months Total contact = min. 4 sessions (2hrs)	
Derwig 2021	Primary care clinics	I: CCHD	245	133 (54.3)	4.1 (0.1)	z score: 1.6 (0.3) BMI: 18.1 (0.4)	12 mo. 10 min dialogue with child, family, clinician. 45 min family guidance Total contact = 55 min	Primary: BMI z score (IOTF) Secondary: Costs for delivery of care, number of extra visits and number of referrals
		C: Standard care	245	132 (53.9)	4.1 (0.1)	z score: 1.6 (0.3) BMI: 18.1 (0.5)	Single visit of a health dialogue of overweight	

* = average age not reported in study

CCHD = Child-Centered Health Dialogue; LAUNCH = Learning about Activity and Understanding Nutrition for Child Health. Developing diet, physical activity, and parenting skills; MEND = Mind, Exercise, Nutrition… Do it; PA = physical activity

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Table 2. Quality of evidence

Outcome	No of Studies	Design	Risk of Bias	Precision	Consistency	Directness	Other Considerations	No of Patients	Effect (Absolute)	Quality	Rating
BMI z-score	8	RCT	Serious Limitations[1]	No serious imprecision	No serious consistency	No uncertainty	None	1491	-0.10 (-0.12, -0.09)	⊕⊕⊕⊖	Moderate
%BMIp95	4	RCT	Serious Limitations[1]	Serious[2]	Serious consistency[3]	Uncertainty[4]	None	349	-1.34 (-3.17, 0.49)	⊕⊕⊖⊖	Low

Footnotes:

1. Studies rated down for attrition bias with over 20% loss to follow-up, high risk of performance bias
2. Rated down due to wide confidence intervals
3. High I^2 (79% for %BMIp95)
4. Indirectness due to differences in setting between interventions (clinic, home, and community centres)

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Figure 1. PRISMA Flow Diagram

Figure 2. Forest plot showing the mean difference in **BMI z score** comparing intervention versus control

Figure 3. Forest plot showing the mean difference in **%BMIp95** comparing intervention versus control

Figure 4. Forest plot showing the mean difference in **BMI** comparing intervention versus control

Figure 5. Forest plot showing the mean difference in **BMI z score** comparing multicomponent educational intervention versus standard care

Figure 6. Forest plot showing the mean difference in **%BMIp95** comparing multicomponent educational intervention versus standard care

Figure 7. Forest plot showing the mean difference in **BMI z score** comparing motivational interviewing versus standard care

Figure 8. Forest plot showing the mean difference in **PedsQL** comparing intervention versus control

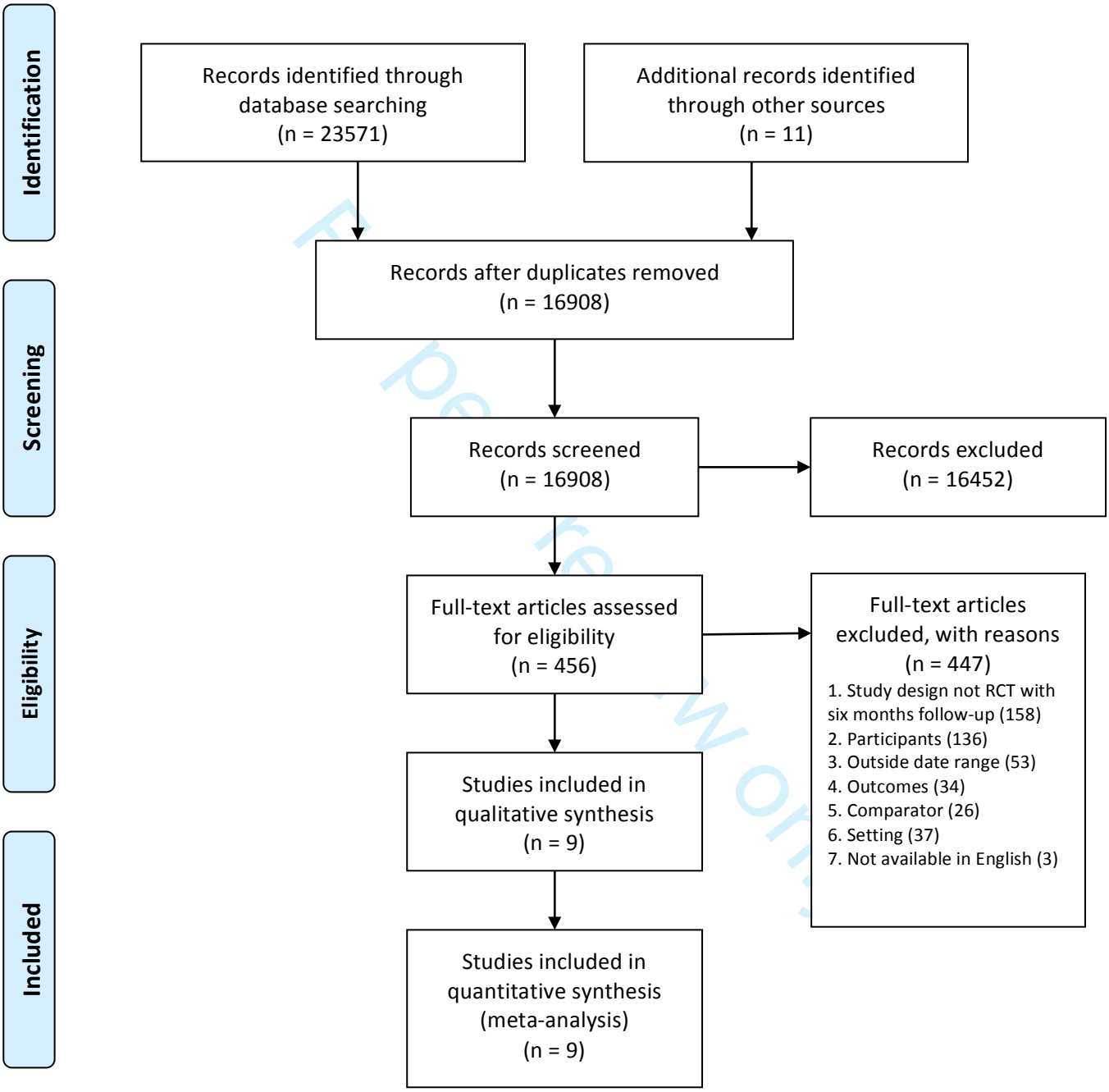
Figure 9. Forest plot showing the mean difference in **Weight** comparing intervention versus control

Figure 10. Methodological quality summary: authors' judgments for each methodological quality item for each included study

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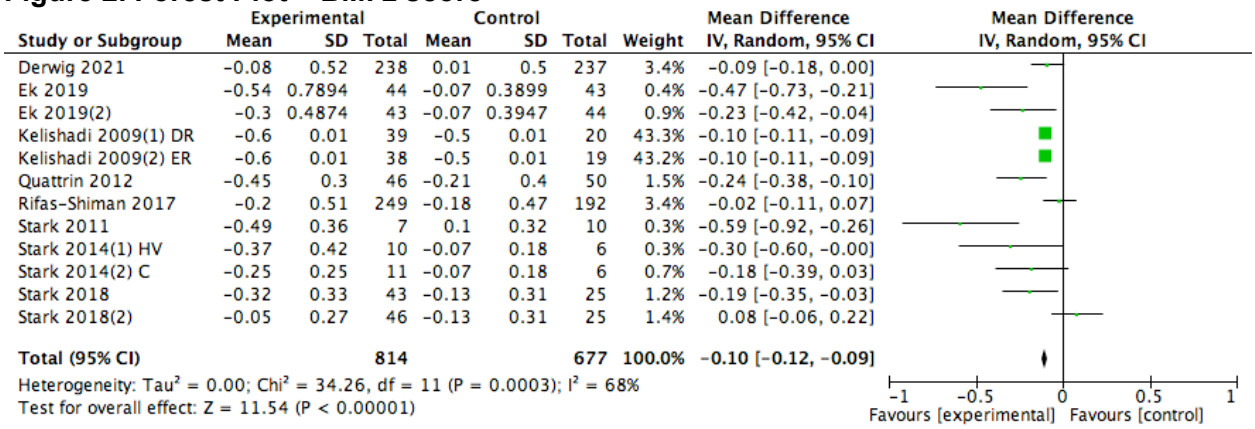
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Figure 1. PRISMA Flow Diagram



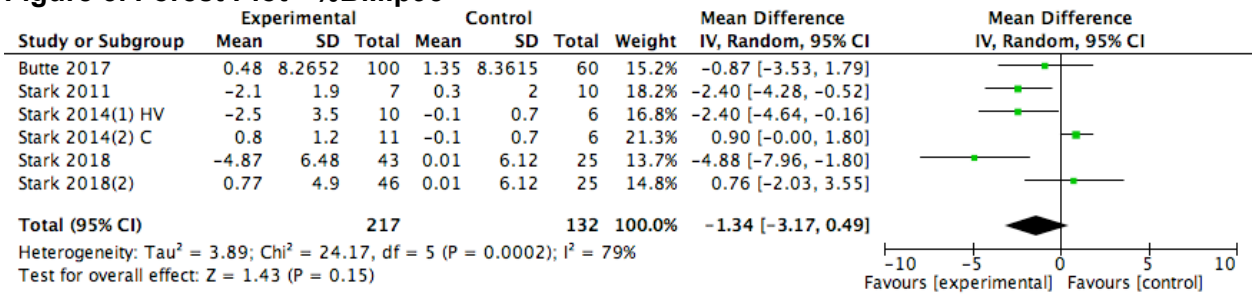
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Figure 2. Forest Plot – BMI z score



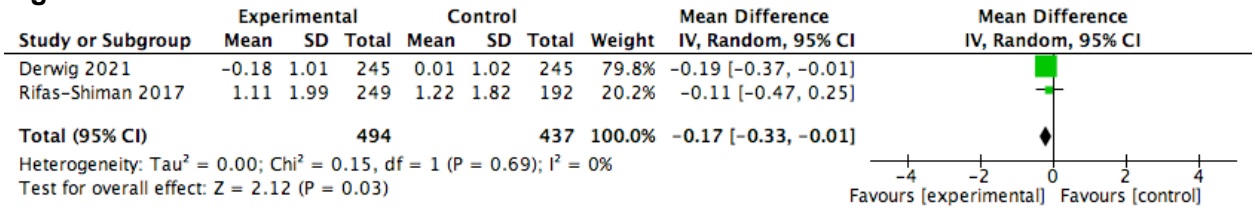
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Figure 3. Forest Plot - %BMIp95



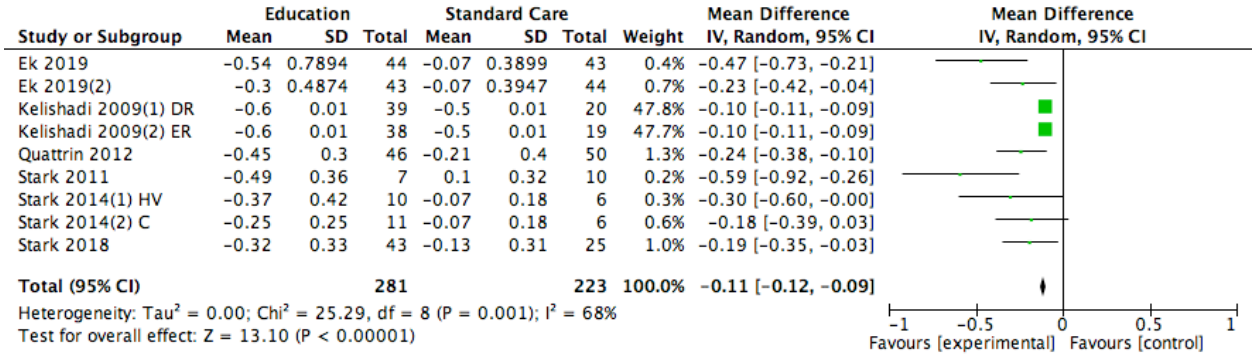
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Figure 4. Forest Plot – BMI



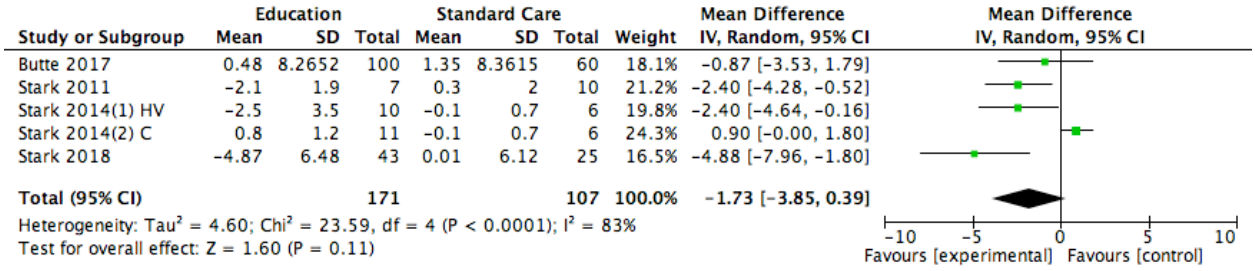
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Figure 5. Forest Plot – BMI z score – subgroup analysis – Multicomponent Educational Interventions



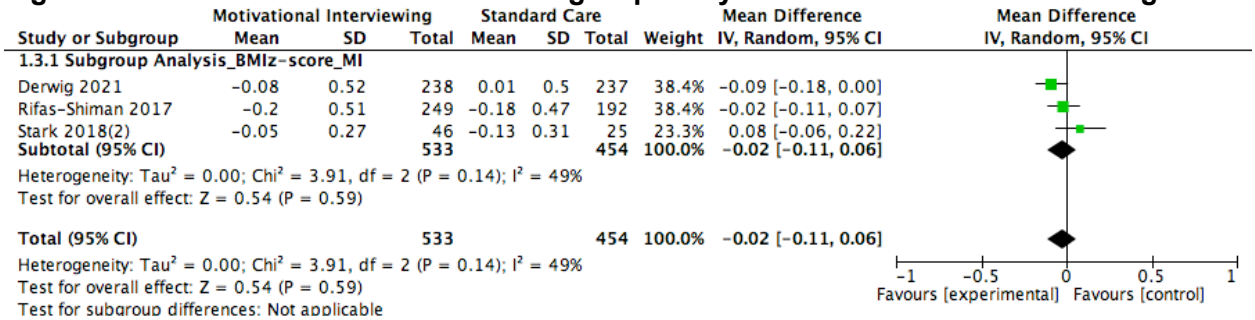
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Figure 6. Forest Plot - %BMIp95 – subgroup analysis – Multicomponent Educational Interventions



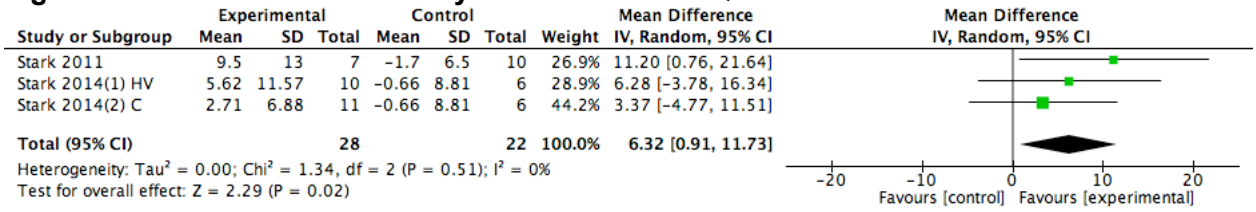
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Figure 7. Forest Plot – BMI z score – subgroup analysis – Motivational Interviewing



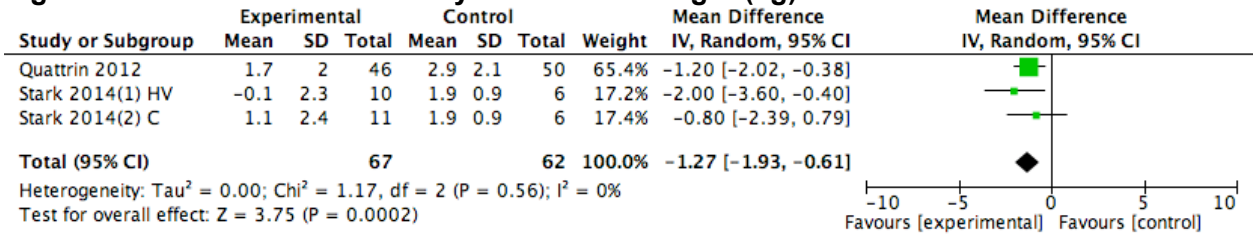
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Figure 8. Forest Plot – Secondary Outcome – PedsQL



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Figure 9. Forest Plot – Secondary Outcome – Weight (kg)



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Medline

Part I: Obesity

1. Obesity/
2. Obesity, Morbid/
3. Obesity, Abdominal/
4. Pediatric Obesity/
5. Overweight/
6. Weight Loss/
7. Body Mass Index/
8. (adipos* or obes*).tw.
9. (overweight* or over weight*).tw.
10. (weight adj1 (reduc* or los* or control* or manage*)).tw.
11. or/1-10

Part II: Intervention

12. exp Behavior Therapy/
13. exp Counseling/
14. Family Therapy/
15. Social Support/
16. Program Evaluation/
17. exp Exercise/
18. exp Exercise Therapy/
19. exp Physical Education/
20. exp Exercise Movement Techniques/
21. Motor Activity/
22. exp Diet/
23. exp Diet Therapy/
24. Patient Education as Topic/
25. Health Education/
26. exp Health Behavior/
27. exp Health Promotion/
28. Life style/
29. ((obesity adj3 intervention) or program or programme or camp?).tw
30. (lifestyle or life style).tw.
31. exercis*.tw.
32. (physic* adj (activ* or fit*)).tw.
33. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
34. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*).tw
35. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
36. ((group or family or cognit* or behav*) adj therap*).tw.
37. counsel?ing.tw.
38. educat*.tw.
39. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
40. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw

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41. or/12-40

Part III: Part I + Part II and additional MeSH/subheading combination

42. 11 and 41
43. Obesity/ or Obesity, Morbid/ or Overweight/ or Weight Loss/
44. diet therapy.fs. or prevention & control.fs. or rehabilitation.fs. or therapy.fs. or psychology.fs
45. 43 and 44
46. 42 or 45

Part IV: Population [adapted from Leclercq 2013]

47. exp Child/
48. Pediatrics/
49. minors.tw.
50. (boy*).tw.
51. girl*.tw.
52. toddler?.tw
53. (kid or kids).tw.
54. (child or childs or children* or childhood* or childcar*)tw
55. (preschool* or pre-school* or toddler* or kindergar* or nursery school*).tw
56. juvenil*.tw.
57. youth*.tw.
58. (underage* or under age*).tw.
59. p?ediatric*.tw.
60. or/47-59

Part V: Part III AND IV and additional MeSH/subheading combination

61. 46 and 60
62. Pediatric Obesity/
63. diet therapy.fs. or prevention & control.fs. or rehabilitation.fs. or therapy.fs. or psychology.fs
64. 62 and 63
65. 61 or 64

Part VI: Study filter [Cochrane Handbook 2008 RCT filter - sensitivity and precision maximizing version]

66. randomized controlled trial.pt.
67. controlled clinical trial.pt.
68. randomi?ed.ab.
69. placebo.ab.
70. clinical trials as topic/
71. randomly.ab.
72. trial.ti.
73. or/66-72
74. exp animals/ not humans/
75. 73 not 74

Part VII: Part V + Part VI

76. 65 and 75

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Embase

Part I: Obesity

1. obesity/
2. morbid obesity/
3. abdominal obesity/
4. childhood obesity/
5. weight reduction/
6. weight control/
7. body mass index/
8. (adipos* or obes*).tw.
9. (overweight* or over weight*).tw.
10. (weight adj1 (reduc* or los* or control* or manage*)).tw.
11. or/1-10

Part II: Intervention

12. behavior therapy/
13. cognitive therapy/
14. exp counseling/
15. family therapy/
16. social support/
17. exp program evaluation/
18. exp exercise/
19. exp physical education/
20. exp physical activity/
21. exp motor activity/
22. training/
23. exp diet/
24. exp diet therapy/
25. nutritional health/
26. child nutrition/
27. feeding behavior/
28. patient education/
29. health promotion/
30. health literacy/
31. nutrition education/
32. health education/
33. lifestyle/
34. lifestyle modification/
35. ((obesity adj3 intervention) or program or programme or camp?).tw
36. (lifestyle or life style).tw.
37. exercis*.tw.
38. (physic* adj (activ* or fit*)).tw.
39. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
40. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*).tw
41. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
42. ((group or family or cognit* or behav*) adj therap*).tw.

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43. counsel?ing.tw.
44. educat*.tw.
45. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
46. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw
47. or/12-45

Part III: Part I + Part II and additional MeSH/subheading combination

48. 11 and 47
49. obesity/ or morbid obesity/
50. pc.fs or rh.fs or th.fs. [prevention.fs. or rehabilitation.fs. or therapy.fs.]
51. 49 and 50
52. 48 or 51

Part IV: Population [adapted from Leclercq 2013]

53. juvenile/
54. child/
55. toddler/
56. preschool child/
57. pediatrics/
58. minors.tw.
59. boy*.tw.
60. girl*.tw.
61. toddler?.tw.
62. (kid or kids).tw.
63. (child or childs or children* or childhood*).tw
64. juvenil*.tw.
65. youth*.tw.
66. (underage* or under age*).tw.
67. p?ediatric*.tw.
68. or/53-67

Part V: Part III AND IV and additional MeSH/subheading combination

69. 52 and 68
70. childhood obesity/
71. pc.fs or rh.fs or th.fs. [prevention.fs. or rehabilitation.fs. or therapy.fs.]
72. 70 and 71
73. 69 or 72

Part VI: Search Filters (Cochrane Handbook)

74. 'crossover procedure':de
75. 'double-blind procedure':de
76. 'randomized controlled trial':de
77. 'single-blind procedure':de
78. (random* OR factorial* OR crossover*
79. cross NEXT/1 over*
80. placebo*
81. doubl* NEAR/1 blind*
82. singl* NEAR/1 blind*
83. assign*

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84. allocat*
85. (volunteer*):de,ab,ti
86. or/74-85

Part VII: Part V + Part VI

87. 73 and 86

PsychINFO

Part I: Obesity

1. exp Overweight
2. exp Body Mass Index/
3. (adipos* or obes*).tw.
4. (overweight* or over weight*).tw.
5. or/1-4

Part II: Intervention

6. Weight Control/
7. Weight Loss/
8. Aerobic Exercise/
9. Diets/
10. exp Exercise/
11. Movement Therapy/
12. Dance Therapy/
13. exp Physical Activity/
14. Physical Fitness/
15. Health Behavior/
16. Health Promotion/
17. Health Knowledge/
18. Health Literacy/
19. Health Education/
20. Client Education/
21. Lifestyle/
22. Physical Education/
23. exp Program Evaluation/
24. Educational Programs/
25. Educational Therapy/
26. exp Program Development/
27. Counseling/
28. Group Counseling/
29. Family Therapy/
30. Support Groups/
31. Social Support/
32. exp Behavior Modification/
33. Cognitive Behavior Therapy/
34. Cognitive Therapy/
35. ((obesity adj3 intervention) or program or programme or camp?).tw

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36. (lifestyle or life style).tw.
37. exercis*.tw.
38. (physic* adj (activ* or fit*)).tw.
39. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
40. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*)
41. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
42. ((group or family or cognit* or behav*) adj therap*).tw.
43. counsel?ing.tw.
44. educat*.tw.
45. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
46. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw
47. or/6-46

Part III: Part I + Part II

48. 5 and 47

Part IV: Population [adapted from Leclercq 2013]

49. minors.tw.
50. (boys*).tw.
51. girl*.tw.
52. toddler?.tw.
53. (kid or kids).tw.
54. (child or childs or children* or childhood*)tw
55. juvenil*.tw.
56. youth*.tw.
57. (underage* or under age*).tw.
58. p?ediatric*.tw.
59. or/49-58

Part V: Part III AND IV and additional MeSH/subheading combination

60. 48 and 59

Part VI: Study filter [Eady 2008filter - BS version]

61. control*.tw. OR random*.tw. OR exp Treatment/

Part VII: Part V + Part VI

62. 60 and 61

CINAHL

Part I: Obesity

1. MH ”Obesity+“
2. MH "Body Mass Index"
3. TX (adipos* or obes*)
4. TX (overweight* or ”over weight*“)
5. S1 OR S2 OR S3 OR S4

Part II: Intervention

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6. MH ”Weight Loss“
7. MH ”Behavior Modification+“
8. MH ”Counseling“
9. MH ”Family Therapy“
10. MH ”Support, Psychosocial“
11. MH ”Support Groups“
12. MH ”Program Evaluation“
13. MH ”Program Implementation“
14. MH ”Exercise+“
15. MH ”Sports+“
16. MH ”Therapeutic Exercise+“
17. MH ”Physical Fitness“
18. MH ”Physical Education and Training+“
19. MH ”Health Education+“
20. MH ”Diet+“
21. MH ”Diet Therapy+“
22. MH ”Health Behavior “
23. MH ”Eating Behavior“
24. MH ”Health Promotion“
25. MH ”Life style changes“
26. MH ”Life style“
27. TX (weight N1 (reduc* or los* or control* or manage*))
28. TX ((obesity N3 intervention) OR program OR programme OR camp#)
29. TX (lifestyle or ”life style“)
30. TX exercis*
31. TX (physic* N1 (activ* or fit*))
32. TX (walk* or jog* or swim* or weight lift* or danc* or aerobics)
33. TX ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) N3 train*)
34. TX (behavio#ral or behavio#r modification or psychoth* or psychosocial)
35. TX ((group or family or cognit* or behav*) N1 therap*)
36. TX counsel#ing
37. TX educat*
38. TX ((parent# or family) N1 (based or focused or directed or centered or only or led))
39. TX (diet* or ”healthy nutrition“ or (nutrition* N1 (knowledge or educat* or therap* or program* or intervention*)))
40. S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR 37 OR 38

Part III: Part I + Part II and additional MeSH/subheading combination

41. S5 AND S40
42. MH ”Obesity+/DH/ED/PC/PF/RH/TH“) [diet therapy or education or prevention & control or psychosocial factors or rehabilitation or therapy]
43. S40 OR S41

Part IV: Population [based on Leclercq 2013]

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44. MH ”Child+“
45. MH ”Pediatrics“
46. TX minors
47. TX (boy*)
48. TX girl*
49. TX toddler#
50. TX (kid OR kids)
51. TX (child OR childs OR children* OR childhood*)
52. TX juvenil*
53. TX youth*
54. TX (underage* or under age*)
55. TX (paediatric* OR pediatric*)
56. S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54 OR S55

Part V: Part III AND IV and additional MeSH/subheading combination

57. S43 AND S56
58. MH ”Pediatric Obesity/DH/ED/PC/PF/RH/TH“) [diet therapy or education or prevention & control or psychosocial factors or rehabilitation or therapy]
59. S57 OR S58

Part VI: Study filter [Wong 2006bfilter - SDSSGS version]

60. MH ”treatment outcomes+“ OR MH ”experimental studies+“ or random*

Part VII: Part V + Part VI

61. S58AND S59

Cochrane Library

Part I: Obesity

1. Obesity.mp
2. Obesity, morbid.mp
3. Obesity, abdominal.mp
4. Pediatric obesity.mp
5. Overweight.mp
6. Weight loss.mp
7. Body mass index.mp
8. (adipos$ or obes$).ti, ab
9. (overweight$ or (over ADJ1 weight$)).ti,ab
10. (weight ADJ1 (reduc$ or los$ or control$ or manage$)).ti,ab
11. {or #1-#10}

Part II: Intervention

12. Behavior therapy.mp
13. Counseling.mp

[mh ”Behavior Therapy“]

13. [mh ”Counseling“]
14. [mh ^”Family Therapy“]
15. [mh ^”Social Support“]

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16. [mh ^"Program Evaluation"]
17. [mh "Exercise"]
18. [mh "Exercise Therapy"]
19. [mh "Physical Education and Training"]
20. [mh "Exercise Movement Techniques"]
21. [mh ^"Motor Activity"]
22. [mh Diet]
23. [mh "Diet Therapy"]
24. [mh ^"Patient Education"]
25. [mh ^"Health Education"]
26. [mh "Health Behavior"]
27. [mh "Health Promotion"]
28. [mh ^"Life style"]
29. (("obesity" near/4 "intervention") or "program" or "programme" or "camp" or "camps"):ti,ab
31. exercis*:ti,ab
32. (physic* next (activ* or fit*)):ti,ab
33. (walk* or jog* or swim* or ("weight" next lift*) or danc* or "aerobics"):ti,ab
34. ((physic* or strength* or resist* or "circuit" or "weight" or aerob* or "cross" or "endurance" or structur*) near/4 train*):ti,ab
35. ("behavioral" or "behavioural" or (("behavior" or "behaviour") next "modification") or psychoth* or "psychosocial"):ti,ab
36. (("group" or "family" or cognit* or behav*) next therap*):ti,ab
37. (counseling or counselling):ti,ab
38. educat*:ti,ab
39. (("parent" or "parents" or "family") next ("based" or "focused" or "directed" or "centered" or "only" or "led")):ti,ab
40. diet* or "healthy nutrition" or (nutrition* next ("knowledge" or educat* or therap* or program* or intervention*))):ti,ab
41. {or #12-#40}

Part III: Part I + Part II and additional MeSH/subheading combination

42. #11 and #40
43. [mh Ôbesity] or [mh ^"Obesity, Morbid"] or [mh Ôverweight]
44. [mh /DH,PC,RH,TH,PX][diet therapy or prevention & control or rehabilitation or therapy or psychology]
45. #43 and #44
46. #42 or #45

Part IV: Population [adapted from Leclercq 2013]

47. [mh Child]
48. [mh ^Pediatrics]
49. "minors":ti,ab
50. "boys*":ti,ab
51. girl*:ti,ab
52. ("kid" or "kids"):ti,ab
53. ("toddler" or "toddlers"):ti,ab
54. ("child" or "childs" or children* or childhood*):ti,ab

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55. juvenil*:ti,ab
56. youth*:ti,ab
57. (underage* or (”under“ next age*)):ti,ab
58. (paediatric* or pediatric*):ti,ab
59. {or #47-#58}

Part V: Part III AND IV and additional MeSH/subheading combination

60. #46 and #59
61. [mh ^”Pediatric Obesity“]
62. [mh /DH,PC,RH,TH,PX]
63. #61 and #62
64. #60 or #63

Web of Science:

Obesity

TS =(obesity)
TS=(obesity, morbid)
TS=(obesity, abdominal)
TS=(pediatric obesity)
TS=(overweight)
TS=(weight loss)
TS=(body mass index)
#7 OR #6 OR #5 OR #4 OR #3 OR #2 OR #1

Intervention:

TS=(behavior therapy)
TS=(counseling)
TS=(family therapy)
TS=(social support)
TS=(program evaluation)
TS=(exercise)
TS=(exercise therapy)
TS=(physical education and training)
TS=(exercise movement techniques)
TS=(motor activity)
TS=(diet)
TS=(diet therapy)
TS=(patient education)
TS=(Health Education)
TS=(Health Behavior)
TS=(Health Promotion)
TS=(Life style)
#9 OR #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21or #22 or #23 or #24 or #25

Part III: Part I + Part II

1. #8 AND #26

Population

TS=(Child*)

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TS=(Pediatric*)
TS=(Preschool child*)
TS=(Toddler*)
TS=(Minor*)
TS=(Youth*)
TS=(Kid*)
#28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34

Study filters

TS= (clinical trial*)
TS= (research design)
TS= (controlled trial*)
TS= (single blind*)
TS= (double blind*)
TS= (follow-up stud*)

Clinicaltrials.gov

Conditions: obese OR overweight OR obesity
Study Type: Interventional Studies
Age Group: Child (birth-17)

WHO Clinical Trial Registry

Activated "search for clinical trials in children"
Title: obes* OR overweight*
OR
Condition: obes* OR overweight*
Recruitment: All

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Article Title	Efficacy of a community- versus primary care-centered program for childhood obesity: TX CORD RCT	
Authors	Butte, N., Hoelscher, D., Barlow, S., Pont, S., Durand, C., Vandewater, E., Liu, Y., Adolph, A., Pérez, A., Wilson, T., Gonzalez, A., Puyau, M., Sharma, S., Byrd-Willams, C., Oluyomi, A., Huang, T., Finkelstein, E., Sacher, P. & Kelder, S.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	TX CORD: Texas Childhood Obesity Demonstration project **Intervention:** MEND2-5 (Mind, Exercise, Nutrition…Do it!) program **Comparison**: Next Steps program	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	The TX CORD project was designed to address childhood obesity through an integrated, systems-oriented model that incorporated primary and secondary obesity interventions. Within this population-level, systems-based intervention was a 12-month randomized controlled trial targeting low-income, ethnically diverse children with overweight or obesity and their families. It is hypothesized that the 12-month, community-centered program would significantly reduce BMI compared to the health care-centered program.	1585 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Intervention :** Conducted at the YMCA within 5 miles of the participants. Education materials surrounding general nutrition, PA and behaviours were needed (not included in paper/protocol). Based on CATCH program. All programs were delivered in English or Spanish. A digital scale and stadiometer for weight and height measurements. Transition program includes the book *A Year of Being Well* (role models stories of low-income families using CDC obesity prevention behaviours), cooking classes	74-76 (2)

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	based on *The Happy Kitchen/La Coicina Alegre* and MED World online/print material. **Control:** Electronic medical record at the primary care physician's office. Counselling materials for the primary care provider. Activity booklet for parents and children. Material was available in both English and Spanish, and available through the AAP website (https://shop.aap.org/next-steps-a-practitioners-guide-for-themed-follow-up-visits-for-their-patients-to-achieve-a-heal/).	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention:** A multicomponent, healthy lifestyle program including behavioural, nutrition and physical activity sessions. Included 10 weekly 90-minute sessions relating to general nutrition, physical activity and behaviours for children including: - 30 minutes of guided active play - 15 minutes of healthy snack time evidence-based, exposure-based technique to promote acceptance and increased intake of fruit and vegetables - 45 minutes where children participate in creative play activities while parents attend an interactive education and skill development session based on group-based parent training principles. The transition program includes monthly 90-minutes post-program sessions that children and parents attend together. Sessions include a brief review of the MEND concepts, role model stories and cooking lessons. **Control:** A clinic-based program that optimizes health care procedures to assess BMI and provides brief counselling materials to support family/child self-improvements in healthy lifestyles. Electronic health record (BMI screening, alerts for BMI ≥ 85th centile, specific diagnostic codes, lab tests and referrals common in obesity management) changes to support childhood obesity visits. Counselling materials were provided to the primary care providers. Next Steps activity booklets for parents and children to work on nutrition and physical activity targets in a self-directed manner. Families were encouraged to seek follow up clinic visits to address	74-75 (2) 75 (2), 1586 (1)

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	childhood obesity if identified as being obese. The brief counselling materials include information on facilitating clinical intervention and supporting educational materials for families.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention**: All sessions were delivered by bilingual allied health professionals (dieticians and health educations) and were assisted by community health workers. Community health workers were trained in MEND, motivational interviewing, nutrition and physical activity programming. **Control:** Program was delivered by primary care physicians in their offices, and self-directed learning by the families/children. No specific training, however counselling material was provided to the primary care physicians Trained research staff measured weight and height (?unclear training)	75 (2) 1586 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Intervention:** The intervention was delivered at the YMCA face-to-face. ? It was not stated whether this was individual sessions or group sessions. Measurements were performed at the Children's Nutrition Research Centre in Houston or YMCAs/offices in Austin. **Control:** Delivered face-to-face in primary care physician's office, as well as workbooks provided to complete individually at home.	1585 (1) 1585 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Intervention:** conducted at YMCAs in Austin and Houston. Measurements were performed at the Children's Nutritional Research Centre in Houston or YMCAs/offices in Austin. ?No information regarding how participants accessed this **Control**: primary care clinics in Texas	1585 (1)

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When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Intervention:** The intensive phase consisted of nine weekly 90-minute sessions. The transition phase offered 90-minutes session including MEND reviews (nine months in total). In total, MEND2-5 offered 27 hours of contact. **Intervention:** A clinic-based program – families were provided with self-directed education material as well as encouraged to seek follow-up clinic visits to address childhood obesity. An estimated 8-hours of contact was provided.	1585 (1), 75 (2) 1585-1586 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	"extensively adapted for the US population" (from British study) - ?did not report how it was adapted "pilot tested program elements and used qualitative data to further tailor the program to our population" TX CORD culturally adapted Next Steps material for low-income, urban, black and Latino children in both English and Spanish versions.	1585 (1) 76 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Fidelity to the session content and participant engagement was assessed using instructor checklists. Compliance in the intervention arm was based on attendance (the number of sessions attended out of the number offered). Retention was based on completion of measurements at 3 and 12 months.	1587 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	Retention rates were 64% for the intervention group compared to 87% for the comparison at 3 months. The retention rate at 12 months was 54% for the intervention group compared to 82% for the comparison group.	1587-1588 (1)

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	During the Intensive Phase, 86% of the sessions were delivered completely as planned. During the Transition Phase, 75% of the sessions were delivered completely as planned. Mean attendance was 46% during the Intensive Phase and 16% during the Transition Phase. Fidelity was not assessed for the comparison group.	1590 (1)

** **Authors -** use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

(1) Butte, NF, Hoelscher, DM, Barlow, S, et al. Efficacy of a community- versus primary care-centered program for childhood obesity: TX CORD RCT. Pediatric Obesity 2017; 25:1584-1593

(2) Hoelscher DM, Butte NF, Barlow S, et al. Incorporating primary and secondary prevention approaches to address childhood obesity prevention and treatment in a low-income, ethnically diverse population: study design and demographic data from the Texas Childhood Obesity Research Demonstration (TX CORD) study. Child Obes 2015;11:71-91.

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Article Title	A parent treatment program for preschoolers with obesity: A randomized controlled trial	
Authors	Ek, A., Chamberlain, K., Sorjonen, K., Hammar, U., Malek, Mahnoush., Sandvik, P., Somaraki, M., Nyman, J., Lindberg, L., Nordin, K., Ejderhamn, J., Fisher, P., Chamberlain, P., Marcus, C., & Nowicka, P.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	More or Less Study: 1) Parent-only group treatment **with** follow-up booster sessions 2) Parent only group treatment **without** booster sessions 3) Standard treatment	2 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Given the significant role of parents in a young child's life, the treatment program created was grounded in evidence-based parenting practices to evaluate whether a focus on parenting would be especially suitable for this age group as suggested by previous papers. It is hypothesized that the parent-only treatment would be more effective compared to standard treatment.	2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Intervention** - Based on the KEEP (Keeping Foster and Kin Parents Supported and Trained) program – a positive parenting program. The More or Less program was developed in collaboration with the KEEP program developers with conceptual and cultural adaptations designed for Swedish population. Involved home assignments (not provided as part of paper/protocol) In collaboration with the KEEP developers, the authors translated and adapted the parent group manual outlining the content of each session and parent handouts (not included in the protocol/paper).	2 (1) 6 (2)

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	Control -receive treatment as usual in an outpatient pediatric unit. A questionnaire was developed to monitor the healthcare providers involved (as it varied), and the number and focus of the visits. The questionnaire was not included as part of the protocol or paper.	6 (2)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention -** ML program consisted of 10 weekly 90-minute sessions with parents only, with childcare provided. Each session centered around a parenting practice as well as a lifestyle component aimed to provide parents with knowledge/skills to support a healthy home environment. The session was finished with a home practice assignment related to the focus of the day. If parents missed a session, material was sent home and key parts were reviewed over the phone by 1 of the group leaders. **More or Less with Booster** – Group leaders called participants every 4-6 weeks for a 30-minute phone call (max 7 calls). The parents were referred back to the discussions had during the program and to the manual for additional support **Control** – based on the action plan for childhood obesity in Stockholm County (not described in the paper). Received at least 4 treatment visits, typically the first with the physician, and subsequent with the pediatric nurse. The visits centered around lifestyle advice. Reasons for treatment are discussed including weight gain and future risk factors. Treatments varied between pediatric units – a questionnaire was included regarding questions about the professions of the involved healthcare providers, number and focus of visits.	3 (1) 6-8 (2) 2(1) 6 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention** – Led by 5 registered dieticians – they received training by the KEEP developers, who also provided weekly supervision on videotaped sessions (translated from Swedish to English) for program certification and fidelity. The initial training was a 5-day experiential training including information on program's theory and practice in the delivery of group sessions. Each session of the ML program was thoroughly discussed, and each trainee role played facilitating several key sessions.	3 (1) 7 (2)

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	The training continued through external supervision from the KEEP developer group. **Control** – first and last visit were with a pediatrician, and the remaining visits were with a pediatric nurse (minimum of 4 visits). Some children were referred to dieticians and/or PTs.	3 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**More or Less without Boosters**– face-to-face group sessions **More or Less with booster** – face-to-face group sessions followed by seven 30-minute phone calls every 4-6 weeks (9 months in total) Neither the group leaders or parents were informed about which families were allocated to the boosters until after the group sessions were finished in order to avoid differential treatment. **Control** – delivered in person during routine visits to pediatrician	3 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Intervention -** Meetings were held in health care facilities. Both parents were invited to participate and childcare was provided. **Control** – Delivered at 14 outpatient pediatric clinics	3 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**No Booster** – 10 weekly 90-minute sessions **Booster** – 10 weekly 90-minute sessions with 30-minute phone calls (max 7) every 4-6 weeks for the following 9 months. **Control** – Were offered ≥ 4 individual visits of around 30 minutes over 12 months. Treatment setup varied between clinics – each clinic completed a protocol at every visit documenting details (*not available in paper*)	3 (1) 6 (2)
Tailoring		

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If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention** – The parent group manual outlining each session and parent handouts was translated and adapted from the original KEEP developers' version. **Control** – treatment set up varied between practitioners and location. Some children were referred to dieticians and/or PTs. Number of appointments with pediatrician/pediatric nurse also varied.	6 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	A 6-month follow-up was added post baseline to better evaluate the treatment program	2 (1)
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Intervention** - Training of the ML group leaders was conducted by the KEEP developers. 5-day experiential training session. The training continued through external supervision from the KEEP group – all parent groups were videotaped (and translated to English). KEEP consultants viewed the recordings, rated them for fidelity and identified areas for reinforcement and feedback Group leaders completed a session review with questions regarding what went well and challenges experienced. Also completed weekly forms on parent attendance and engagement ratings – these measures informed the consultation process **Control -** Because the treatments vary between pediatric units, a questionnaire including questions about the profession of healthcare providers involved as well as the number and focus of visits	7 (2) 6 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**Intervention**– Not reported in the final paper. **Control** – Each clinic filled out a protocol at every visit documenting what treatment the family had attended and on how many occasions.	3 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

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(1) Ek A, Lewis Chamberlain K, Sorjonen K, et al. A Parent Treatment Program for Preschoolers With Obesity: A Randomized Controlled Trial. Pediatrics. 2019;144(2): e20183457

(2) Ek A., Lewis Chamberlain K., Ejderhamn J., et al. The more or less study: A randomized controlled trial testing different approaches to treat obesity in preschoolers. Public Health. 2015. 15:735: DOI 10.1186/s12889-015-1912-1

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Article Title	Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study (1)	
Authors	Sheryl L. Rifas-Shiman, Elsie M. Taveras, Steven L. Gortmaker, Katherine H. Hohman, Christine M. Horan, Ken P. Kleinman, Kathleen Mitchell, Sarah Price, Lisa A. Prosser and Matthew W. Gillman	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	High Five for Kids primary care-based pre-schooler obesity intervention	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Targeting preschool-age children may be beneficial because children of this age visit their primary care clinicians frequently for routine medical care and parents have more control over their children's health-related behaviours, and these behaviours may be more malleable. The purpose of the study was to assess the effectiveness of the primary care-based obesity intervention over a 2-year period. The overarching model for this intervention was the Chronic Care Model which posits that changes in primary care to produce functional patient outcomes requires changes for all members of the practice team.	24 (2) 2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where	**Intervention:** Posters were developed for the waiting rooms that encouraged the five target behaviours of the High Five. An image of this poster is available at: https://pubmed.ncbi.nlm.nih.gov/21464376/. Educational modules were developed to target fast-food and sugar-sweetened beverage consumption, as well as television viewing (https://www.researchgate.net/publication/261773967_Health_Behavior_Change_A_Guide_For_Practitioners). A list of local resources for physical activity was also available but not provided in the study write-up, as was a website with recipes and educational materials. Small physical incentives (water bottles, snack containers, books) were provided by nurse practitioners to participants, as was an	2-3 (2)

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the materials can be accessed (e.g. online appendix, URL).	electronic tool to monitor television-viewing time (more information on this tool, including where to get it, was not provided in the study). The electronic medical record was enhanced to help physicians make decisions, track patients, follow-up, schedule and bill.	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention:** Intervention was based on the Chronic Care Model (3), which requires change from all members of the health care team. Participants (including families) received motivational interviewing by nurse practitioners and educational modules targeting certain behavioural modifications. Specifically, the interviewing and modules targeted reducing sugar-sweetened beverage intake, reducing television viewing time, increasing fruit and vegetable consumption, decreasing fast-food consumption and encouraging more time in active play. **Usual care:** Participants got the standard of care offered by their pediatric practice. This included the baseline and annual well-child care visits as well as follow ups with the paediatrician or subspecialist (ex. Dietician).	2 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention:** Pediatric nurse practitioners were trained for their role as the central intervening clinician. They also received training in motivational interviewing. This training included enhancing motivation for change, enhancing self-efficacy, how to help recognize inconsistencies between desired and actual behaviours, and how to help reduce this dissonance. Medical assistants were trained on how to conduct research-standard anthropometric measurements (for collecting height and weight data). Medical receptionists were trained to schedule the initial and follow-up visits with the nurse practitioners in accordance with the study protocol. Primary care pediatricians were trained in obesity management and to use brief, focused negotiation skills. **Usual care:** Participants got the standard of care offered by their pediatric practice. This included the baseline and annual well-child care visits as well as follow-ups with the pediatrician or subspecialist (ex. Dietician).	2 (2)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone)	**Intervention:** In the first year, four visits were face-to-face and two were by telephone. In the second year, both visits were face-to-face. Visits were with individual families. **Usual care:** At the discretion of the participants regular pediatrician.	2 (2)

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of the intervention and whether it was provided individually or in a group.		
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Done at 10 pediatric primary care practices within the Harvard Vanguard Medical Associates, which is a multi-site group practice in Massachusetts. Half of the practices provided usual care and half provided the intervention care.	2 (2)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Intervention:** Four in person (each 25 minutes) visits with the clinician over the first year, as well as two phone calls (15 minutes each). Two in-person visits in the maintenance year (length not described). **Usual care:** Frequency of usual care visits was determined by the routine care procedures for each particular pediatric practice.	2 (2)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention:** During their visits, the families could choose to work on one or more of the behavioural targets (previously listed).	4 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	
How Well		

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Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Intervention:** Adherence was assessed by completion of the various visits/phone calls, during data analysis. No strategies were used to improve fidelity.	5 (2), 3 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**Intervention:** In the first year (the intervention period), only 56% of participants in the intervention completed at least 2 of the 6 intervention activities (in person or phone visits). During the second year (the maintenance period), only 17% went to one visit and only 21% of participants went to both visits. Overall adherence was thus low. **Usual care:** adherence was not measured for usual care participants.	5 (2), 3 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Rifas-Shiman, S. L., Taveras, E. M., Gortmaker, S. L., Hohman, K. H., Horan, C. M., Kleinman, K. P., ... & Gillman, M. W. (2017). Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study. *Pediatric obesity*, *12*(3), e24-e27.

2. Taveras, E. M., Gortmaker, S. L., Hohman, K. H., Horan, C. M., Kleinman, K. P., Mitchell, K., ... & Gillman, M. W. (2011). Randomized controlled trial to improve primary care to prevent and manage childhood obesity: the High Five for Kids study. *Archives of pediatrics & adolescent medicine*, *165*(8), 714-722.

3. Wagner, E. H. (1998). Chronic disease management: what will it take to improve care for chronic illness?. *Effective clinical practice*, *1*(1).

Article Title	Clinic and Home-Based Behavioral Intervention for Obesity in Preschoolers: A	

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	Randomized Trial (1)	
Authors	Lori J. Stark, PhD, Stephanie Spear Filigno, PhD, Christopher Bolling, MD, Megan B. Ratcliff, PhD, MPH, Jessica C. Kichler, PhD, Shannon M. Robson, PhD, MPH, RD, Stacey L. Simon, PhD, Mary Beth McCullough, PhD, Lisa M. Clifford, PhD, Cathleen Odar Stough, PhD, Cynthia Zion, BS, and Richard F. Ittenbach, PhD	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Comparing LAUNCH (Learning about Activity and Understanding Nutrition for Child Health), motivational interviewing and standard care as in intervention for pre-school obesity	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Obesity in preschool years increases the risk of being overweight and obese in later childhood through to adulthood. Focusing on children with obesity is important as they are at higher risk for mortality compared with overweight peers. This study hypothesizes that an skills-based behavioural family clinic in combination with a home-based intervention would reduce BMIz score compared with motivational interviewing and standard care in preschool age children with obesity.	1,2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**LAUNCH**: Caregivers were given an iPod Touch so that they could use the "DietTracker" application to track the child's food and beverage intake. Pedometers were also provided to the children and their caregivers, to help caregivers monitor physical activity levels. Caregivers were also given a two-week supply of fresh vegetables at each of the clinic sessions to help expose children to new vegetables. **Motivational Interviewing:** Caregivers were given a pack of informational handouts from the Let's Go 5-2- 1-0 program (https://www.mainehealth.org/Lets-Go/Childrens-Program/Parents). A motivational interviewing (MI) manual was used	13, 16 (2), 135 (4)

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	for MI review by interviewers and study staff (3). **Standard Care:** N/A	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**LAUNCH**: Sessions alternated between clinic and home-based sessions. During the clinic sessions, the caregiver(s) and child were in different groups. Caregiver groups received education and problem-solving skills on a number of diet and physical activity, as well as strategies to manage the behaviour of their child. A list of topics by session is available in Table 1 (2). In following the teachings of the sessions, families aimed to decrease the child's screen time, increase physical activity (for which they had daily step goals for both the parents and the child), stay within the prescribed daily calorie goals for the child as well as choose healthier foods, and to record child's food and beverage intake. The behavioural strategies that they learned to help achieve these goals included praising their child's healthy dietary and physical activity choices, modelling healthy behaviours, managing tantrums by ignoring and using time-outs, contingency management, as well as how to best set up the food environment to encourage healthy choices (ex. Not having unhealthy food available). More information about the teaching sessions and strategies learned is available here (4). The child group included education about healthy eating and opportunities for physical exercise. Sessions also included a light nutritious meal provided to caregivers and children, which served as exposure to fruits and vegetables for the children. During home visits, families got a chance to practice and receive feedback on the skills they had learned in clinic, as well as to problem solve any barriers they were facing in implementing those strategies in the home setting. **Motivational Interviewing:** MI was a caregiver-only intervention. During the first session, caregivers were asked about their concerns for their child's diet, weight, or physical activity habits and then given a chance to set a goal that was meaningful to	4-5 (1), 13-15 (2), 135 (4)

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	them. The subsequent sessions (either by phone or in person) involved a discussion of the previous goals, how their implementation was going, enhancing motivation, working on establishing the desired behaviours and choosing new goals to work towards. **Standard Care:** No intervention was provided outside of the routine pediatric care the participant was already receiving, other than to be notified of their weight.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**LAUNCH**: The LAUNCH interventionists were therapists trained on the protocol in pilot studies (more information on training was not provided). A new LAUNCH interventionist was brought on midway through the study. She had extensive previous training in pediatric behavioural weight management and was trained on the LAUNCH protocol by shadowing and co-leading sessions with one of the previous interventionists. Home visits were completed by postdoctoral fellows, who were trained by reviewing the LAUNCH manual (not provided), listening to audiotapes of home sessions, watching videotapes of child group sessions, administering role-played sessions, and shadowing a second year fellow during home visits for one cycle before being assigned their own visits. The group-sessions in the clinic were done by a licensed clinical psychologist. Fidelity coders had PhDs and expertise in obesity treatment. **Motivational Interviewing:** Interviewers were either pediatricians or PhD level psychologists. The first session was done by the pediatrician and the following sessions were done by the psychologists. Interviewers were selected for their previous MI training and experience; each interviewer had approximately 10 years of MI experience and had participated in previous structured MI training. **Standard Care:** Standard care was provided by the participant's regular pediatrician.	13, 16 (2)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or	**LAUNCH**: All sessions were done in-person. The clinic sessions were done in groups, with one group for the parents and one group for the children. Home-based sessions	13 (2)

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telephone) of the intervention and whether it was provided individually or in a group.	were done with individual families. **Motivational Interviewing:** There were 4 in-person sessions (sessions 2, 12, 16) and 14 sessions done over the phone. All sessions were done with individual caregivers. **Standard Care:** Participants in the standard care group received only a notification of their child's weight, provided by phone call and during the initial (in-person) assessment.	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**LAUNCH**: The home sessions were done in the participants' homes, which were all within 50 miles of the treatment centre. The clinic sessions were done at a medical centre (the treatment centre). **Motivational Interviewing:** The initial assessment interview was done at the medical centre, but subsequent sessions were done at the families' homes or over the telephone. **Standard Care:** At the discretion of the participants' regular pediatricians.	117 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**LAUNCH**: Intervention sessions alternated between clinic (10 sessions) and home sessions (8 sessions), for a total of 18 sessions. For the first three months there were weekly sessions, and for the following three months sessions were every second week. The clinic-based sessions were each 90 minutes and the home visits were 60 minutes. Sessions were scheduled for the late afternoons/early evenings to minimize scheduling conflicts/barriers. **Motivational Interviewing:** The 18 sessions were delivered in alignment with the LAUNCH session schedule. Phone sessions varied in length depending on caregiver preference, but were generally between 15-35 minutes long. Home visits were all scheduled for 60 minutes.	13, 15 (2)

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	Standard Care: Length and frequency of appointments was determined by the participant's normal pediatrician, in line with their practice policies.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**LAUNCH**: Home visits provided the opportunity to tailor the program for individual children/families. If there was a barrier to using a clinic-taught strategy in the home, the home visit provided an opportunity to problem-solve. It was also an opportunity for targeted feedback, since therapist was able to witness the taught behavioural strategies being used in vivo. Caregivers also received weekly written feedback on their child's food and beverage consumption, including feedback on their progress towards their goals. Feedback was also provided on activity levels, based on the number of steps recorded on the pedometers. **Motivational Interviewing:** Caregivers were asked to select a specific nutrition or physical activity goal from the Let's Go 5-2-1-0 (5 fruits or vegetables a day, limit recreational screen time to 2 hours or less, get 1 or more hours of physical activity a day, limit or eliminate sugary drinks) and AAP recommendations (https://pediatrics.aappublications.org/content/120/Supplement_4/S164). This was then used as the focus of the motivational interviewing sessions. **Standard Care:** Care was personalized as appropriate by the pediatrician providing care.	13 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	**LAUNCH**: N/A **Motivational Interviewing:** N/A **Standard Care:** N/A	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	All LAUNCH and MI sessions were videotaped and/or audiotaped. **LAUNCH**: Fidelity coders reviewed the first session of the first cycle of LAUNCH and provided feedback to the interventionist. One quarter of all LAUNCH sessions (both clinic and home-based) were compared to a LAUNCH checklist by a PhD level	16 (2)

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	consultant with experience in weight modification treatment. If fidelity was found to drop below 90%, the therapist was provided with feedback on the components that needed improvement. **Motivational Interviewing:** For the first three cycles of interviewing, the two interviewers met together to review audiotapes of each other's MI interviews and provide feedback on MI fidelity. After the third interviewer joined the study, her audiotapes were reviewed and given regular feedback by the other two interviewers. MI fidelity coders also reviewed audiotapes regularly throughout the study and provided feedback to each interviewer on a quarterly basis. 25% of MI sessions were compared to a MI checklist (3) to assess fidelity. If fidelity was found to drop below 90%, the therapist was provided with feedback on the components that needed improvement. **Standard Care:** N/A	
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**LAUNCH**: Fidelity was assessed as planned – see previous box. **Motivational Interviewing:** Fidelity was assessed as planned – see previous box. **Standard Care:** N/A	16 (2)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark, L. J., Filigno, S. S., Bolling, C., Ratcliff, M. B., Kichler, J. C., Robson, S. M., ... & Zion, C. (2018). Clinic and home-based behavioral intervention for obesity in preschoolers: a randomized trial. *The Journal of pediatrics*, *192*, 115-121.

2. Stark, L. J., Filigno, S. S., Bolling, C., Ratcliff, M. B., Kichler, J. C., Robson, S. L., ... & Zion, C. (2017). Learning about Activity and Understanding Nutrition for Child Health (LAUNCH): Rationale, design, and implementation of a randomized clinical trial of a family-based pediatric weight management program for preschoolers. *Contemporary clinical trials*, *52*, 10-19.

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3. Moyers, T. B., Martin, T., Manuel, J. K., Miller, W. R., & Ernst, D. (2010). Revised global scales: Motivational interviewing treatment integrity 3.1. 1 (MITI 3.1. 1). *Unpublished manuscript, University of New Mexico, Albuquerque, NM*.

4. Stark, L. J., Spear, S., Boles, R., Kuhl, E., Ratcliff, M., Scharf, C., ... & Rausch, J. (2011). A pilot randomized controlled trial of a clinic and home-based behavioral intervention to decrease obesity in preschoolers. *Obesity*, *19*(1), 134-141.

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Article Title	A child-centered health dialogue for the prevention of obesity in child health services in Sweden – A randomized controlled trial including an economic evaluation	
Authors	Derwig, M., Tiberg, I., Björk, J., Welander Tärneberg, A., & Kristensson	

	Hallström, I.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Child-Centered Health Dialogue (CCHD)	Pg. 2 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	The attention paid to dietary and physical activity behaviours in children is infrequent at the 4-years health visit in Sweden and raises concern about the efficacy of prevention efforts against child obesity. Nurses working in child health services lack guidance for the management of pre-school children with overweight, which results in a number of consequences that vary from unnecessary extra visits (continuous observation) to not identifying overweight at all. Need for continuous training in the use of structure person-centered family counselling empowering families when a child is identified with overweight. A structured model was developed for the prevention of obesity that can be implemented in routine practices in CHS.	Pg. 2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Control:** Guided using the digital National Handbook for Child Health (*get reference)*. Illustrations were developed for the CCHD were published in the national handbook. A questionnaire to describe usual care was distributed at the start of the RCT. **Intervention:** Illustrations (*not provided*). BMI growth charts were also used in discussion with children and their parents.	Pg. 3 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Control:** Received the 4-years health visit including a health dialogue and identification of overweight using the IOTF definition. The nurses working in child health services are guided using the digital National Handbook of Child Health Services. Illustrations were published in the national handbook to help guide education – overview of the illustrations available (2). A questionnaire to describe the usual care was distributed before the start of the RCT. As child health services is	Pg. 3 (1), 387 (2)

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	not responsible for treatment of diseases, children with obesity are referred to a multidisciplinary team consisting of a specialized nurse, dietician and a pediatrician for treatment outside the primary care setting. **Intervention:** 4-year-old children received CCHD at the 4-years health visit. Children with overweight were offered the targeted part of CCHD and extra visits or a referral to an overweight team, dietician or CHS psychologist at the caregiver's request. Children with obesity were referred to specialized care (described under usual care). CCHD is a low-intense multicomponent child-centered intervention based on the Logic model (2). Builds on a family-therapeutic solution-focused approach while emphasizing a relationship based on trust and facilitation of the family to make healthy choices. Consists of 2 parts: Universal part directed at all 4-year-olds regardless of weight and their families, and a targeted part offered to families when the child is identified with overweight. *Universal:* 10-minute structured dialogue between the child, caregiver and nurse using 8 animated illustrations (2) based on healthy choices (fruit/veg consumption drinking water, portion sizes, physical activity, tooth brushing and bedtime routines). As well as a neutral discussion re child's growth using BMI growth charts. *Targeted:* 45-minute sessions, Family Guidance is offered 1-3 weeks after the universal health visit, offered to caregivers, the child and other adults important to the family. Inspired by the Standardized Obesity Family Therapy – an evidence-based treatment model that engages family members through collaborative family support in either identifying protective factors or finding sustainable changes in their lifestyle.	Pg. 3(1), 386 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Control:** Guided using the National Handbook for Child Health Services and illustrations developed for CCHD. However, they were not trained in the CCHD approach. Received a traditional 120-minute lecture on child overweight and the BMI growth chart with no possibility for discussion or reflection.	Pg. 3 (1)

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	Intervention: Received an interactive 1-day training in small groups focused on how to promote child participation and how to apply a family-therapeutic solution focused approach. Main components of training available (3). Was followed by 4 tutorial sessions of 1 hour every 2 months to reflect on how to build trust with the family, how to contextualize and re-frame into positive reflections and how to handle the challenges encountered in the early communication about weight.	Pg 3 (1), 823 (3)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Control & Intervention:** Started at the 4-year health visit, assumed to be in-person.	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Control & Intervention:** Occurred at 37 Child Health Centres in a southern county in Sweden.	Pg. 2 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Control:** A one-time visit at the 4 year old mark **Intervention:** 2 visits. A 10-minuted structured dialogue for the universal part, and a 45-minute Family Guidance visit. *?any other visits*	Pg 3-4 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention:** The Family Guidance targeted session engaged family members in identifying protective factors or finding sustainable lifestyle changes.	Pg. 4 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		

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Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Fidelity to the study protocol was monitored through the collection of data from the child's records on delivery of care for children with overweight at baseline and through a digital questionnaire containing 11 questionnaires distributed to all nurses at baseline.	Pg. 9 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	Throughout the RCT, 26 nurses stopped working in the Child Health Services: 11 nurses in the control group and 16 nurses in the intervention group. The intervention group recruited 4 new nurses trained in CCHD, while 5 nurses could not be trained, resulting in the withdrawal of 285 children in the intervention group that met a nurse not trained in CCHD, of which 31 children had overweight.	Pg. 9 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

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1. Derwig, M., Tiberg, I., Björk, J., Welander Tärneberg, A., & Kristensson Hallström, I. (2021). A child-entered health dialogue for the prevention of obesity in child health services in Sweden - A randomized controlled trial including an economic evaluation . Obesity Science and Practice, 1–14.
2. Derwig M, Tiberg I, Björk J, Hallström I. Child-Centred health dialogue for primary prevention of obesity in child health services – a feasibility study. Scand J Public Health. 2021;49(4):384-392. https:// doi.org/10.1177/1403494819891025
3. Castor C, Derwig M, Borg SJ, Ollhage ME, Tiberg I. A challenging balancing act to engage children and their families in a healthy lifestyle – nurses' experiences of child-centred health dialogue in child health services in Sweden. J Clin Nurs. 2021;30(5/6):819-829. https://doi.org/10.1111/jocn.15622

Article Title	Can a dairy-rich diet be effective in long-term weight control of young children	
Authors	Kelishadi R, Zemel MB, Hashemipour M, Hosseini M, Mohammadifard N, Poursafa P.	
		Page Number

		(Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	NR	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	A longitudinal study on the effect of dairy intake in early childhood on the acquisition of body fat throughout childhood showed that suboptimal dairy intake during preschool was associated with greater gains in body fat throughout childhood. The objective of the current study was to determine the short- and long-term results of a randomized controlled trial of a dairy-rich diet on generalized and abdominal obesity, and the components of metabolic syndrome among obese pre-pubescent children.	Pg 602 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	All participants attended family-centered education sessions on healthy lifestyle (no materials provided)	Pg. 603 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**All Groups:** All participants attended 6 consecutive monthly family-centered education sessions about healthy lifestyle (healthy nutrition and increasing physical activity) conducted by a pediatrician and nutritionist. **Dairy Rich Group (DR):** A dairy rich diet of >800 mg of calcium per day with no change in energy or macronutrient intake was recommended to the children of this group. Were recommended to obtain most of their calcium from low-fat and regular milk, cheese and yogurt, as well as liquid and sold curt. **Energy-Restricted Group (ER)**: Participated in similar nutrition education session as	Pg. 603 (1)

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	the DR group. Were placed on a caloric restriction regimen with an energy content restricted to the caloric requirement for height. To ensure equal treatment intensity between the 2 intervention groups, the same team provided diet recommendations through educational sessions of the same number, duration and interval. **Control:** Received no dietary recommendations aside from what was discussed during the healthy lifestyle education sessions.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	The healthy lifestyle education sessions were provided by a pediatrician and nutritionist It is not clear who provided the DR and ER groups education, or whether this was done in conjunction with the healthy lifestyle education sessions.	Pg. 603 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	It is not stated whether the healthy lifestyle education sessions were provided as a group or individually. It is assumed that the groups were done in person, but not explicitly stated	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	The participants had been referred to the Obesity Research Clinic, Preventive Pediatric Cardiology Department, Isfahan Cardiovascular Research Centre (ICRC), a collaborating centre of the WHO in Isfahan, Iran.	Pg. 602 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their	All participants attended 6 consecutive monthly education sessions. Not stated duration of the sessions, and whether the intervention groups education sessions	Pg. 603 (1)

For peer review only

duration, intensity or dose.	ran in conjunction with the general education sessions.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	None reported	
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Intervention adherence was monitored by retention and attrition rates. No strategies regarding fidelity were discussed.	Pg. 604 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	120 children with obesity were randomized, and of these 95 participants completed the 36-month follow-up period (75%).	Pg 604 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Kelishadi R, Zemel MB, Hashemipour M, Hosseini M, Mohammadifard N, Poursafa P. Can a dairy-rich diet be effective in long-term weight control of young children?. J Am Coll Nutr 2009;28(5):601–10.

Article Title	Efficacy of family-based weight control program for preschool children in primary care	
Authors	Quattrin T, Roemmich JN, Paluch R, Yu J, Epstein LH, Ecker MA.	
		Page Number

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		(Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Buffalo Healthy Tots	Pg 684 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Goal was to test the efficacy of an innovative family-based intervention program designed for treating overweight/obese children (aged 2-5 years) and an overweight parent in the primary care setting. Hypothesized that children and parents randomized to a family-based behavioural intervention would have greater reductions in BMI percentile and BMI as the contributions of parental eating habits and obesity status is well linked to their child's weight status.	661 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Control & Intervention** – standardized letter updating parents on the progress of their child, not provided **Intervention:** Icon diary used for documenting physical activity and sedentary behaviours. This was not provided	661 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Information Control:** ten 60-minute sessions over 6 months in which a consistent group leader delivery dietary, physical and sedentary activities education. Children were engaged in active games. Parents received 8 phone calls between meetings by an assigned coach. Were given a child weight loss goal of 0.5 to 1lbs/week. Follow-up regarding progress was done with a standardized letter (*not provided)* at 3 months and at a well-child visit at 6 months. **Intervention:** Same as the information control. The group leader also emphasized parenting and behavioural strategies to promote child and parent-behaviour change. Parents also attended a 1:1 meeting at the time of group sessions with an assigned coach who assisted parents in shaping behaviour goals. Parents were instructed to record child and parent food intake, as well as physical and sedentary	661 (1)

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	behaviours using an icon-based diary (*not provided)*, this allowed for shaping of goals. Were asked to weight themselves and their child twice per week with a weight loss goal of 1-2 lbs/week.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Information Control/Intervention:** trained group leader (*unclear role & training received*) as well as an assigned "coach" who provided mid-way point phone check ins (*unclear training of "coach"*) Pediatricians blind to the group assignments reviewed the child's progress at 3 and 6 month interval. Trained research assistant measured child's weight at every session	661 (1) 662 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Control & Intervention** – Group sessions, presumed to be in-person with individual phone calls by coach in between meetings. **Intervention -** Same as control with the addition of a 1:1 meeting at each group session, presumed to be in person	661-662 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Primary care setting → unclear location	660 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Control –** 53 randomized to control. Had ten 60-minute sessions over 6 months (4 weekly, 2 bimonthly and 4 monthly). Received 8 phone calls between meetings by their assigned "coach". **Intervention -** 52 randomized to the intervention. Also had ten 60-minute sessons over 6 months (4 weekly, 2 bimonthly and 4 monthly). Received 8 phone calls between meetings by their assigned "coach". Also met 1:1 with their coach either before or after each of the meetings	661,663 (1)
Tailoring		

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If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention –** Parents attended 1:1 meetings with an assigned coach who assisted parents in shaping behavioural goals after having reviewed food/activity diaries.	662 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Control & Intervention:** Group leaders were closely supervised by investigators. No other strategies were discussed regarding maintaining fidelity.	662 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	None reported	

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Quattrin T, Roemmich JN, Paluch R, Yu J, Epstein LH, Ecker MA. Efficacy of family-based weight control program for preschool children in primary care. Pediatrics 2012;130 (4):660–6.

Article Title	A pilot randomized controlled trial of a clinic and home-based behavioural intervention to decrease obesity in preschoolers	
Authors	Stark LJ, Spear S, Boles R, Kuhl E, Ratcliff M, Scharf C, et al. (2011)	
		Page Number (Location in Text)

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Brief Name		
Provide the name or a phrase that describes the intervention.	LAUNCH – Learning about Activity and Understanding Nutrition for Child Health	134 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Developed a 6-month behavioural intervention for preschool obesity based on social cognitive theory and modeled on successful behavioural family-based intervention and modeled on successful behavioural family-based interventions for school-aged children. The program will specifically target preschool years, and was found to have a decrease in BMI z-score which was maintained at 12 months post-intervention. LAUNCH was designed to produce small decreases or stabilize the rate of children's weight gain, consistent with the current recommendations for treatment of preschool obesity.	134, 135 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	*Standard of Care:* Pediatrician used a scripted manual (not provided) to delivery dietary and physical activity recommendations outlined by the AAP (article with recommendations referenced). Each family was given a 1-page healthy food and activity brochure created by the Collaboration for Health Ohio (http://www.healthyohioprogram.org/healthylife/nutri2/nutrikids2/ounce.aspx). *LAUNCH*: Pedometer and diet diary were used as feedback tools. Parents were given a 14-day supply of vegetables at each of the six clinic visits. A written manual (not provided) was used by the clinical psychologist to conduct the parent group.	135 (1) (2)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	*Standard of Care*: Dietary and physical activity recommendations (outlined by AAP) were provided by a pediatrician using a scripted manual. Each family met with the pediatrician for a one-time 45-minute visit to review the growth chart and BMI. Each family was given a 1-page handout on healthy food and activity. *LAUNCH:* 6-month intervention consisting of 2 phases. Phase 1 (intensive phase) was 12 weekly sessions alternating between group-based clinic sessions (parents and	

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	child attending concurrent groups), and individual home visits. Phase 2 (maintenance) was 12 weeks of every other week sessions, alternating between group sessions in clinic and home sessions. Parent-group sessions addressed dietary education, physical activity and parenting skills. Separate sessions were done for snacks/beverages, breakfast/lunch and dinner. Sessions were also done on decreasing screen time, increasing physical activity. Education regarding child behaviour management skills was also taught. Parents were given a 14-day supply of vegetables at each of the six clinic visits. Children attended a group session concurrently – nutrition education done through various activities as well as completed 15 minutes of moderate to vigorous activity. In-home sessions supported generalization of clinic-taught skills to the home environment. Phase 2 (maintenance) focused on helping families continue to make/maintain changes in eating and physical activity.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	*Control:* Pediatricians *LAUNCH:* The parent group during the intensive phase was conducted by a licensed clinical psychologist following a written manual. Child groups were conducted by a pediatric psychology postdoctoral fellow and a research coordinator. In-home sessions were conducted by psychology postdoctoral fellows. No mentioned of specific training for implementation of the intervention. Anthropometric measurements were done by trained personnel from the research centre → did not mention what training required.	135-136 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Control**: Delivered via a clinic appointment in person **LAUNCH:** Group sessions for both parents and children were done in person. Home visits were individual and done in person. Unclear if the maintenance phase (phase	

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	2) were done in-person or whether done as a group/individual session.	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Control:** Pediatrician office **LAUNCH:** Group sessions were done in a clinic location. Home sessions done in participant's houses. Did not specify where phase 2 occurred.	
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Control:** One 45 minutes session **LAUNCH:** Phase 1 consisted for 12 weekly sessions alternating between group sessions and home visits. Group sessions were 90 minutes in duration and covered 1 topic relating to diet and physical activity. In-home sessions lasted 60-90 minutes.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**LAUNCH:** Unclear whether home visits were personalized to address clinic-taught skills	
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Control –** The pediatricians followed a scripted manual **LAUNCH** – Parent group was conducted by a licensed clinical psychologist following a written manual. Inter-rater reliability was conducted by an independent coder for home food environment assessments. No other outcomes were reported to have this. No other information regarding planned fidelity	
Actual: If intervention adherence or fidelity was		

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assessed, describe the extent to which the intervention was delivered as planned.	Not reported	

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark LJ, Spear S, Boles R, Kuhl E, Ratcliff M, Scharf C, et al. A pilot randomized controlled trial of a clinic and home-based behavioral intervention to decrease obesity in preschoolers. Obesity (Silver Spring) 2011;19(1):134–41.
2. Spear BA, Barlow SE, Ervin C et al. Recommendations for treatment of child and adolescent overweight and obesity. Pediatrics 2007;120 Suppl 4:S254–S288.

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Article Title	A Pilot Randomized Controlled Trial of a Behavioral Family-Based Intervention With and Without Home Visits to Decrease Obesity in Preschoolers	
Authors	Stark LJ, Clifford LM, Towner EK, Filigno SS, Zion C, Bolling C, et al. (2014)	

		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Learning about Activity and Understanding Nutrition for Child Health (LAUNCH)	Pg 1002 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	To address the unique aspects of treating obesity in preschoolers (food neophobia or unwillingness to try new foods, tantrumming peaks in this age group). Changing the diet of preschoolers would necessitate changing parents' interaction with their child and their home environment. Developed and piloted a behavioural intervention for preschool obesity modelled on successful behaviour family-based intervention for school-age children, while addressing the unique challenges of preschoolers. Sought to examine a version of LAUNCH that conforms to Expert Committee guidelines for Stage 3 intervention on the number of sessions, structure, and support compared with a condition similar to Stage 1 intervention.	Pg. 1002 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**LAUNCH-Home Visits (HV) & Clinic:** Required a diet diary, vegetables. LAUNCH-clinic required a home clean-out box. **Control:** A 1-page healthy food and activity brochure created by the Collaboration for Healthy Ohio (https://www.nationwidechildrens.org/specialties/center-for-healthy-weight-and-nutrition/services-we-offer/ounce-of-prevention)	Pg. 1003, 1005 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**LAUNCH-HV**: Manualized intervention consisting of two phases (Intensive & Maintenance), alternating between group clinic and individual home sessions. *Phase 1:* Months 1-3 targeted lifestyle behaviour modifications and improving parenting skills (supplementary material available online for treatment flow and session topics). Weeks 2-7 included education on diet, weeks 8-12 education on physical activity and parenting skills at all visits. Parents kept a 7-day diet diary for themselves and their child. They were also provided with vegetables at each session	Pg. 1003-1005 (1)

For peer review only

	for daily taste tests between sessions. Children participated in a concurrent manualized group-based intervention paralleling the topics covered in the parent sessions focusing on education about healthy education, opportunities to try new foods and engage in physical activity. Home sessions followed a manualized protocol to support generalization of clinic-taught skills to the home environment through instruction, therapist modeling and parent rehearsal of dietary changes, physical activity and behaviour techniques. "Home clean-outs" were conducted following each clinic session on diet. *Phase 2:* Months 4-6 focused on helping families maintain treatment gains by engaging parents in long-term planning, problem-solving and use of parenting skills to promote maintenance of diet and activity changes. **LAUNCH-C:** Intervention content was identical to LAUNCH-HV clinic sessions. Instead of home-visits, parents were provided a "home clean-out" box to use on their own to eliminate high-calorie low-nutrient foods from the home. **Standard of Care:** Manualized intervention designed to deliver dietary and physical activity recommendations outlined by the AAP (2). Pediatrician explained BMI, BMI percentile and reviewed the child's growth chart. Pediatrician recommended screen time ≤ 2 hours; active play ≥ 60 mins daily, eliminating soda and ≤ 4 oz daily; fruits and vegetables ≥ 5 servings daily, limiting eating out; and appropriate portion size for preschoolers. They were also given a 1-page healthy food and activity brochure (3).	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**LAUNCH-HV:** Parent-group clinic sessions were conducted by a licensed clinical psychologist or 2nd year psychology postdoctoral fellow. The kid groups were led by a pediatric psychology postdoctoral fellow and a research coordinator. Home sessions were conducted by a pediatric psychology fellow.	Pg. 1002-1003 (1)

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	LAUNCH-C: Same as the LAUNCH-HV **Standard of Care:** Pediatrician	
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**LAUNCH-HV**: The education sessions were provided in the clinic as a group. The home-visits were individual. **LAUNCH-C:** Education sessions were in-person group sessions. **Standard of Care:** Individual in-person sessions.	Pg. 1002-1003 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Conducted at the Cincinnati Children's Hospital Medical Centre as well as home visits (LAUNCH-HV)	Pg. 1003 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**LAUNCH-HV:** 18-sessions, 6-month intervention consisting of two phases. Phase 1 was 3 months in duration consisting of 12 weekly sessions alternating between group-based clinic sessions and individual home visits. The parent group sessions were 90 minutes in duration. Home visits lasted 6-90 minutes. Phase 2 consisted of 12 weeks of every-other-week sessions alternating between group clinic sessions and individual home visits. **LAUNCH-C:** Sessions were conducted every other week during months 1-3 and monthly during months 4-6 of the intervention, for 10 treatment sessions. **Standard of Care:** Pediatrician met with each family individually for one 45 minutes visit.	Pg. 1003-1005 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why,	**LAUNCH-HV/C:** The maintenance phase focusing on helping to maintain treatment gains focused on problem-solving around individual barriers, and use of parenting	Pg. 1004 (1)

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when, and how.	skills.	
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	No planned fidelity assessments were discussed. Intervention adherence was monitored via attrition	Pg. 1006 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	15 families were randomized to LAUNCH-HV, 14 to LAUNCH-C, and 13 to standard of care. Of those, 3 families in LAUNCH-HV, 1 in LAUNCH-C and 1 in standard of care, dropped out before completing all the treatment sessions. The 1 in standard of care dropped out after completing the session.	Pg. 1006 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark LJ, Clifford LM, Towner EK, Filigno SS, Zion C, Bolling C, et al. A pilot randomized controlled trial of a behavioral family-based intervention with and without home visits to decrease obesity in preschoolers. J Pediatr Psychol 2014;39(9):1001–12.

2. Spear, B. A., Barlow, S. E., Ervin, C., Ludwig, D. S.,Saelens, B. E., Schetzina, K. E., & Taveras, E. M.(2007). Recommendations for treatment of child and adolescent overweight and obesity. Pediatrics, 120(Suppl. 4), S254–S288. doi:10.1542/peds.2007-2329F

3. Toolkit (2007). Ounce of prevention is worth a pound: Preventing childhood obesity starting from birth. Retrieved December 26, 2021. https://www.nationwidechildrens.org/specialties/center-for-healthy-weight-and-nutrition/services-we-offer/ounce-of-prevention

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RITES TOOL

Explanatory Design: determines whether an intervention produces the expected result under ideal research circumstances and are intended to provide evidence on the *efficacy* of an intervention.

Pragmatic Design: determines the degree of beneficial effect under "real world" clinical conditions and are intended to provide evidence on the *effectiveness* of an intervention.

Criteria		Descriptor for trial with maximum *effectiveness* (pragmatic)	Descriptor for trial with maximum *efficacy* (explanatory)
Participants	**Characteristics/Description***	The participants are representative of the population who would receive the experimental intervention if it was part of usual care. ** They are of similar age, severity of illness and comorbidities to those patients who would be candidates for the intervention in usual care setting. They reflect diversity along parameters that could impact adherence.	The participants are a homogenous population and are markedly different from those seen in usual care. May have been deliberately selected to comply with treatment, respond to treatment or demonstrate the efficacy of the experimental intervention. There may be other exclusions that would not be seen in usual care ** (e.g. exclusion of participants with comorbidities).
	Eligibility	How similar are the participants to those who would receive the intervention if it was part of usual care?	There are many exclusions (e.g. those who don't comply, respond to treatment, or are not at high risk for the primary outcome, etc.), or the trial uses many selection tests not used in usual care.
	Recruitment	Recruitment is through usual appointments or clinics	Recruitment is through targeted information letters, advertising in newspapers, radio plus incentives and other routes that would not be used in usual care.
Trial Characteristics	**Setting**	The overall setting of the trial is similar to usual care – and includes diverse sub-settings common in usual care for this intervention (e.g. primary care, specialized care)	The setting is selected to maximize the ability to carry out the trial and identify an intervention effect if there is one. The setting may be more specialized than the setting in which the intervention would be delivered in usual care (e.g. a single center, specialized clinics or specialized academic centres or only providers with high levels of experience).

	Organization	The resources, provider expertise and the organization of care delivery in the intervention arm are identical to usual care	The trial increases staff levels, gives additional training, requires more than the usual experience or certification, or certification and increase resources.
	Comparison intervention	The control typically is applied by the full range of practitioners and in the full range of clinical settings, with only ordinary attention to the training, experience and performance of the practitioners.	The practitioner expertise is standardized in the control to maximize the chances of detecting whatever comparative benefits the experimental intervention might have
Flexibility	**Delivery**	Flexibility in the intervention is identical to that in usual care	Intervention delivery is less flexible than usual care. There is a strict protocol, monitoring and measures to improve compliance, with specific advice on allowed co-interventions and complications.
	Adherence	There is no more than usual encouragement to adhere to the intervention	Exclusion based on adherence and measures to improve adherence if found wanting.
	Practitioner adherence to study protocol	There is unobtrusive (or no) measure of practitioner adherence and no special strategies to maintain or improve it	There is close monitoring of how well the participating clinicians and centres are adhering to even the minute details in the trial protocol and "manual of procedures".
	Flexibility of the control	"Usual practice" or the best available alternative management strategy, offering practitioners considerable leeway in deciding how to apply it	Restrictive flexibility of the comparison intervention
Clinical relevance of experimental and comparison interventions	**Clinical Relevance**	Both the experimental and comparison intervention have the potential to be "best practice"	One or both of the experimental and comparison intervention is not a clinically relevant treatment (e.g. placebo, sub-clinical doses)
	Primary outcome	The primary outcome is of obvious importance to participants.	The primary outcome uses a surrogate, physiological outcome, central adjudication or uses assessment expertise that is not available in usual care, or the outcome is measured at an earlier time than in usual care.
	Primary analysis	The most pragmatic approach is to use intention to treat with all available data	The most explanatory analysis is one that excludes ineligible post-randomization participants and includes only those following the treatment protocol.

* Participants included in the study should be judged according to whether they are representative of the general population of those with the condition of interest who would receive usual care in the geographic area in which the trial is carried out

** Usual care describes the type of care routinely received by patients in the geographic area in which the trial is carried out. The care provided can vary by patient characteristics (eg. age, sex, education), the setting (inpatient/outpatient, primary care/specialized care) in which the patient is seen, individual providers and insurance plans.

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5-point Likert Scale:

1 = Strong Emphasis on Efficacy
2 = Rather strong emphasis on efficacy
3 = Balanced emphasis on both efficacy and effectiveness
4 = Rather strong emphasis on effectiveness
5 = Strong emphasis on effectiveness

Study Title: A Parent Treatment Program for Preschoolers with Obesity: A Randomized Control Trial

Authors: Ek A., Chamberlain, K., Sorjonen, K., Hammar, U., Malek, M., Sandvik, P., Somaraki, M., Nyman, J., Lindberg L., Nordin, K., Ejderhamn, J., Fisher, P., Chamberlain, P., Marcus, C., Nowicka, P.

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☒	☐	☐	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: Efficacy of a Community- Versus Primary Care-Centered Program for Childhood Obesity: TX CORD RCT

Authors: Butte, N., Hoelscher, D., Barlow, S., Pont, S., Durand, C., Vandewater, E., Liu, Y., Adolph, A., Pérez, A., Wilson, T., Gonzalez, A., Puyau, M., Sharma, S., Byrd-Williams, C., Oluyomi, A., Huang, T., Finkelstein, E., Sacher, P., Kelder, S.,

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☐	☒	☐	☐
Flexibility of Intervention	☐	☐	☐	☒	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

For peer review only

Study Title: Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study

Authors: Sheryl L. Rifas-Shiman, Elsie M. Taveras, Steven L. Gortmaker, Katherine H. Hohman, Christine M. Horan, Ken P. Kleinman, Kathleen Mitchell, Sarah Price, Lisa A. Prosser and Matthew W. Gillman

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☒	☐	☐
Trial Setting	☐	☐	☒	☐	☐
Flexibility of Intervention	☐	☒	☐	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: Clinic and Home-Based Behavioral Intervention for Obesity in Preschoolers: A Randomized Control Trial

Authors: Stark, L., Filigno, S., Bolling, C., Ratcliff, M., Kichler, J., Robson, S., Simon, S., McCullough, M., Clifford, L., Stough, C., Zion, C., Ittenbach, R.

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: A pilot randomized controlled trial of a behavioral family-based intervention with and without home visits to decrease obesity in preschoolers (2014)

Authors: Stark, L., Clifford, L., Towner, E., Filigno, S., Zion, C., Bolling, C., & Rausch, J.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☒	☐	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☒	☐	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

For peer review only

Study Title: A pilot randomized controlled trial of a clinic and home-based behavioural intervention to decrease obesity in preschoolers (Stark 2011)

Authors: Stark, L., Spear, S., Boles, R., Kuhl, E., Ratcliff, M., Scharf, C., Bolling, C. & Rausch, J.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☒	☐	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

For peer review only

Study Title: Efficacy of family-based weight control program for preschool children in primary care (Quattrin 2012)

Author: Quattrin, T., Roemmich, J., Paluch, R., Yu, J., Epstein, L., & Ecker, M.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☒	☐	☐
Trial Setting	☐	☐	☒	☐	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

For peer review only

Study Title: Can a dairy-rich diet be effective in long-term weight control of young children

Author: Kelishadi, R., Zemel, M., Hashemipour, M., Hosseini, M., Mohammadifard, N. & Poursafa, P.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☐	☐	☒	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: A child-centered health dialogue for the prevention of obesity in child health services in Sweden – A randomized controlled trial including an economic evaluation

Authors: Derwig, M., Tiberg, I., Björk, J., Welander Tärneberg, A., & Kristensson Hallström, I.

Reviewer: SN, RG

	Strong Emphasis on Efficacy ❶	Rather strong emphasis on efficacy ❷	Balanced emphasis on both efficacy and effectiveness ❸	Rather strong emphasis on effectiveness ❹	Strong emphasis on effectiveness ❺
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☐	☐	☒	☐
Flexibility of Intervention	☐	☐	☐	☒	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P.1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P.2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P.4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P.5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P.5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P.6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P.6-7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P.7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P.6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P.6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P.7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P.8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P.8-9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P.8-9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P.8-9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P.8-9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P.8-9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P.7-9
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P.7-9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P.9 Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	P.9 Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P.11 Figure 7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P.9-11 Figures 2-6
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P.11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P.9-11 Figures 2-6
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P.9-11 Figures 2-6
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P.11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P.13-15
	23b	Discuss any limitations of the evidence included in the review.	P.14-15
	23c	Discuss any limitations of the review processes used.	P.14-15
	23d	Discuss implications of the results for practice, policy, and future research.	P.15-16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P.3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P.3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P.16
Competing interests	26	Declare any competing interests of review authors.	P.16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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BMJ Open

Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis

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Secondary Subject Heading:	Paediatrics
Keywords:	PAEDIATRICS, STATISTICS & RESEARCH METHODS, Paediatric endocrinology < PAEDIATRICS

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Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis

Authors: *Samantha V. Nordlund, MD,[a] *Patrick G. McPhee, PhD,[a,b] Ramy Gabarin, BHSc,[c] Charlotte Deacon, BHSc,[d] Lawrence Mbuagbaw, MD, MPH, PhD,[e-h] Katherine M. Morrison, MD,[a,b]

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ABSTRACT

Objectives: (1) To summarize the literature on the impact of pediatric weight management interventions on health outcomes in preschool age children with overweight or obesity. (2) To evaluate the completeness of intervention description and real-world applicability using validated tools.

Design: Systematic review and meta-analysis.

Data sources: Medline, EMBASE, CINAHL, Cochrane Library and PsychInfo were searched between 10 March 2015 through 21 November 2021.

Eligibility criteria: Randomized controlled trials (RCTs) addressing weight management in preschool children (two to six years) with overweight or obesity.

Data extraction and synthesis: Two reviewers independently extracted key information from each study and assessed risk of bias. Random-effects meta-analysis was performed where there was evidence for homogeneous effects. The certainty of evidence was assessed by the Grading of Recommendations Assessment, Development and Evaluation.

Results: Of the 16908 studies retrieved, nine trials (1687 participants) met the inclusion criteria. These interventions utilized motivational interviewing (MI) or multicomponent educational interventions related to health behaviour approaches and were 6 to 12 months in duration. All studies contained some risk of bias. A difference was found in the intervention groups compared to controls for BMI z score (mean difference (MD) -0.10, 95% CI -0.12 to -0.09; 8 trials, 1491 participants; $p<0.001$; I^2 68%), though there was substantial heterogeneity. There were no subgroup effects between studies using MI compared to studies using multicomponent interventions. The certainty of the evidence was considered low. The trials were reported in sufficient detail and were considered pragmatic.

Conclusions: Pediatric weight management interventions delivered to the parents of young children with obesity result in small declines in BMI z score. The results should be interpreted cautiously as they were inconsistent and the quality of the evidence was low.

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PROSPERO registration number: CRD42020166843.

Strengths and limitations of this study

- This review updates our understanding of the effectiveness of weight management interventions on health outcomes of preschool aged children over the past six years.
- This review highlights a knowledge gap for the best practice for pediatric weight management in preschool children.
- This review evaluates the description of interventions and their real-world applicability by using the TIDieR checklist and RITES tool, respectfully.
- This review investigated children ages two to six years old with overweight or obesity and not those with a medical condition known to affect weight status or syndromic causes for obesity.

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INTRODUCTION

The prevalence of overweight and obesity in children has been increasing globally.[1] There is estimated to be 124 million children and adolescents with obesity worldwide.[2] Nationally representative data from 144 countries suggests that 43 million children under five years of age have obesity.[3] Children with obesity carry an increased risk for cardiometabolic disease and poor psychosocial health.[4-7] Childhood obesity often persists into adulthood and is associated with increased risk of morbidity throughout life.[8-10] These alarming rates and health risks underscore a need for effective interventions to treat obesity in early childhood and to prevent continued obesity related health issues across the lifespan.

Family-based behaviour modification remains the foundation of pediatric weight management programs.[11] In a 2016 Cochrane systematic review, multicomponent interventions tested with randomized controlled trials (RCTs) in preschool children with obesity aged two to six years old were effective in reducing body mass index (BMI) z score, a common metric to classify obesity in children.[12] Interventions typically consisted of lifestyle behaviour modification, including dietary and physical activity advice, along with parenting skills. Four trials (n=210 participants) were included in that analysis, and a mean change of -0.3 units (95% CI -0.4 to -0.2) in BMI z score was reported. However, the quality of evidence was deemed to be low and a high risk of bias was identified. Since then, there has been increased interest in this field given the increasing prevalence of pediatric obesity.[13] This systematic review will update the effectiveness of weight management interventions on preschool age children since the Cochrane review in 2016.[12]

Although systematic reviews can be invaluable in facilitating clinical decision-making by providing a critical evaluation of the evidence on a specific topic or condition,[14] to do so the intervention must be adequately described. Further, the pragmatism of the trial will inform the potential for 'real-world' implementation. The Template for Intervention Description and Replication (TIDieR) checklist was created to ensure interventions are described in sufficient

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detail to allow for replication.[15] The Rating of Included Trials on the Efficacy-Effectiveness Spectrum (RITES) tool is used to assess the real-world applicability and pragmatism of trials.[16]

The objectives of this systematic review were: (1) to assess the impact of pediatric weight management interventions on health outcomes in preschool children aged two to six with overweight or obesity; and (2) to evaluate the description of these interventions using the TIDieR checklist and their real-world applicability using the RITES tool. This knowledge will help inform future clinical practice guidelines and the planning of health programming for the treatment of young children with obesity.

METHODS

This review was conducted and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses.[17] A PRISMA study flow diagram was completed to present a summary of searching, screening, and inclusion of studies in this systematic review.[17] The review protocol for this study was registered with PROSPERO (CRD42020166843).

Patient and public involvement statement

No individual participant data were involved in the study; therefore, patients or the public were not involved in the design, conduct, reporting, or dissemination plans of our research.

Eligibility

Study designs eligible for inclusion were RCTs with the aim of weight management in preschool children with overweight or obesity. Any type of intervention of at least six months duration and delivered in a clinical setting was included. The interventions could be delivered by a single professional or be multidisciplinary or multicomponent. Multicomponent interventions typically address health behaviours including diet and physical activity. Participation of family members was not mandatory. Interventions for prevention of obesity, studies with observational designs and studies with self-reported height and weight were excluded. Participants included children with overweight or obesity who were between the ages of two and six years at time of enrolment into a weight management intervention. Studies of children with obesity who had a medical

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condition known to affect weight status or syndromic causes for obesity were excluded (e.g., genetic syndromes such as Prader-Willi Syndrome and Bardet-Biedl Syndrome). The control group in eligible RCTs was defined as no intervention or usual or standard care.

The primary outcomes were change in BMI or change in BMI z score or BMI percentile. BMI changes with age during normal growth,[6] and so age and sex-specific BMI z scores, based on a normative standard were included.[18] Additional secondary health outcomes were included in the search terms: body weight, waist circumference, waist-to-height ratio, skin fold thickness, body fat percentage measured with bioelectric impedance (BIA) or dual-energy x-ray absorptiometry, lipid profile (including LDL, HDL, triglycerides, and total cholesterol), blood pressure, glycemia (fasting plasma glucose, glucose level at 2 hours in a standard oral glucose tolerance test, Hemoglobin A1C) and quality of life. Adverse events were defined as any adverse outcome occurring during or after the intervention, including events not necessarily caused by the intervention.

Search of studies

We conducted a comprehensive literature search in the following electronic databases: Ovid Medline, Embase, CINAHL, Cochrane Library, PsychInfo, WHO International Clinical Trials Registry Platform, Web of Science, and Clinicaltrials.gov. These databases were first searched from March 10 2015 (the end date of the 2016 Cochrane review)[12] to January 31, 2020 and this search was updated on November 21, 2021. The search strategy was developed in consultation with a research librarian using the following major themes: ‘preschool’, ‘overweight or obesity’, and ‘intervention’. Key terms within the search strategy were aligned to medical subject headings (MeSH) in Ovid Medline and were expanded to include more descriptive terms. The search strategy for each database is presented in Appendix 1. Reference lists of included publications and reviews and of other systematic reviews or meta-analyses were manually searched to identify other studies that might not have been identified through the database

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searches. The database searches resulted in a combined total of 16908 studies after removal of duplicates (Figure 1).

Study selection

Titles, abstracts, and full texts were reviewed independently for eligibility by two groups of two authors (SN and CD, PM and RG) using a pre-tested, standardized screening form. Publications selected for inclusion were reviewed by an additional author and expert in the field and compared against the inclusion criteria. Only papers published in English were included. Every attempt was made to identify duplicate publications or multiple reports/companion pieces based on the same study and, if that occurred, the publication with the longest follow-up period for the primary outcome was included.

Data extraction

Key information from each included study was extracted in duplicate using a screening form and data extraction template. If data were missing, attempts were made to obtain the data from the corresponding author noted in the publication. Any disagreement during screening or extraction was resolved by discussion or in consultation with a third author.

Risk of bias and certainty of evidence

Risk of bias of each of the included studies was assessed using the Cochrane Risk of Bias tool.[14] The overall certainty of the evidence for each outcome was judged using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. The certainty of the evidence was assessed across the domains of risk of bias, precision, consistency and publication bias (internal validity) as well as directness (external validity). The quality of evidence was rated for each outcome (high, moderate, low or very low). Certainty was judged as high (further search is very unlikely to change the confidence in the effect estimate), moderate (further research is likely to have an important impact on the confidence in the effect estimate and may change the estimate), low (further research is very likely to have an important impact on the confidence of the effect estimate and is likely to change the estimate), or very low

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(very uncertain about the estimate of the effect). A checklist was created and used to help ensure consistency and reproducibility of the GRADE assessments.[14]

The TIDieR checklist was completed to ascertain if the intervention was reported in sufficient detail to allow for reproduction of the study.[15] The RITES tool was completed to evaluate the pragmatism of the intervention; this tool uses a 5-point Likert scale in four domains: (1) Participant characteristics, (2) Trial setting, (3) Flexibility of the intervention, and (4) Clinical relevance.[16] Two reviewers independently classified each trial on the Likert scale. Disagreements were resolved with a discussion to achieve consensus. All items were summed for each study to provide an efficacy/effectiveness score. The score is a continuum from maximum efficacy (score of 4) on one end to maximum effectiveness (or most pragmatic) at the other (score of 20). A low total score (i.e. 4) suggests a strong emphasis on efficacy and a high total score (i.e. 20) is consistent with a strong emphasis on effectiveness or a more pragmatic trial.

Statistical analysis

Statistical analyses were performed using Review Manager5 (RevMan 2013) software package (Cochrane Training, Cochrane). For continuous data measured on the same scale, the mean difference (MD) was calculated. Overall effects were reported and classified as small, medium, or large based on Cohen's d effect sizes.[19] We used a random-effects model taking into account the interstudy variation which provided a more conservative effect than a fixed model. All effect sizes were reported with 95% CIs. The statistic I^2 was calculated to investigate the variation in effect size due to heterogeneity across studies. 0% to 40% corresponded with low heterogeneity, 30% to 60% corresponded with moderate heterogeneity, 50% to 90% corresponded with substantial heterogeneity, and >75% corresponded with considerable heterogeneity.[20] For three-arm trials (two interventions with a control group), the control group was split for meta-analyses to compare interventions to control (Cochrane Handbook 5.1 Ch:16.5.4).[21]

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Synthesis of results

Review Manager5 was used to combine and calculate each outcome (RevMan 2013). If there was good evidence for homogeneous effects (i.e. tests of heterogeneity were not significant) across trials, a random effects meta-analysis was performed. If there was substantial heterogeneity, subgroup analyses based on sex, differences in BMI z score at baseline (e.g., overweight vs. obesity), or intervention setting were to be performed. If substantial heterogeneity remained, a qualitative synthesis was conducted.

Sensitivity analyses excluding trials with high risk of bias, or with high rate of participant attrition (>20%) were to be performed (PROSPERO CRD42020166843).

RESULTS

Search results

After title and abstract screening, 456 studies remained and underwent full text screening. After full text review and removal of studies that did not meet the inclusion criteria and considering articles and ongoing trials from the previous Cochrane review, 9 studies remained. A manual search of the articles in the reference lists from the included studies did not generate any additional relevant studies. Therefore, a total of 9 RCTs were included in this review.

Current study versus previous Cochrane systematic review

The current study is an updated systematic review and meta-analysis of the previous study by Colquitt et al.[12] The previous study included seven RCTs and identified four ongoing trials that might be relevant in future meta-analyses. Upon review and consideration of these previous studies, four of the seven previously analyzed studies were also included in the current meta-analysis.[22-25] Three studies from the previous review were excluded: Bocca 2012 was a 4-month intervention (<6 months);[26] Lanigan 2013 included an age range of 1-5 years old (<2 years old);[27] Taveras 2011[28] included the same participants as Rifas-Shiman 2017.[29] Consistent with our inclusion criteria, we selected the study with the longest follow-up period[29] and excluded Taveras 2011.[28] Of the four ongoing trials identified in the previous study,[12] one status was

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unknown (NCT00916318), another was terminated due to no funding to launch RCT (NCT01698606), and two were unidentifiable in the literature (NCT02292602; Reifsnider 2012).

Summary of included studies

The nine included studies, published between 2009 and 2021, were conducted either in the USA (n=6),[29-31; 22; 24; 25] in Sweden (n=2),[32; 33] or in Iran (n=1).[23] Of the 1687 participants, 937 were randomized to an intervention and 750 to a comparator group. Five studies were multi-center trials,[29; 30; 32; 22; 33] while four studies were conducted at a single center.[31; 24; 25; 23] Five studies were delivered in a primary care setting,[31; 29; 32; 33; 30] one study was delivered in community-centered programs and primary care clinics,[30] while four studies were implemented in the outpatient care setting under the guidance of healthcare professionals.[22; 24; 25; 23] The interventions included motivational interviewing (MI) for parents[31] or families,[29; 33] dietary interventions with an educational component[23] or education on the principles of good nutrition and activity.[31; 30; 32; 24; 25] (Table 1). Interventions ranged from six[31] to 12 months[29; 30; 32] in duration; three studies included a 12-month follow-up period.[24; 29; 25] and one study had a three year follow-up.[23] Interventions varied in intensity, from less than 1 hour contact time[33] to 3 hrs total contact time (MI)[29] to upwards of 27 hrs total contact time (education).[30]

Effects of interventions

The BMI outcome was BMI z score for eight studies and %BMIp95 was reported for four trials. A difference in BMI z score between intervention and control groups was identified at the end of intervention (mean difference (MD) –0.10, 95% CI -0.12 to -0.09; 8 trials, 1491 participants; p=<0.001; I^2 68%) (Figure 2). No between group difference in %BMIp95 was identified (MD -1.34, 95% CI -3.17 to 0.49; 4 trials, 349 participants; p=0.15; I^2 79%) (Figure 3). Of note, the heterogeneity in both analyses exceeded 50% indicating substantial heterogeneity. Two RCTs included BMI as a primary outcome. There was a small improvement in BMI between intervention and usual care (MD -0.17, 95% CI -0.33 to -0.01; 2 trials, 931 participants; p=0.03, I^2 0%) (Figure 4).

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Subgroup analyses

An exploratory subgroup analysis was completed based on intervention modality (multicomponent educational interventions vs. MI) for BMI z score. We saw no subgroup effects when we compared the studies that used MI (MD -0.02, 95% CI -0.11, 0.06; 3 trials, 987 participants; p=0.59; I^2 49%) compared to the others (Figure 5).or when we compared multicomponent educational interventions (MD -0.11, 95% CI -0.12 to -0.09; 6 trials, 504 participants; p<0.001; I^2 68%) to the others (Figure 6). None of these subgroup analyses explained the heterogeneity. None of the preplanned subgroup analyses were possible due to missing or insufficient reporting of data.

Sensitivity analysis

All the included studies were at high risk of bias and therefore conducting sensitivity analysis based on overall risk of bias was not possible. However, the results from BMI z score in the studies at low risk of bias due to attrition had a similar direction of effect (reducing BMI z score) (i.e., attrition <20%; mean difference (MD) –0.10, 95% CI -0.11 to -0.09; 4 trials, 704 participants; p=<0.001; I^2 67%), despite substantial heterogeneity (Figure 7).

Secondary outcomes

Six of the nine trials included secondary outcomes. Unfortunately, not all data were able to be interpreted in the context of a meta-analysis due to missing results (i.e., no mean standard change reported).[32; 30] Kelishadi 2009 included many secondary outcomes including waist circumference, body fat, lipid profile and blood pressure, however this was the only RCT to include these outcomes.[23] Two of the trials[24; 25] included pediatric quality of life, and noted an improvement in quality of life (MD 6.32, 95% CI 0.91 to 11.73; 2 trials, 50 participants; p= 0.02; I^2 0%) (Figure 8). Weight (kg) was also reported in two trials,[22; 25] with a clinically significant improvement (MD -1.27, 95% CI -1.93 to -0.61; 2 trials, 129 participants; p<0.001; I^2 0%) (Figure 9).

Adverse events

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Two trials reported no adverse events,[31; 32] , while Derwig 2021 reported no adverse events aside from worsening weight status of several participants.[33] Adverse events from the other trials were not reported.

Risk of bias

All trials reported adequate sequence generation and allocation concealment and used a form of computer random generator for randomization. As noted in the risk of bias assessment presented in Figure 10, all but one trial[22] did not blind participants or healthcare providers to the intervention or control groups resulting in a high risk for performance bias for study outcomes. Three of the trials did not mention blinding for data collectors or outcome assessors; these trials were rated as uncertain risk for detection bias.[29; 30; 33] Four of the RCTs blinded study staff collecting outcome measures to participant assignment and therefore were judged as low risk for detection bias.[22; 23; 25; 31] Ek 2019 stated that data were collected by child healthcare professionals participating in the study and was therefore rated high risk for detection bias.[32] The drop-out rates were unbalanced across studies: only 38% of intervention participants in the study by Rifas-Shiman 2017 completed more than 1 visit.[29] Of the remaining trials, drop-out rates for the intervention groups ranged from 2.5%[23] to 36%.[30] Only four RCTs reported less than a 20% drop out rate, making these studies low risk for attrition bias.[22-24; 33] The remaining five RCTs reported at least a 20% drop out rate resulting in a high risk for attrition bias. We judged six of nine trials to have low risk of reporting bias.[29-33; 24] We did not draw funnel plots or perform Egger's test to assess reporting bias due to limited number of trials per outcome (n=8; BMI z score).

Certainty of the evidence

The overall quality of the evidence was considered low as a result of the high risk for performance bias and attrition bias (Table 2). We did not downgrade for blinding of participants when examining overall quality of evidence for the primary outcomes.

Template for Intervention Description and Replication (TIDieR)

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Most of the studies included adequate information regarding the intervention (Appendix 2). In three of the studies,[30; 32; 22] the intervention was well described, although some of the materials required to conduct the intervention were not provided or were not described in sufficient detail to reproduce the intervention. Three of the included trials were variations on the same intervention: Stark has published three RCTs using their LAUNCH intervention.[24; 25; 31] While the papers did include well-described interventions, there is extensive training required to effectively administer the intervention, and training information was not included. The lack of information regarding the training or education for those implementing the intervention was common throughout many of the included studies.[30; 24; 25; 31; 23; 22] In some cases, the reviewers were required to search for previously described protocols to gain an understanding of the materials and training details.[24; 25; 31; 30] Derwig 2021 did require training to implement their intervention, however they did provide an additional paper describing the training and materials in detail.[33]

While Tidier is not typically used for describing the standard of care or usual treatment, in the case of preschool aged obesity treatment, many of the studies included interventions beyond the current standard of care. As such, this review also included the TIDiER checklist for the control intervention. Both Kelshadi 2009 and Quattrin 2012 control groups received the same education sessions as the intervention groups, which was significantly more than the standard of care. Butte 2017 and Stark also included additional interventions for their control groups, over and above standard of care. The standard of care was not always adequately described for all the included studies. Ek 2019 described standard of care based on a specific action plan but did not include more detailed information of what this entailed. Butte 2017 described their standard treatment control with additional material, although the extent to which the material was utilized was unreported.

Rating of Included Trials on the Efficacy-Effectiveness Spectrum (RITES)

The RITES tool data for all nine trials are presented in Appendix 3. All of the included trials scored between 12 to 16, indicating more pragmatic trials. Five of the trials[30; 32; 31; 24; 25] had more

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explanatory trial settings as they required extensive training for implementation of the intervention and/or ongoing supervision to ensure the precise implementation. All the included studies were found to be highly effective in terms of their clinical relevance. However, all the trials, aside from Derwig 2021,[33] excluded participants with underlying conditions or comorbidities, resulting in greater emphasis on efficacy for participant characteristics pertaining to overweight or obesity.

DISCUSSION

Summary of evidence

Since the previous review published in 2016, five RCTs examining obesity treatment interventions in young children have been published. This review included RCTs from the original Cochrane study that were consistent with this review's inclusion criteria. This resulted in an additional four studies being included. Seven of the nine RCTs included an educational component as part of the intervention, however, one of those had a primarily dietary intervention.[23] The other two interventions included family guidance (family-based therapy) and MI.

In the previous Cochrane review, multicomponent interventions were found to be more successful at reducing BMI z score compared to controls.[12] When grouped together, the trials from the Cochrane review along with the updated search in this current review had high heterogeneity with an overall small effect in reducing BMI z score. Most of the trials' interventions were well described using the TIDiER checklist, although some did not include the materials or staff training required for the intervention. The control group did lack some detail in their description. All the trials were noted to be relatively pragmatic, suggesting that they could be applied in real-world settings.

Deviations from protocol

We were unable to perform subgroup analyses based on sex or differences in BMI z score at baseline due to missing data. Similarly, descriptions of outpatient intervention settings

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were vague and therefore were not compared to other settings (e.g., Primary care clinics). In the protocol, we planned to report a narrative synthesis in the presence of substantial heterogeneity. However, given the direction of effect, and the use of random effects models to incorporate heterogeneity, we have reported a summary measure, but do advise caution when interpreting these results.

An exploratory subgroup analysis considering intervention modality (multicomponent educational interventions vs MI) was completed due to substantial heterogeneity in the primary analysis. The six RCTs with a multicomponent educational intervention utilizing BMI z score as their primary outcome showed consistent effectiveness in reducing BMI z score (8 out of 9 experimental groups showed that multicomponent educational interventions reduced BMI z scpre). MI, either focused on the parents, child or the family did not show consistent effectiveness for the treatment of overweight and obesity in preschool aged children. Two of the interventions included MI over the telephone if parents missed a session. The implications of providing an intervention virtually are largely unknown and may have affected the overall impact of the intervention.

It is important to note that the multicomponent interventions had greater contact time with the families compared with MI or family guidance. A previous systematic review concluded that at least 26 hours of multicomponent intervention over a minimum of 6 months improved BMI z score.[34] Of the included RCTs, only three reported at least 26 hours of intervention time.[30; 24; 25] Kelishadi 2009 did not document total intervention contact time. On the contrary, MI interventions were limited to 3 to 7.5 hours of total contact time, while family guidance had very limited contact time at 1 hour.[33] Given these differences, one cannot suggest with confidence that it was the intervention type (i.e. education vs MI) and not the 'intensity' of the intervention that led to differences in effectiveness.

Overall, the trials included in this study described their interventions in sufficient detail in the methods sections, or via reference to a published protocol paper. However, the standard

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care provided to the control group was not often described in sufficient detail to allow replication. Trials need to be fully transparent when describing the details of usual or standard care control groups, to truly understand the effectiveness of interventions in the real-world setting. This is an important area for further exploration in future studies and reviews.

To the best of our knowledge, this was the first systematic review in children with obesity to utilize the RITES tool. Generally, the trials were more pragmatic than efficacious on the effectiveness-efficacy scale. Effective or pragmatic trials are similar to routine clinical practices and indicate a more 'real-world' setting. The fact that the majority of trials were conducted in the primary care setting, or a combination of primary care and the community, supports a pragmatic approach; however, the level of training and resources required to implement the interventions in real-world scenarios may pose a challenge. Additionally, most trials recruited participants following stringent criteria, excluding those with comorbidities, and resulting in less pragmatism.[35] This limits the generalizability of the findings beyond preschool age children with overweight or obesity only.

Limitations

The search strategy for this systematic review was influenced by the previously published Cochrane review.[12] Bias was introduced as the searches performed and trials included in this review were restricted to English. Splitting the control group in three arm trials limits the overall variability within trials; however, it is a recommended approach in the Cochrane Handbook to analyze such multicomponent interventions.[36] Further, the methodological heterogeneity between trials weakens our confidence in the conclusion regarding an overall effect of interventions on improving BMI z score. The subgroup analysis based on intervention modality did not result in a change in heterogeneity, which remained >50% and consistent with substantial heterogeneity. As such, we downgraded the certainty of evidence due to inconsistency. Although there was a greater decline in BMI z score for those undergoing educational interventions, it was not statistically significant compared to non educational

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interventions, and the certainty of the evidence for this outcome was considered low. The main methodological concern from review authors was the substantial attrition bias in most trials (>20%). Unfortunately this is common in pediatric weight management.[37] We found that trials with low risk of bias due to attrition had a similar direction of effect (reducing BMI z score), but substantial heterogeneity remained. Most trials declared that the participants and healthcare providers were not blinded to their intervention or control group, leading to concerns of performance bias. However, weight management interventions involving services from healthcare providers make it difficult to blind personnel. Due to this, we did not downgrade for blinding when examining overall quality of evidence for the primary outcomes.

Future directions

There continues to be limited evidence surrounding the best practice for pediatric weight management in young children. Furthermore, efficacy has been almost entirely based on anthropometric health outcomes alone. Other health outcomes are inconsistently reported and further evidence of the impact of these intervention programs on these outcomes including measures of cardiometabolic health and health related quality of life, will be important given their relevance to clinical practice. Future interventions should also consistently consider potential adverse health outcomes.

Fundamentally, emphasis on a continued search for novel pathways to intervene upon or new approaches to current paradigms must be undertaken given the modest outcomes to date, while also considering the relevance and potential impact of observational evidence. Furthermore, although modest changes in BMI z score were noted, the persistence of the impact of these interventions over time remains uncertain and is an important area for further research.

Conclusions

There continues to be limited evidence regarding the treatment of obesity in children of preschool age. In this updated systematic review, there was limited evidence of the benefit of MI as a treatment option, though the interventions were quite brief. There is some evidence

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education-based programs involving parents from the start of the intervention appear to assist with BMI z score reduction, but further research is required to improve on the care of young children with obesity.

Contributors

SN and PGM have equally contributed as the leads of this study and share first authorship. SN, PGM, RG, LM and KMM contributed to the conceptualization of the study. SN, PGM, RG, and CD led the data searches, data extraction, and quality appraisal. PGM was responsible for data analysis with guidance from LM. SN, PGM, and RG drafted the initial manuscript. All authors contributed to the interpretation of findings, critically reviewed and provided feedback on multiple iterations of the manuscript and approved the final version of the manuscript for submission.

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Competing interests

KMM is an advisory board member for Novo Nordisk and a member of a Data Safety Monitoring Board for Novartis.

Ethics statement

This meta-analysis did not involve individual participant data; therefore, Institutional Review Board approval was not required for this study.

Data availability statement

Data are available upon reasonable request from the corresponding author (kmorrison@mcmaster.ca).

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Table 1. Characteristics of included studies

Source	Setting	Intervention / comparator	Sample size	Sex (F, (%))	Age (mean (SD))	BMI measure (baseline)	Intervention (duration)	Effect Size (95% CI)
Kelishadi 2009	Outpatient	I1: Calcium rich diet	40	NR	5.4 (0.2)	z score: 2.4 (0.01) BMI: 22.1 (0.9)	6 mo. with 3-year follow-up >800mg calcium/day Total contact = 6 education sessions	-0.1 (-0.11, -0.09)
		I2: Energy-restricted group	40	NR	5.5 (0.7)	z score: 2.3 (0.04) BMI: 22.7 (0.8)	6 mo. with 3-year follow-up energy content restricted Total contact = 6 education sessions	-0.1 (-0.11, -0.09)
		C: Control	40	NR	5.7 (0.3)	z score: 2.4 (0.01) BMI: 22.4 (0.5)	No dietary recommendation	
Stark 2011	Outpatient	I: LAUNCH	8	2 (25.0)	4.4 (0.9)	BMI %ile: 99.0 (0.9)	6 mo. with 12 mo. follow-up Phase 1: 12 weekly sessions (90 min) Phase 2: 12 biweekly sessions, alternating between group and home Total contact = ≥30 hrs	-0.6 (-0.92, -0.26)
		C: Control	10	4 (40.0)	3.9 (1.1)	BMI %ile: 97.7 (2.5)	45 min: diet and PA recommendations	
Quattrin 2012	Outpatient	I: Family-based weight control	46	31 (67.4)	4.6 (1.1)	z score: 2.2 (0.8) BMI %: 32.4 (22.4)	12 mo. 60 min sessions/mo. (13 total). Control + behavioural modification Total contact = 13hrs (+8 phone calls)	-0.2 (-0.38, -0.10)
		C: Information control	50	33 (66.0)	4.4 (1.1)	z score: 2.1 (0.7) BMI %: 29.8 (17.1)	60 min sessions/mo. (13 total). Dietary and PA education Total contact = 13hrs	
Stark 2014	Outpatient	I1: LAUNCH – home + clinic	10	8 (80)	4.7 (1.3)	z score: 2.1 (0.2)	6 mo. with 12 mo. follow-up Phase 1: 12 sessions/week (90 min) Phase 2: 12 weeks group clinic + home Total contact = ≥30 hrs	-0.3 (-0.60, -0.00)
		I2: LAUNCH - clinic	11	7 (64)	4.2 (1.1)	z score: 2.5 (0.8)	Phase 1: Biweekly sessions months 1-3 Phase 2: Monthly clinic visits for months 4-6 for 10 total sessions Total contact = 10 hrs	
		C: Control	12	8 (67)	4.8 (0.7)	z score: 2.4 (0.4)	45 min: diet and PA recommendations	
Rifas-Shiman 2017	Primary care clinics	I: High five for kids (motivational interviewing)	253	121 (47.8)	4.8 (1.2)	z score: 1.9 (0.7) BMI: 19.2 (2.6)	12 mo. 4 (25 min) in-person visits + 2 (15 min) phone calls 12 mo. maintenance: 2 in-person visits Total contact = 8 sessions (3hrs)	-0.0 (-0.11, 0.07)

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		C: Usual care	192	94 (49.0)	5.2 (1.1)	z score: 1.8 (0.6) BMI: 19.1 (2.0)	Baseline + annual well child care visits	
Butte 2017	Primary care clinics + Community programs	I: MEND	100	51 (51.0)	2–5*	BMI %ile: 113.0 (15.2) BMI: 20.5 (0.3)	3 mo. intense: 9 (90 min) sessions/week 9 mo. transition: (90 min) sessions/mo. Total contact = 18 sessions (27hrs)	-0.9 (-3.53, 1.79)
		C: 12-month clinic-based program	60	29 (49.0)	2–5*	BMI %ile: 110.2 (13.7) BMI: 20.0 (0.3)	12 mo. clinic based program Total contact = 8hrs	
Stark 2018	Primary care clinics	I1: LAUNCH	47	25 (53.2)	4.6 (1.0)	z score: 2.4 (0.5) BMI %ile: 98.6 (1.2)	3 mo. intense: 90/60 min sessions/week 3 mo. maintenance: biweekly sessions Total contact = 18 sessions (23hrs)	-0.2 (-0.35, -0.03)
		I2: Motivational interviewing	50	29 (58.0)	4.6 (0.9)	z score: 2.4 (0.6) BMI %ile: 98.5 (1.3)	3 mo. intense: weekly sessions 3 mo. maintenance: biweekly sessions Total contact = 18 sessions (7.5hrs)	0.1 (-0.06, 0.22)
		C: Control	54	32 (59.3)	4.6 (0.9)	z score: 2.5 (0.7) BMI %ile: 98.6 (1.3)	Informed caregivers of child's weight	
Ek 2019	Primary care clinics	I1: No booster	43	23 (53.2)	5.2 (0.9)	z score: 3.1 (0.7) BMI: 21.9 (2.3)	12 mo. 10 (90 min) sessions Total contact = 10 sessions (15hrs)	-0.5 (-0.73, -0.21)
		I2: Booster	44	19 (43.2)	5.2 (0.8)	z score: 3.0 (0.5) BMI: 21.4 (1.5)	12 mo. 10 (90 min) sessions + Booster (7 (30 min) phone calls) Total contact = 17 sessions (18.5hrs)	-0.2 (-0.42, -0.04)
		C: Standard care	87	56 (64.4)	5.3 (0.7)	z score: 2.9 (0.6) BMI: 21.3 (1.7)	≥4 visits of 30 minutes over 12 months Total contact = min. 4 sessions (2hrs)	
Derwig 2021	Primary care clinics	I: CCHD	245	133 (54.3)	4.1 (0.1)	z score: 1.6 (0.3) BMI: 18.1 (0.4)	12 mo. 10 min dialogue with child, family, clinician. 45 min family guidance Total contact = 55 min	-0.1 (-0.18, 0.00)
		C: Standard care	245	132 (53.9)	4.1 (0.1)	z score: 1.6 (0.3) BMI: 18.1 (0.5)	Single visit of a health dialogue of overweight	

* = average age not reported in study

CCHD = Child-Centered Health Dialogue; LAUNCH = Learning about Activity and Understanding Nutrition for Child Health. Developing diet, physical activity, and parenting skills; MEND = Mind, Exercise, Nutrition… Do it; PA = physical activity

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Table 2. Quality of evidence

Outcome	No of Studies	Design	Risk of Bias	Precision	Consistency	Directness	Other Considerations	No of Patients	Effect (Absolute)	Quality	Rating
BMI z score	8	RCT	Serious Limitations[1]	No serious imprecision	Serious consistency[3]	No uncertainty	None	1491	-0.10 (-0.12, -0.09)	⊕⊕⊖⊖	Low
%BMIp95	4	RCT	Serious Limitations[1]	Serious[2]	Serious consistency[3]	Uncertainty[4]	None	349	-1.34 (-3.17, 0.49)	⊕⊕⊖⊖	Low

Footnotes:

1. Studies rated down for attrition bias with over 20% loss to follow-up, high risk of performance bias
2. Rated down due to wide confidence intervals
3. High I^2 (68% for BMI z score; 79% for %BMIp95)
4. Indirectness due to differences in setting between interventions (clinic, home, and community centres)

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Figure 1. PRISMA Flow Diagram

Figure 2. Forest plot showing the mean difference in **BMI z score** comparing intervention versus control

Figure 3. Forest plot showing the mean difference in **%BMIp95** comparing intervention versus control

Figure 4. Forest plot showing the mean difference in **BMI** comparing intervention versus control

Figure 5. Forest plot showing the mean difference in **BMI z score** comparing motivational interviewing versus other trials

Figure 6. Forest plot showing the mean difference in **BMI z score** comparing multicomponent educational intervention versus other trials

Figure 7. Forest plot showing the mean difference in **BMI z score** comparing high versus low attrition bias

Figure 8. Forest plot showing the mean difference in **PedsQL** comparing intervention versus control

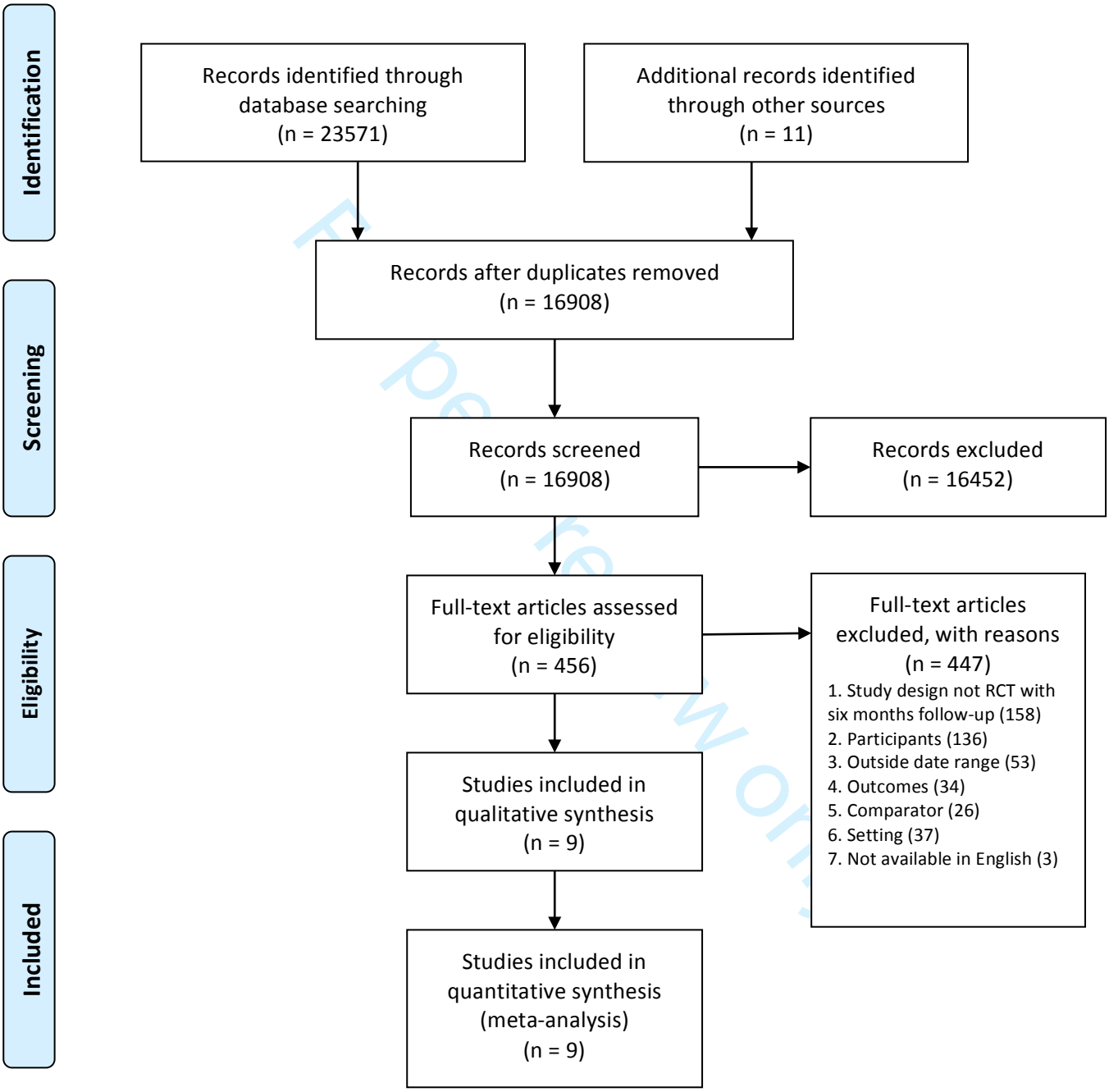
Figure 9. Forest plot showing the mean difference in **Weight** comparing intervention versus control

Figure 10. Methodological quality summary: authors' judgments for each methodological quality item for each included study

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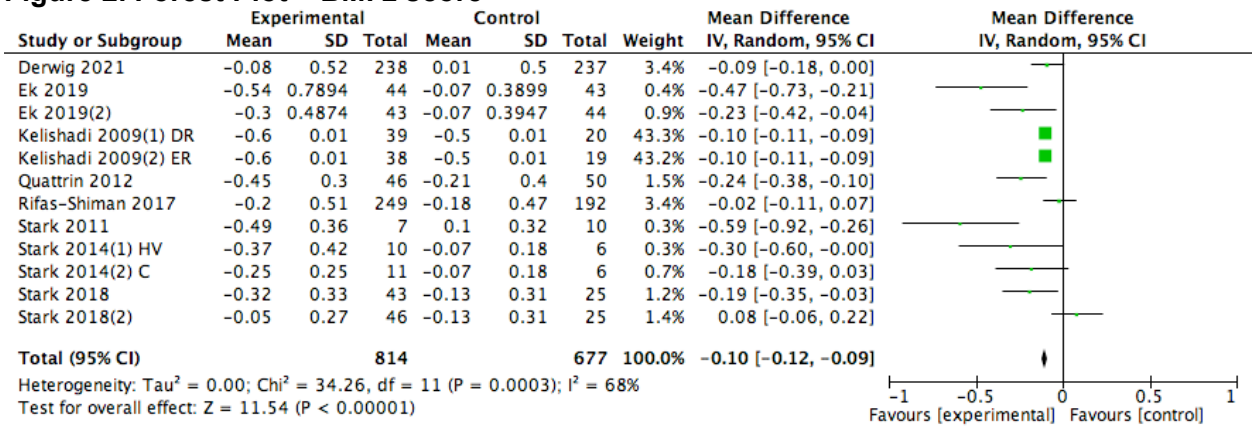
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Figure 1. PRISMA Flow Diagram



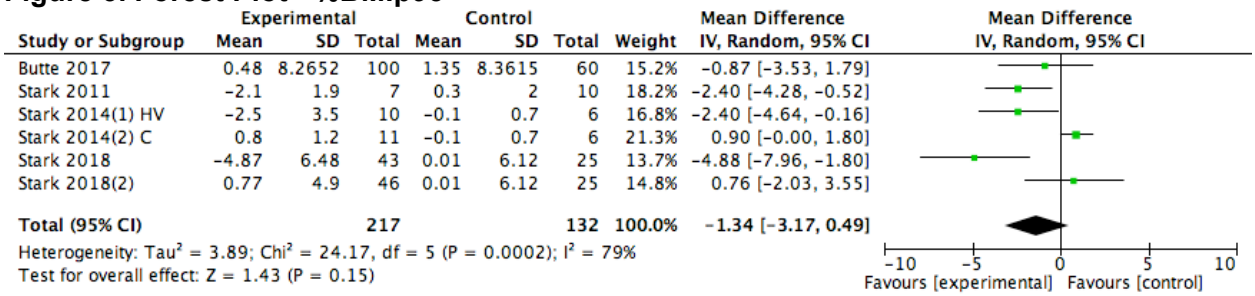
For peer review only

Figure 2. Forest Plot – BMI z score



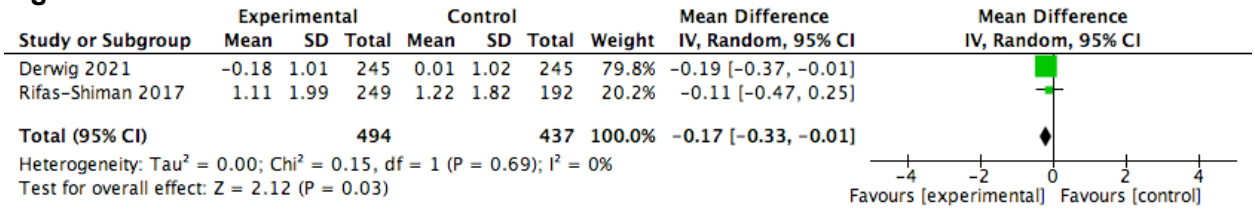
For peer review only

Figure 3. Forest Plot - %BMIp95



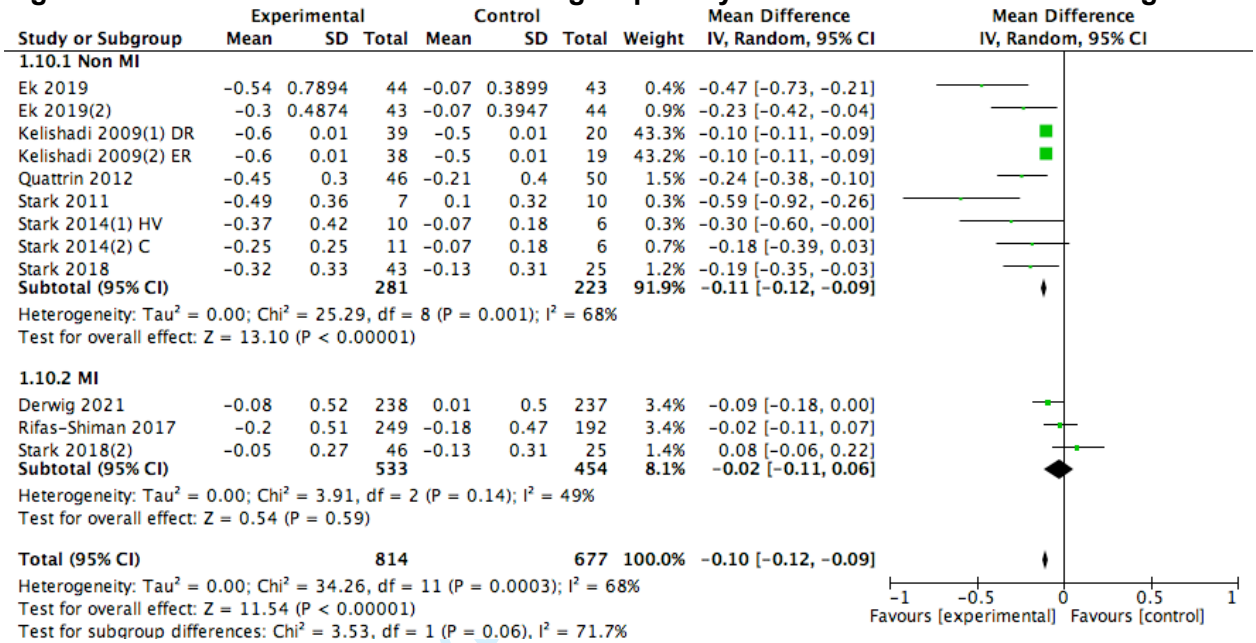
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Figure 4. Forest Plot – BMI



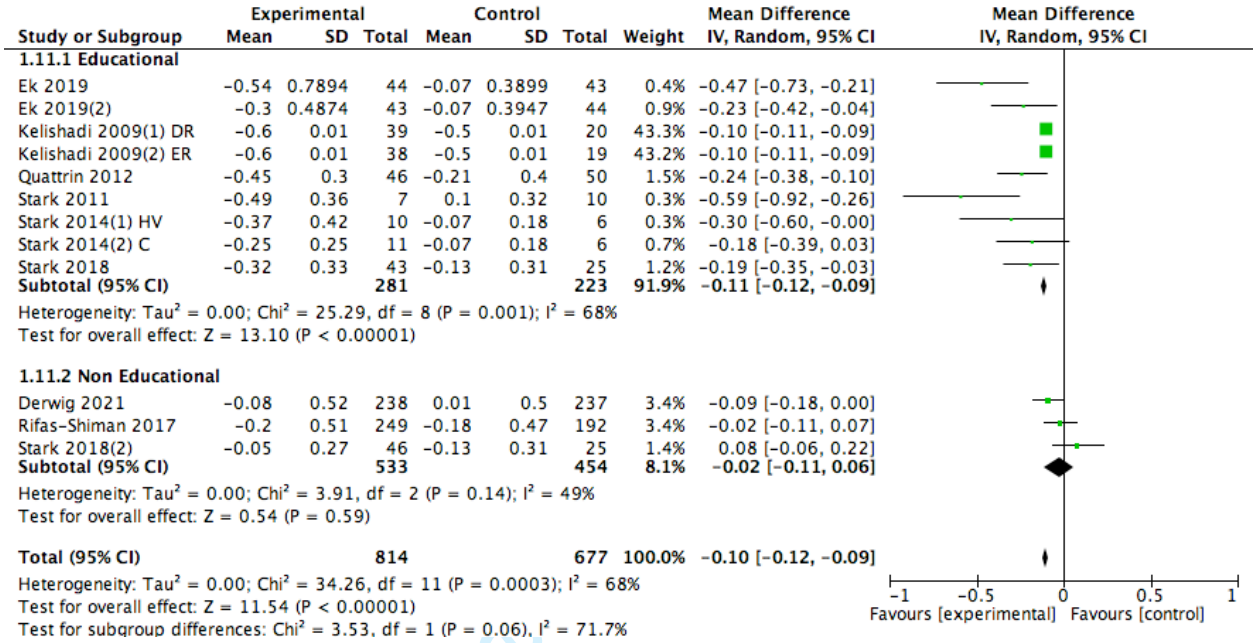
For peer review only

Figure 5. Forest Plot – BMI z score – subgroup analysis – Motivational Interviewing



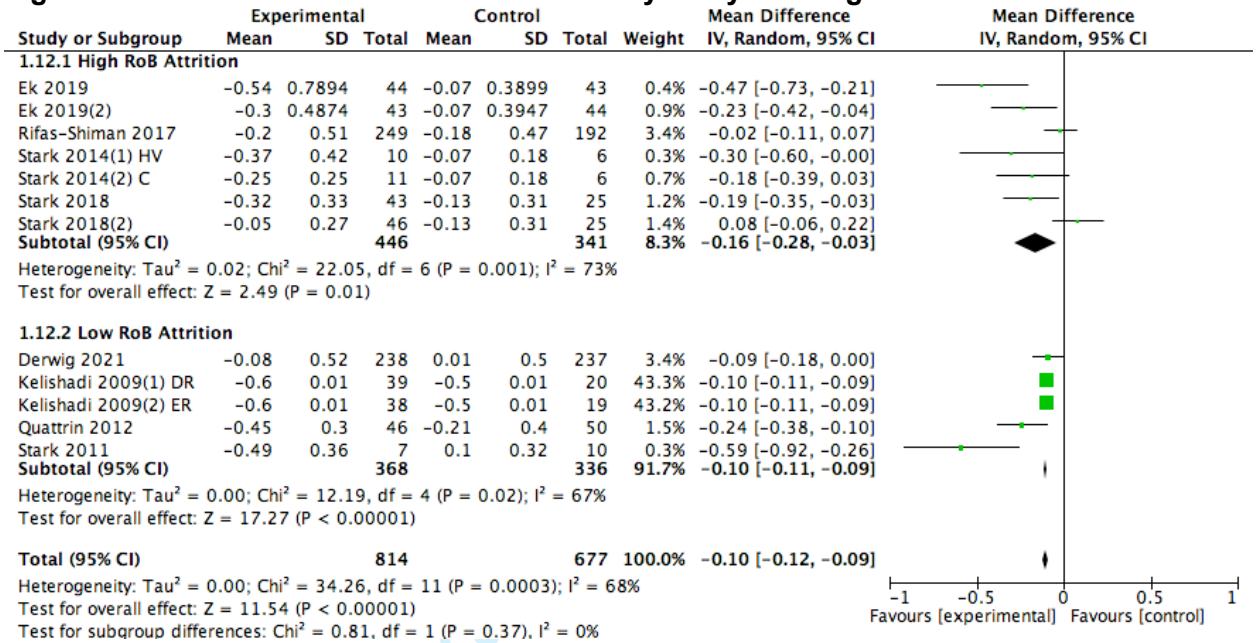
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Figure 6. Forest Plot – BMI z score – subgroup analysis – Multicomponent Educational Interventions



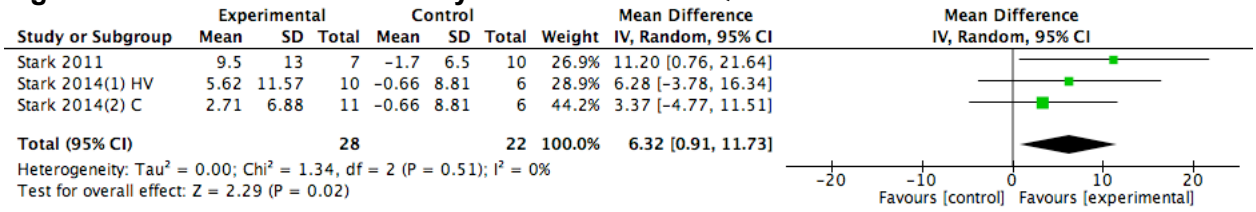
For peer review only

Figure 7. Forest Plot – BMI z score – sensitivity analysis – High versus low attrition bias



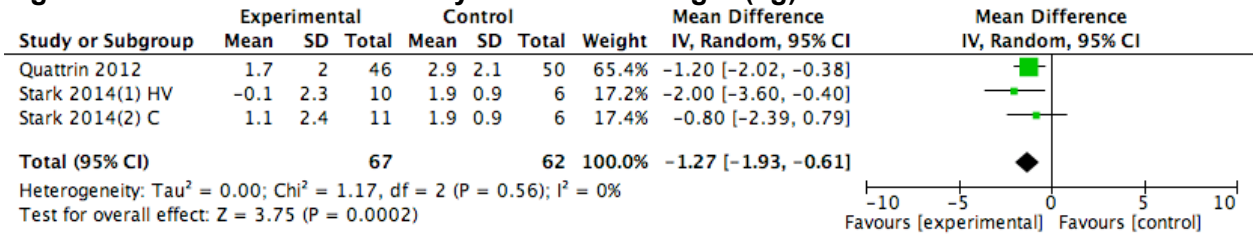
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Figure 8. Forest Plot – Secondary Outcome – PedsQL



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Figure 9. Forest Plot – Secondary Outcome – Weight (kg)



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Medline

Part I: Obesity

1. Obesity/
2. Obesity, Morbid/
3. Obesity, Abdominal/
4. Pediatric Obesity/
5. Overweight/
6. Weight Loss/
7. Body Mass Index/
8. (adipos* or obes*).tw.
9. (overweight* or over weight*).tw.
10. (weight adj1 (reduc* or los* or control* or manage*)).tw.
11. or/1-10

Part II: Intervention

12. exp Behavior Therapy/
13. exp Counseling/
14. Family Therapy/
15. Social Support/
16. Program Evaluation/
17. exp Exercise/
18. exp Exercise Therapy/
19. exp Physical Education/
20. exp Exercise Movement Techniques/
21. Motor Activity/
22. exp Diet/
23. exp Diet Therapy/
24. Patient Education as Topic/
25. Health Education/
26. exp Health Behavior/
27. exp Health Promotion/
28. Life style/
29. ((obesity adj3 intervention) or program or programme or camp?).tw
30. (lifestyle or life style).tw.
31. exercis*.tw.
32. (physic* adj (activ* or fit*)).tw.
33. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
34. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*).tw
35. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
36. ((group or family or cognit* or behav*) adj therap*).tw.
37. counsel?ing.tw.
38. educat*.tw.
39. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
40. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw

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41. or/12-40

Part III: Part I + Part II and additional MeSH/subheading combination

42. 11 and 41
43. Obesity/ or Obesity, Morbid/ or Overweight/ or Weight Loss/
44. diet therapy.fs. or prevention & control.fs. or rehabilitation.fs. or therapy.fs. or psychology.fs
45. 43 and 44
46. 42 or 45

Part IV: Population [adapted from Leclercq 2013]

47. exp Child/
48. Pediatrics/
49. minors.tw.
50. (boy*).tw.
51. girl*.tw.
52. toddler?.tw
53. (kid or kids).tw.
54. (child or childs or children* or childhood* or childcar*)tw
55. (preschool* or pre-school* or toddler* or kindergar* or nursery school*).tw
56. juvenil*.tw.
57. youth*.tw.
58. (underage* or under age*).tw.
59. p?ediatric*.tw.
60. or/47-59

Part V: Part III AND IV and additional MeSH/subheading combination

61. 46 and 60
62. Pediatric Obesity/
63. diet therapy.fs. or prevention & control.fs. or rehabilitation.fs. or therapy.fs. or psychology.fs
64. 62 and 63
65. 61 or 64

Part VI: Study filter [Cochrane Handbook 2008 RCT filter - sensitivity and precision maximizing version]

66. randomized controlled trial.pt.
67. controlled clinical trial.pt.
68. randomi?ed.ab.
69. placebo.ab.
70. clinical trials as topic/
71. randomly.ab.
72. trial.ti.
73. or/66-72
74. exp animals/ not humans/
75. 73 not 74

Part VII: Part V + Part VI

76. 65 and 75

For peer review only

Embase

Part I: Obesity

1. obesity/
2. morbid obesity/
3. abdominal obesity/
4. childhood obesity/
5. weight reduction/
6. weight control/
7. body mass index/
8. (adipos* or obes*).tw.
9. (overweight* or over weight*).tw.
10. (weight adj1 (reduc* or los* or control* or manage*)).tw.
11. or/1-10

Part II: Intervention

12. behavior therapy/
13. cognitive therapy/
14. exp counseling/
15. family therapy/
16. social support/
17. exp program evaluation/
18. exp exercise/
19. exp physical education/
20. exp physical activity/
21. exp motor activity/
22. training/
23. exp diet/
24. exp diet therapy/
25. nutritional health/
26. child nutrition/
27. feeding behavior/
28. patient education/
29. health promotion/
30. health literacy/
31. nutrition education/
32. health education/
33. lifestyle/
34. lifestyle modification/
35. ((obesity adj3 intervention) or program or programme or camp?).tw
36. (lifestyle or life style).tw.
37. exercis*.tw.
38. (physic* adj (activ* or fit*)).tw.
39. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
40. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*).tw
41. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
42. ((group or family or cognit* or behav*) adj therap*).tw.

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43. counsel?ing.tw.
44. educat*.tw.
45. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
46. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw
47. or/12-45

Part III: Part I + Part II and additional MeSH/subheading combination

48. 11 and 47
49. obesity/ or morbid obesity/
50. pc.fs or rh.fs or th.fs. [prevention.fs. or rehabilitation.fs. or therapy.fs.]
51. 49 and 50
52. 48 or 51

Part IV: Population [adapted from Leclercq 2013]

53. juvenile/
54. child/
55. toddler/
56. preschool child/
57. pediatrics/
58. minors.tw.
59. boy*.tw.
60. girl*.tw.
61. toddler?.tw.
62. (kid or kids).tw.
63. (child or childs or children* or childhood*).tw
64. juvenil*.tw.
65. youth*.tw.
66. (underage* or under age*).tw.
67. p?ediatric*.tw.
68. or/53-67

Part V: Part III AND IV and additional MeSH/subheading combination

69. 52 and 68
70. childhood obesity/
71. pc.fs or rh.fs or th.fs. [prevention.fs. or rehabilitation.fs. or therapy.fs.]
72. 70 and 71
73. 69 or 72

Part VI: Search Filters (Cochrane Handbook)

74. 'crossover procedure':de
75. 'double-blind procedure':de
76. 'randomized controlled trial':de
77. 'single-blind procedure':de
78. (random* OR factorial* OR crossover*
79. cross NEXT/1 over*
80. placebo*
81. doubl* NEAR/1 blind*
82. singl* NEAR/1 blind*
83. assign*

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84. allocat*
85. (volunteer*):de,ab,ti
86. or/74-85

Part VII: Part V + Part VI

87. 73 and 86

PsychINFO

Part I: Obesity

1. exp Overweight
2. exp Body Mass Index/
3. (adipos* or obes*).tw.
4. (overweight* or over weight*).tw.
5. or/1-4

Part II: Intervention

6. Weight Control/
7. Weight Loss/
8. Aerobic Exercise/
9. Diets/
10. exp Exercise/
11. Movement Therapy/
12. Dance Therapy/
13. exp Physical Activity/
14. Physical Fitness/
15. Health Behavior/
16. Health Promotion/
17. Health Knowledge/
18. Health Literacy/
19. Health Education/
20. Client Education/
21. Lifestyle/
22. Physical Education/
23. exp Program Evaluation/
24. Educational Programs/
25. Educational Therapy/
26. exp Program Development/
27. Counseling/
28. Group Counseling/
29. Family Therapy/
30. Support Groups/
31. Social Support/
32. exp Behavior Modification/
33. Cognitive Behavior Therapy/
34. Cognitive Therapy/
35. ((obesity adj3 intervention) or program or programme or camp?).tw

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36. (lifestyle or life style).tw.
37. exercis*.tw.
38. (physic* adj (activ* or fit*)).tw.
39. (walk* or jog* or swim* or weight lift* or danc* or aerobics).tw
40. ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) adj3 train*)
41. (behavio?ral or behavio?r modification or psychoth* or psychosocial).tw
42. ((group or family or cognit* or behav*) adj therap*).tw.
43. counsel?ing.tw.
44. educat*.tw.
45. ((parent? or family) adj (based or focused or directed or centered or only or led)).tw
46. (diet* or healthy nutrition or (nutrition* adj (knowledge or educat* or therap* or program* or intervention*))).tw
47. or/6-46

Part III: Part I + Part II

48. 5 and 47

Part IV: Population [adapted from Leclercq 2013]

49. minors.tw.
50. (boys*).tw.
51. girl*.tw.
52. toddler?.tw.
53. (kid or kids).tw.
54. (child or childs or children* or childhood*)tw
55. juvenil*.tw.
56. youth*.tw.
57. (underage* or under age*).tw.
58. p?ediatric*.tw.
59. or/49-58

Part V: Part III AND IV and additional MeSH/subheading combination

60. 48 and 59

Part VI: Study filter [Eady 2008filter - BS version]

61. control*.tw. OR random*.tw. OR exp Treatment/

Part VII: Part V + Part VI

62. 60 and 61

CINAHL

Part I: Obesity

1. MH ”Obesity+“
2. MH "Body Mass Index"
3. TX (adipos* or obes*)
4. TX (overweight* or ”over weight*“)
5. S1 OR S2 OR S3 OR S4

Part II: Intervention

For peer review only

6. MH ”Weight Loss“
7. MH ”Behavior Modification+“
8. MH ”Counseling“
9. MH ”Family Therapy“
10. MH ”Support, Psychosocial“
11. MH ”Support Groups“
12. MH ”Program Evaluation“
13. MH ”Program Implementation“
14. MH ”Exercise+“
15. MH ”Sports+“
16. MH ”Therapeutic Exercise+“
17. MH ”Physical Fitness“
18. MH ”Physical Education and Training+“
19. MH ”Health Education+“
20. MH ”Diet+“
21. MH ”Diet Therapy+“
22. MH ”Health Behavior “
23. MH ”Eating Behavior“
24. MH ”Health Promotion“
25. MH ”Life style changes“
26. MH ”Life style“
27. TX (weight N1 (reduc* or los* or control* or manage*))
28. TX ((obesity N3 intervention) OR program OR programme OR camp#)
29. TX (lifestyle or ”life style“)
30. TX exercis*
31. TX (physic* N1 (activ* or fit*))
32. TX (walk* or jog* or swim* or weight lift* or danc* or aerobics)
33. TX ((physic* or strength* or resist* or circuit or weight or aerob* or cross or endurance or structur*) N3 train*)
34. TX (behavio#ral or behavio#r modification or psychoth* or psychosocial)
35. TX ((group or family or cognit* or behav*) N1 therap*)
36. TX counsel#ing
37. TX educat*
38. TX ((parent# or family) N1 (based or focused or directed or centered or only or led))
39. TX (diet* or ”healthy nutrition“ or (nutrition* N1 (knowledge or educat* or therap* or program* or intervention*)))
40. S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR 37 OR 38

Part III: Part I + Part II and additional MeSH/subheading combination

41. S5 AND S40
42. MH ”Obesity+/DH/ED/PC/PF/RH/TH“) [diet therapy or education or prevention & control or psychosocial factors or rehabilitation or therapy]
43. S40 OR S41

Part IV: Population [based on Leclercq 2013]

For peer review only

44. MH ”Child+“
45. MH ”Pediatrics“
46. TX minors
47. TX (boy*)
48. TX girl*
49. TX toddler#
50. TX (kid OR kids)
51. TX (child OR childs OR children* OR childhood*)
52. TX juvenil*
53. TX youth*
54. TX (underage* or under age*)
55. TX (paediatric* OR pediatric*)
56. S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54 OR S55

Part V: Part III AND IV and additional MeSH/subheading combination

57. S43 AND S56
58. MH ”Pediatric Obesity/DH/ED/PC/PF/RH/TH“) [diet therapy or education or prevention & control or psychosocial factors or rehabilitation or therapy]
59. S57 OR S58

Part VI: Study filter [Wong 2006bfilter - SDSSGS version]

60. MH ”treatment outcomes+“ OR MH ”experimental studies+“ or random*

Part VII: Part V + Part VI

61. S58AND S59

Cochrane Library

Part I: Obesity

1. Obesity.mp
2. Obesity, morbid.mp
3. Obesity, abdominal.mp
4. Pediatric obesity.mp
5. Overweight.mp
6. Weight loss.mp
7. Body mass index.mp
8. (adipos$ or obes$).ti, ab
9. (overweight$ or (over ADJ1 weight$)).ti,ab
10. (weight ADJ1 (reduc$ or los$ or control$ or manage$)).ti,ab
11. {or #1-#10}

Part II: Intervention

12. Behavior therapy.mp
13. Counseling.mp

[mh ”Behavior Therapy“]

13. [mh ”Counseling“]
14. [mh ^”Family Therapy“]
15. [mh ^”Social Support“]

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16. [mh ^"Program Evaluation"]
17. [mh "Exercise"]
18. [mh "Exercise Therapy"]
19. [mh "Physical Education and Training"]
20. [mh "Exercise Movement Techniques"]
21. [mh ^"Motor Activity"]
22. [mh Diet]
23. [mh "Diet Therapy"]
24. [mh ^"Patient Education"]
25. [mh ^"Health Education"]
26. [mh "Health Behavior"]
27. [mh "Health Promotion"]
28. [mh ^"Life style"]
29. (("obesity" near/4 "intervention") or "program" or "programme" or "camp" or "camps"):ti,ab
31. exercis*:ti,ab
32. (physic* next (activ* or fit*)):ti,ab
33. (walk* or jog* or swim* or ("weight" next lift*) or danc* or "aerobics"):ti,ab
34. ((physic* or strength* or resist* or "circuit" or "weight" or aerob* or "cross" or "endurance" or structur*) near/4 train*):ti,ab
35. ("behavioral" or "behavioural" or (("behavior" or "behaviour") next "modification") or psychoth* or "psychosocial"):ti,ab
36. (("group" or "family" or cognit* or behav*) next therap*):ti,ab
37. (counseling or counselling):ti,ab
38. educat*:ti,ab
39. (("parent" or "parents" or "family") next ("based" or "focused" or "directed" or "centered" or "only" or "led")):ti,ab
40. diet* or "healthy nutrition" or (nutrition* next ("knowledge" or educat* or therap* or program* or intervention*))):ti,ab
41. {or #12-#40}

Part III: Part I + Part II and additional MeSH/subheading combination

42. #11 and #40
43. [mh Ôbesity] or [mh ^"Obesity, Morbid"] or [mh Ôverweight]
44. [mh /DH,PC,RH,TH,PX][diet therapy or prevention & control or rehabilitation or therapy or psychology]
45. #43 and #44
46. #42 or #45

Part IV: Population [adapted from Leclercq 2013]

47. [mh Child]
48. [mh ^Pediatrics]
49. "minors":ti,ab
50. "boys*":ti,ab
51. girl*:ti,ab
52. ("kid" or "kids"):ti,ab
53. ("toddler" or "toddlers"):ti,ab
54. ("child" or "childs" or children* or childhood*):ti,ab

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55. juvenil*:ti,ab
56. youth*:ti,ab
57. (underage* or (”under“ next age*)):ti,ab
58. (paediatric* or pediatric*):ti,ab
59. {or #47-#58}

Part V: Part III AND IV and additional MeSH/subheading combination

60. #46 and #59
61. [mh ^”Pediatric Obesity“]
62. [mh /DH,PC,RH,TH,PX]
63. #61 and #62
64. #60 or #63

Web of Science:

Obesity

TS =(obesity)
TS=(obesity, morbid)
TS=(obesity, abdominal)
TS=(pediatric obesity)
TS=(overweight)
TS=(weight loss)
TS=(body mass index)
#7 OR #6 OR #5 OR #4 OR #3 OR #2 OR #1

Intervention:

TS=(behavior therapy)
TS=(counseling)
TS=(family therapy)
TS=(social support)
TS=(program evaluation)
TS=(exercise)
TS=(exercise therapy)
TS=(physical education and training)
TS=(exercise movement techniques)
TS=(motor activity)
TS=(diet)
TS=(diet therapy)
TS=(patient education)
TS=(Health Education)
TS=(Health Behavior)
TS=(Health Promotion)
TS=(Life style)
#9 OR #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21or #22 or #23 or #24 or #25

Part III: Part I + Part II

1. #8 AND #26

Population

TS=(Child*)

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TS=(Pediatric*)
TS=(Preschool child*)
TS=(Toddler*)
TS=(Minor*)
TS=(Youth*)
TS=(Kid*)
#28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34

Study filters

TS= (clinical trial*)
TS= (research design)
TS= (controlled trial*)
TS= (single blind*)
TS= (double blind*)
TS= (follow-up stud*)

Clinicaltrials.gov

Conditions: obese OR overweight OR obesity
Study Type: Interventional Studies
Age Group: Child (birth-17)

WHO Clinical Trial Registry

Activated "search for clinical trials in children"
Title: obes* OR overweight*
OR
Condition: obes* OR overweight*
Recruitment: All

For peer review only

Article Title	Efficacy of a community- versus primary care-centered program for childhood obesity: TX CORD RCT	
Authors	Butte, N., Hoelscher, D., Barlow, S., Pont, S., Durand, C., Vandewater, E., Liu, Y., Adolph, A., Pérez, A., Wilson, T., Gonzalez, A., Puyau, M., Sharma, S., Byrd-Willams, C., Oluyomi, A., Huang, T., Finkelstein, E., Sacher, P. & Kelder, S.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	TX CORD: Texas Childhood Obesity Demonstration project **Intervention:** MEND2-5 (Mind, Exercise, Nutrition…Do it!) program **Comparison**: Next Steps program	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	The TX CORD project was designed to address childhood obesity through an integrated, systems-oriented model that incorporated primary and secondary obesity interventions. Within this population-level, systems-based intervention was a 12-month randomized controlled trial targeting low-income, ethnically diverse children with overweight or obesity and their families. It is hypothesized that the 12-month, community-centered program would significantly reduce BMI compared to the health care-centered program.	1585 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Intervention :** Conducted at the YMCA within 5 miles of the participants. Education materials surrounding general nutrition, PA and behaviours were needed (not included in paper/protocol). Based on CATCH program. All programs were delivered in English or Spanish. A digital scale and stadiometer for weight and height measurements. Transition program includes the book *A Year of Being Well* (role models stories of low-income families using CDC obesity prevention behaviours), cooking classes	74-76 (2)

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	based on *The Happy Kitchen/La Coicina Alegre* and MED World online/print material. **Control:** Electronic medical record at the primary care physician's office. Counselling materials for the primary care provider. Activity booklet for parents and children. Material was available in both English and Spanish, and available through the AAP website (https://shop.aap.org/next-steps-a-practitioners-guide-for-themed-follow-up-visits-for-their-patients-to-achieve-a-heal/).	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention:** A multicomponent, healthy lifestyle program including behavioural, nutrition and physical activity sessions. Included 10 weekly 90-minute sessions relating to general nutrition, physical activity and behaviours for children including: - 30 minutes of guided active play - 15 minutes of healthy snack time evidence-based, exposure-based technique to promote acceptance and increased intake of fruit and vegetables - 45 minutes where children participate in creative play activities while parents attend an interactive education and skill development session based on group-based parent training principles. The transition program includes monthly 90-minutes post-program sessions that children and parents attend together. Sessions include a brief review of the MEND concepts, role model stories and cooking lessons. **Control:** A clinic-based program that optimizes health care procedures to assess BMI and provides brief counselling materials to support family/child self-improvements in healthy lifestyles. Electronic health record (BMI screening, alerts for BMI ≥ 85th centile, specific diagnostic codes, lab tests and referrals common in obesity management) changes to support childhood obesity visits. Counselling materials were provided to the primary care providers. Next Steps activity booklets for parents and children to work on nutrition and physical activity targets in a self-directed manner. Families were encouraged to seek follow up clinic visits to address	74-75 (2) 75 (2), 1586 (1)

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	childhood obesity if identified as being obese. The brief counselling materials include information on facilitating clinical intervention and supporting educational materials for families.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention**: All sessions were delivered by bilingual allied health professionals (dieticians and health educations) and were assisted by community health workers. Community health workers were trained in MEND, motivational interviewing, nutrition and physical activity programming. **Control:** Program was delivered by primary care physicians in their offices, and self-directed learning by the families/children. No specific training, however counselling material was provided to the primary care physicians Trained research staff measured weight and height (?unclear training)	75 (2) 1586 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Intervention:** The intervention was delivered at the YMCA face-to-face. ? It was not stated whether this was individual sessions or group sessions. Measurements were performed at the Children's Nutrition Research Centre in Houston or YMCAs/offices in Austin. **Control:** Delivered face-to-face in primary care physician's office, as well as workbooks provided to complete individually at home.	1585 (1) 1585 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Intervention:** conducted at YMCAs in Austin and Houston. Measurements were performed at the Children's Nutritional Research Centre in Houston or YMCAs/offices in Austin. ?No information regarding how participants accessed this **Control**: primary care clinics in Texas	1585 (1)

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When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Intervention:** The intensive phase consisted of nine weekly 90-minute sessions. The transition phase offered 90-minutes session including MEND reviews (nine months in total). In total, MEND2-5 offered 27 hours of contact. **Intervention:** A clinic-based program – families were provided with self-directed education material as well as encouraged to seek follow-up clinic visits to address childhood obesity. An estimated 8-hours of contact was provided.	1585 (1), 75 (2) 1585-1586 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	"extensively adapted for the US population" (from British study) - ?did not report how it was adapted "pilot tested program elements and used qualitative data to further tailor the program to our population" TX CORD culturally adapted Next Steps material for low-income, urban, black and Latino children in both English and Spanish versions.	1585 (1) 76 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Fidelity to the session content and participant engagement was assessed using instructor checklists. Compliance in the intervention arm was based on attendance (the number of sessions attended out of the number offered). Retention was based on completion of measurements at 3 and 12 months.	1587 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	Retention rates were 64% for the intervention group compared to 87% for the comparison at 3 months. The retention rate at 12 months was 54% for the intervention group compared to 82% for the comparison group.	1587-1588 (1)

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	During the Intensive Phase, 86% of the sessions were delivered completely as planned. During the Transition Phase, 75% of the sessions were delivered completely as planned. Mean attendance was 46% during the Intensive Phase and 16% during the Transition Phase. Fidelity was not assessed for the comparison group.	1590 (1)

** **Authors -** use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

(1) Butte, NF, Hoelscher, DM, Barlow, S, et al. Efficacy of a community- versus primary care-centered program for childhood obesity: TX CORD RCT. Pediatric Obesity 2017; 25:1584-1593

(2) Hoelscher DM, Butte NF, Barlow S, et al. Incorporating primary and secondary prevention approaches to address childhood obesity prevention and treatment in a low-income, ethnically diverse population: study design and demographic data from the Texas Childhood Obesity Research Demonstration (TX CORD) study. Child Obes 2015;11:71-91.

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Article Title	A parent treatment program for preschoolers with obesity: A randomized controlled trial	
Authors	Ek, A., Chamberlain, K., Sorjonen, K., Hammar, U., Malek, Mahnoush., Sandvik, P., Somaraki, M., Nyman, J., Lindberg, L., Nordin, K., Ejderhamn, J., Fisher, P., Chamberlain, P., Marcus, C., & Nowicka, P.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	More or Less Study: 1) Parent-only group treatment **with** follow-up booster sessions 2) Parent only group treatment **without** booster sessions 3) Standard treatment	2 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Given the significant role of parents in a young child's life, the treatment program created was grounded in evidence-based parenting practices to evaluate whether a focus on parenting would be especially suitable for this age group as suggested by previous papers. It is hypothesized that the parent-only treatment would be more effective compared to standard treatment.	2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Intervention** - Based on the KEEP (Keeping Foster and Kin Parents Supported and Trained) program – a positive parenting program. The More or Less program was developed in collaboration with the KEEP program developers with conceptual and cultural adaptations designed for Swedish population. Involved home assignments (not provided as part of paper/protocol) In collaboration with the KEEP developers, the authors translated and adapted the parent group manual outlining the content of each session and parent handouts (not included in the protocol/paper).	2 (1) 6 (2)

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	Control -receive treatment as usual in an outpatient pediatric unit. A questionnaire was developed to monitor the healthcare providers involved (as it varied), and the number and focus of the visits. The questionnaire was not included as part of the protocol or paper.	6 (2)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention -** ML program consisted of 10 weekly 90-minute sessions with parents only, with childcare provided. Each session centered around a parenting practice as well as a lifestyle component aimed to provide parents with knowledge/skills to support a healthy home environment. The session was finished with a home practice assignment related to the focus of the day. If parents missed a session, material was sent home and key parts were reviewed over the phone by 1 of the group leaders. **More or Less with Booster** – Group leaders called participants every 4-6 weeks for a 30-minute phone call (max 7 calls). The parents were referred back to the discussions had during the program and to the manual for additional support **Control** – based on the action plan for childhood obesity in Stockholm County (not described in the paper). Received at least 4 treatment visits, typically the first with the physician, and subsequent with the pediatric nurse. The visits centered around lifestyle advice. Reasons for treatment are discussed including weight gain and future risk factors. Treatments varied between pediatric units – a questionnaire was included regarding questions about the professions of the involved healthcare providers, number and focus of visits.	3 (1) 6-8 (2) 2(1) 6 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention** – Led by 5 registered dieticians – they received training by the KEEP developers, who also provided weekly supervision on videotaped sessions (translated from Swedish to English) for program certification and fidelity. The initial training was a 5-day experiential training including information on program's theory and practice in the delivery of group sessions. Each session of the ML program was thoroughly discussed, and each trainee role played facilitating several key sessions.	3 (1) 7 (2)

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	The training continued through external supervision from the KEEP developer group. **Control** – first and last visit were with a pediatrician, and the remaining visits were with a pediatric nurse (minimum of 4 visits). Some children were referred to dieticians and/or PTs.	3 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**More or Less without Boosters**– face-to-face group sessions **More or Less with booster** – face-to-face group sessions followed by seven 30-minute phone calls every 4-6 weeks (9 months in total) Neither the group leaders or parents were informed about which families were allocated to the boosters until after the group sessions were finished in order to avoid differential treatment. **Control** – delivered in person during routine visits to pediatrician	3 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Intervention -** Meetings were held in health care facilities. Both parents were invited to participate and childcare was provided. **Control** – Delivered at 14 outpatient pediatric clinics	3 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**No Booster** – 10 weekly 90-minute sessions **Booster** – 10 weekly 90-minute sessions with 30-minute phone calls (max 7) every 4-6 weeks for the following 9 months. **Control** – Were offered ≥ 4 individual visits of around 30 minutes over 12 months. Treatment setup varied between clinics – each clinic completed a protocol at every visit documenting details (*not available in paper*)	3 (1) 6 (2)
Tailoring		

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If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention** – The parent group manual outlining each session and parent handouts was translated and adapted from the original KEEP developers' version. **Control** – treatment set up varied between practitioners and location. Some children were referred to dieticians and/or PTs. Number of appointments with pediatrician/pediatric nurse also varied.	6 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	A 6-month follow-up was added post baseline to better evaluate the treatment program	2 (1)
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Intervention** - Training of the ML group leaders was conducted by the KEEP developers. 5-day experiential training session. The training continued through external supervision from the KEEP group – all parent groups were videotaped (and translated to English). KEEP consultants viewed the recordings, rated them for fidelity and identified areas for reinforcement and feedback Group leaders completed a session review with questions regarding what went well and challenges experienced. Also completed weekly forms on parent attendance and engagement ratings – these measures informed the consultation process **Control -** Because the treatments vary between pediatric units, a questionnaire including questions about the profession of healthcare providers involved as well as the number and focus of visits	7 (2) 6 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**Intervention**– Not reported in the final paper. **Control** – Each clinic filled out a protocol at every visit documenting what treatment the family had attended and on how many occasions.	3 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

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(1) Ek A, Lewis Chamberlain K, Sorjonen K, et al. A Parent Treatment Program for Preschoolers With Obesity: A Randomized Controlled Trial. Pediatrics. 2019;144(2): e20183457

(2) Ek A., Lewis Chamberlain K., Ejderhamn J., et al. The more or less study: A randomized controlled trial testing different approaches to treat obesity in preschoolers. Public Health. 2015. 15:735: DOI 10.1186/s12889-015-1912-1

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Article Title	Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study (1)	
Authors	Sheryl L. Rifas-Shiman, Elsie M. Taveras, Steven L. Gortmaker, Katherine H. Hohman, Christine M. Horan, Ken P. Kleinman, Kathleen Mitchell, Sarah Price, Lisa A. Prosser and Matthew W. Gillman	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	High Five for Kids primary care-based pre-schooler obesity intervention	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Targeting preschool-age children may be beneficial because children of this age visit their primary care clinicians frequently for routine medical care and parents have more control over their children's health-related behaviours, and these behaviours may be more malleable. The purpose of the study was to assess the effectiveness of the primary care-based obesity intervention over a 2-year period. The overarching model for this intervention was the Chronic Care Model which posits that changes in primary care to produce functional patient outcomes requires changes for all members of the practice team.	24 (2) 2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where	**Intervention:** Posters were developed for the waiting rooms that encouraged the five target behaviours of the High Five. An image of this poster is available at: https://pubmed.ncbi.nlm.nih.gov/21464376/. Educational modules were developed to target fast-food and sugar-sweetened beverage consumption, as well as television viewing (https://www.researchgate.net/publication/261773967_Health_Behavior_Change_A_Guide_For_Practitioners). A list of local resources for physical activity was also available but not provided in the study write-up, as was a website with recipes and educational materials. Small physical incentives (water bottles, snack containers, books) were provided by nurse practitioners to participants, as was an	2-3 (2)

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the materials can be accessed (e.g. online appendix, URL).	electronic tool to monitor television-viewing time (more information on this tool, including where to get it, was not provided in the study). The electronic medical record was enhanced to help physicians make decisions, track patients, follow-up, schedule and bill.	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Intervention:** Intervention was based on the Chronic Care Model (3), which requires change from all members of the health care team. Participants (including families) received motivational interviewing by nurse practitioners and educational modules targeting certain behavioural modifications. Specifically, the interviewing and modules targeted reducing sugar-sweetened beverage intake, reducing television viewing time, increasing fruit and vegetable consumption, decreasing fast-food consumption and encouraging more time in active play. **Usual care:** Participants got the standard of care offered by their pediatric practice. This included the baseline and annual well-child care visits as well as follow ups with the paediatrician or subspecialist (ex. Dietician).	2 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Intervention:** Pediatric nurse practitioners were trained for their role as the central intervening clinician. They also received training in motivational interviewing. This training included enhancing motivation for change, enhancing self-efficacy, how to help recognize inconsistencies between desired and actual behaviours, and how to help reduce this dissonance. Medical assistants were trained on how to conduct research-standard anthropometric measurements (for collecting height and weight data). Medical receptionists were trained to schedule the initial and follow-up visits with the nurse practitioners in accordance with the study protocol. Primary care pediatricians were trained in obesity management and to use brief, focused negotiation skills. **Usual care:** Participants got the standard of care offered by their pediatric practice. This included the baseline and annual well-child care visits as well as follow-ups with the pediatrician or subspecialist (ex. Dietician).	2 (2)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone)	**Intervention:** In the first year, four visits were face-to-face and two were by telephone. In the second year, both visits were face-to-face. Visits were with individual families. **Usual care:** At the discretion of the participants regular pediatrician.	2 (2)

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of the intervention and whether it was provided individually or in a group.		
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Done at 10 pediatric primary care practices within the Harvard Vanguard Medical Associates, which is a multi-site group practice in Massachusetts. Half of the practices provided usual care and half provided the intervention care.	2 (2)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Intervention:** Four in person (each 25 minutes) visits with the clinician over the first year, as well as two phone calls (15 minutes each). Two in-person visits in the maintenance year (length not described). **Usual care:** Frequency of usual care visits was determined by the routine care procedures for each particular pediatric practice.	2 (2)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention:** During their visits, the families could choose to work on one or more of the behavioural targets (previously listed).	4 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	
How Well		

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Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Intervention:** Adherence was assessed by completion of the various visits/phone calls, during data analysis. No strategies were used to improve fidelity.	5 (2), 3 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**Intervention:** In the first year (the intervention period), only 56% of participants in the intervention completed at least 2 of the 6 intervention activities (in person or phone visits). During the second year (the maintenance period), only 17% went to one visit and only 21% of participants went to both visits. Overall adherence was thus low. **Usual care:** adherence was not measured for usual care participants.	5 (2), 3 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Rifas-Shiman, S. L., Taveras, E. M., Gortmaker, S. L., Hohman, K. H., Horan, C. M., Kleinman, K. P., ... & Gillman, M. W. (2017). Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study. *Pediatric obesity*, *12*(3), e24-e27.

2. Taveras, E. M., Gortmaker, S. L., Hohman, K. H., Horan, C. M., Kleinman, K. P., Mitchell, K., ... & Gillman, M. W. (2011). Randomized controlled trial to improve primary care to prevent and manage childhood obesity: the High Five for Kids study. *Archives of pediatrics & adolescent medicine*, *165*(8), 714-722.

3. Wagner, E. H. (1998). Chronic disease management: what will it take to improve care for chronic illness?. *Effective clinical practice*, *1*(1).

Article Title	Clinic and Home-Based Behavioral Intervention for Obesity in Preschoolers: A	

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	Randomized Trial (1)	
Authors	Lori J. Stark, PhD, Stephanie Spear Filigno, PhD, Christopher Bolling, MD, Megan B. Ratcliff, PhD, MPH, Jessica C. Kichler, PhD, Shannon M. Robson, PhD, MPH, RD, Stacey L. Simon, PhD, Mary Beth McCullough, PhD, Lisa M. Clifford, PhD, Cathleen Odar Stough, PhD, Cynthia Zion, BS, and Richard F. Ittenbach, PhD	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Comparing LAUNCH (Learning about Activity and Understanding Nutrition for Child Health), motivational interviewing and standard care as in intervention for pre-school obesity	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Obesity in preschool years increases the risk of being overweight and obese in later childhood through to adulthood. Focusing on children with obesity is important as they are at higher risk for mortality compared with overweight peers. This study hypothesizes that an skills-based behavioural family clinic in combination with a home-based intervention would reduce BMIz score compared with motivational interviewing and standard care in preschool age children with obesity.	1,2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**LAUNCH**: Caregivers were given an iPod Touch so that they could use the "DietTracker" application to track the child's food and beverage intake. Pedometers were also provided to the children and their caregivers, to help caregivers monitor physical activity levels. Caregivers were also given a two-week supply of fresh vegetables at each of the clinic sessions to help expose children to new vegetables. **Motivational Interviewing:** Caregivers were given a pack of informational handouts from the Let's Go 5-2- 1-0 program (https://www.mainehealth.org/Lets-Go/Childrens-Program/Parents). A motivational interviewing (MI) manual was used	13, 16 (2), 135 (4)

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	for MI review by interviewers and study staff (3). **Standard Care:** N/A	
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**LAUNCH**: Sessions alternated between clinic and home-based sessions. During the clinic sessions, the caregiver(s) and child were in different groups. Caregiver groups received education and problem-solving skills on a number of diet and physical activity, as well as strategies to manage the behaviour of their child. A list of topics by session is available in Table 1 (2). In following the teachings of the sessions, families aimed to decrease the child's screen time, increase physical activity (for which they had daily step goals for both the parents and the child), stay within the prescribed daily calorie goals for the child as well as choose healthier foods, and to record child's food and beverage intake. The behavioural strategies that they learned to help achieve these goals included praising their child's healthy dietary and physical activity choices, modelling healthy behaviours, managing tantrums by ignoring and using time-outs, contingency management, as well as how to best set up the food environment to encourage healthy choices (ex. Not having unhealthy food available). More information about the teaching sessions and strategies learned is available here (4). The child group included education about healthy eating and opportunities for physical exercise. Sessions also included a light nutritious meal provided to caregivers and children, which served as exposure to fruits and vegetables for the children. During home visits, families got a chance to practice and receive feedback on the skills they had learned in clinic, as well as to problem solve any barriers they were facing in implementing those strategies in the home setting. **Motivational Interviewing:** MI was a caregiver-only intervention. During the first session, caregivers were asked about their concerns for their child's diet, weight, or physical activity habits and then given a chance to set a goal that was meaningful to	4-5 (1), 13-15 (2), 135 (4)

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	them. The subsequent sessions (either by phone or in person) involved a discussion of the previous goals, how their implementation was going, enhancing motivation, working on establishing the desired behaviours and choosing new goals to work towards. **Standard Care:** No intervention was provided outside of the routine pediatric care the participant was already receiving, other than to be notified of their weight.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**LAUNCH**: The LAUNCH interventionists were therapists trained on the protocol in pilot studies (more information on training was not provided). A new LAUNCH interventionist was brought on midway through the study. She had extensive previous training in pediatric behavioural weight management and was trained on the LAUNCH protocol by shadowing and co-leading sessions with one of the previous interventionists. Home visits were completed by postdoctoral fellows, who were trained by reviewing the LAUNCH manual (not provided), listening to audiotapes of home sessions, watching videotapes of child group sessions, administering role-played sessions, and shadowing a second year fellow during home visits for one cycle before being assigned their own visits. The group-sessions in the clinic were done by a licensed clinical psychologist. Fidelity coders had PhDs and expertise in obesity treatment. **Motivational Interviewing:** Interviewers were either pediatricians or PhD level psychologists. The first session was done by the pediatrician and the following sessions were done by the psychologists. Interviewers were selected for their previous MI training and experience; each interviewer had approximately 10 years of MI experience and had participated in previous structured MI training. **Standard Care:** Standard care was provided by the participant's regular pediatrician.	13, 16 (2)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or	**LAUNCH**: All sessions were done in-person. The clinic sessions were done in groups, with one group for the parents and one group for the children. Home-based sessions	13 (2)

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telephone) of the intervention and whether it was provided individually or in a group.	were done with individual families. **Motivational Interviewing:** There were 4 in-person sessions (sessions 2, 12, 16) and 14 sessions done over the phone. All sessions were done with individual caregivers. **Standard Care:** Participants in the standard care group received only a notification of their child's weight, provided by phone call and during the initial (in-person) assessment.	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**LAUNCH**: The home sessions were done in the participants' homes, which were all within 50 miles of the treatment centre. The clinic sessions were done at a medical centre (the treatment centre). **Motivational Interviewing:** The initial assessment interview was done at the medical centre, but subsequent sessions were done at the families' homes or over the telephone. **Standard Care:** At the discretion of the participants' regular pediatricians.	117 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**LAUNCH**: Intervention sessions alternated between clinic (10 sessions) and home sessions (8 sessions), for a total of 18 sessions. For the first three months there were weekly sessions, and for the following three months sessions were every second week. The clinic-based sessions were each 90 minutes and the home visits were 60 minutes. Sessions were scheduled for the late afternoons/early evenings to minimize scheduling conflicts/barriers. **Motivational Interviewing:** The 18 sessions were delivered in alignment with the LAUNCH session schedule. Phone sessions varied in length depending on caregiver preference, but were generally between 15-35 minutes long. Home visits were all scheduled for 60 minutes.	13, 15 (2)

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	Standard Care: Length and frequency of appointments was determined by the participant's normal pediatrician, in line with their practice policies.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**LAUNCH**: Home visits provided the opportunity to tailor the program for individual children/families. If there was a barrier to using a clinic-taught strategy in the home, the home visit provided an opportunity to problem-solve. It was also an opportunity for targeted feedback, since therapist was able to witness the taught behavioural strategies being used in vivo. Caregivers also received weekly written feedback on their child's food and beverage consumption, including feedback on their progress towards their goals. Feedback was also provided on activity levels, based on the number of steps recorded on the pedometers. **Motivational Interviewing:** Caregivers were asked to select a specific nutrition or physical activity goal from the Let's Go 5-2-1-0 (5 fruits or vegetables a day, limit recreational screen time to 2 hours or less, get 1 or more hours of physical activity a day, limit or eliminate sugary drinks) and AAP recommendations (https://pediatrics.aappublications.org/content/120/Supplement_4/S164). This was then used as the focus of the motivational interviewing sessions. **Standard Care:** Care was personalized as appropriate by the pediatrician providing care.	13 (2)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	**LAUNCH**: N/A **Motivational Interviewing:** N/A **Standard Care:** N/A	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	All LAUNCH and MI sessions were videotaped and/or audiotaped. **LAUNCH**: Fidelity coders reviewed the first session of the first cycle of LAUNCH and provided feedback to the interventionist. One quarter of all LAUNCH sessions (both clinic and home-based) were compared to a LAUNCH checklist by a PhD level	16 (2)

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	consultant with experience in weight modification treatment. If fidelity was found to drop below 90%, the therapist was provided with feedback on the components that needed improvement. **Motivational Interviewing:** For the first three cycles of interviewing, the two interviewers met together to review audiotapes of each other's MI interviews and provide feedback on MI fidelity. After the third interviewer joined the study, her audiotapes were reviewed and given regular feedback by the other two interviewers. MI fidelity coders also reviewed audiotapes regularly throughout the study and provided feedback to each interviewer on a quarterly basis. 25% of MI sessions were compared to a MI checklist (3) to assess fidelity. If fidelity was found to drop below 90%, the therapist was provided with feedback on the components that needed improvement. **Standard Care:** N/A	
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	**LAUNCH**: Fidelity was assessed as planned – see previous box. **Motivational Interviewing:** Fidelity was assessed as planned – see previous box. **Standard Care:** N/A	16 (2)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark, L. J., Filigno, S. S., Bolling, C., Ratcliff, M. B., Kichler, J. C., Robson, S. M., ... & Zion, C. (2018). Clinic and home-based behavioral intervention for obesity in preschoolers: a randomized trial. *The Journal of pediatrics*, *192*, 115-121.

2. Stark, L. J., Filigno, S. S., Bolling, C., Ratcliff, M. B., Kichler, J. C., Robson, S. L., ... & Zion, C. (2017). Learning about Activity and Understanding Nutrition for Child Health (LAUNCH): Rationale, design, and implementation of a randomized clinical trial of a family-based pediatric weight management program for preschoolers. *Contemporary clinical trials*, *52*, 10-19.

For peer review only

3. Moyers, T. B., Martin, T., Manuel, J. K., Miller, W. R., & Ernst, D. (2010). Revised global scales: Motivational interviewing treatment integrity 3.1. 1 (MITI 3.1. 1). *Unpublished manuscript, University of New Mexico, Albuquerque, NM*.

4. Stark, L. J., Spear, S., Boles, R., Kuhl, E., Ratcliff, M., Scharf, C., ... & Rausch, J. (2011). A pilot randomized controlled trial of a clinic and home-based behavioral intervention to decrease obesity in preschoolers. *Obesity*, *19*(1), 134-141.

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Article Title	A child-centered health dialogue for the prevention of obesity in child health services in Sweden – A randomized controlled trial including an economic evaluation	
Authors	Derwig, M., Tiberg, I., Björk, J., Welander Tärneberg, A., & Kristensson	

	Hallström, I.	
		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Child-Centered Health Dialogue (CCHD)	Pg. 2 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	The attention paid to dietary and physical activity behaviours in children is infrequent at the 4-years health visit in Sweden and raises concern about the efficacy of prevention efforts against child obesity. Nurses working in child health services lack guidance for the management of pre-school children with overweight, which results in a number of consequences that vary from unnecessary extra visits (continuous observation) to not identifying overweight at all. Need for continuous training in the use of structure person-centered family counselling empowering families when a child is identified with overweight. A structured model was developed for the prevention of obesity that can be implemented in routine practices in CHS.	Pg. 2 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Control:** Guided using the digital National Handbook for Child Health (*get reference)*. Illustrations were developed for the CCHD were published in the national handbook. A questionnaire to describe usual care was distributed at the start of the RCT. **Intervention:** Illustrations (*not provided*). BMI growth charts were also used in discussion with children and their parents.	Pg. 3 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Control:** Received the 4-years health visit including a health dialogue and identification of overweight using the IOTF definition. The nurses working in child health services are guided using the digital National Handbook of Child Health Services. Illustrations were published in the national handbook to help guide education – overview of the illustrations available (2). A questionnaire to describe the usual care was distributed before the start of the RCT. As child health services is	Pg. 3 (1), 387 (2)

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	not responsible for treatment of diseases, children with obesity are referred to a multidisciplinary team consisting of a specialized nurse, dietician and a pediatrician for treatment outside the primary care setting. **Intervention:** 4-year-old children received CCHD at the 4-years health visit. Children with overweight were offered the targeted part of CCHD and extra visits or a referral to an overweight team, dietician or CHS psychologist at the caregiver's request. Children with obesity were referred to specialized care (described under usual care). CCHD is a low-intense multicomponent child-centered intervention based on the Logic model (2). Builds on a family-therapeutic solution-focused approach while emphasizing a relationship based on trust and facilitation of the family to make healthy choices. Consists of 2 parts: Universal part directed at all 4-year-olds regardless of weight and their families, and a targeted part offered to families when the child is identified with overweight. *Universal:* 10-minute structured dialogue between the child, caregiver and nurse using 8 animated illustrations (2) based on healthy choices (fruit/veg consumption drinking water, portion sizes, physical activity, tooth brushing and bedtime routines). As well as a neutral discussion re child's growth using BMI growth charts. *Targeted:* 45-minute sessions, Family Guidance is offered 1-3 weeks after the universal health visit, offered to caregivers, the child and other adults important to the family. Inspired by the Standardized Obesity Family Therapy – an evidence-based treatment model that engages family members through collaborative family support in either identifying protective factors or finding sustainable changes in their lifestyle.	Pg. 3(1), 386 (2)
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Control:** Guided using the National Handbook for Child Health Services and illustrations developed for CCHD. However, they were not trained in the CCHD approach. Received a traditional 120-minute lecture on child overweight and the BMI growth chart with no possibility for discussion or reflection.	Pg. 3 (1)

For peer review only

	Intervention: Received an interactive 1-day training in small groups focused on how to promote child participation and how to apply a family-therapeutic solution focused approach. Main components of training available (3). Was followed by 4 tutorial sessions of 1 hour every 2 months to reflect on how to build trust with the family, how to contextualize and re-frame into positive reflections and how to handle the challenges encountered in the early communication about weight.	Pg 3 (1), 823 (3)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Control & Intervention:** Started at the 4-year health visit, assumed to be in-person.	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Control & Intervention:** Occurred at 37 Child Health Centres in a southern county in Sweden.	Pg. 2 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Control:** A one-time visit at the 4 year old mark **Intervention:** 2 visits. A 10-minuted structured dialogue for the universal part, and a 45-minute Family Guidance visit. *?any other visits*	Pg 3-4 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention:** The Family Guidance targeted session engaged family members in identifying protective factors or finding sustainable lifestyle changes.	Pg. 4 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		

For peer review only

Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Fidelity to the study protocol was monitored through the collection of data from the child's records on delivery of care for children with overweight at baseline and through a digital questionnaire containing 11 questionnaires distributed to all nurses at baseline.	Pg. 9 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	Throughout the RCT, 26 nurses stopped working in the Child Health Services: 11 nurses in the control group and 16 nurses in the intervention group. The intervention group recruited 4 new nurses trained in CCHD, while 5 nurses could not be trained, resulting in the withdrawal of 285 children in the intervention group that met a nurse not trained in CCHD, of which 31 children had overweight.	Pg. 9 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

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1. Derwig, M., Tiberg, I., Björk, J., Welander Tärneberg, A., & Kristensson Hallström, I. (2021). A child-entered health dialogue for the prevention of obesity in child health services in Sweden - A randomized controlled trial including an economic evaluation . Obesity Science and Practice, 1–14.
2. Derwig M, Tiberg I, Björk J, Hallström I. Child-Centred health dialogue for primary prevention of obesity in child health services – a feasibility study. Scand J Public Health. 2021;49(4):384-392. https:// doi.org/10.1177/1403494819891025
3. Castor C, Derwig M, Borg SJ, Ollhage ME, Tiberg I. A challenging balancing act to engage children and their families in a healthy lifestyle – nurses' experiences of child-centred health dialogue in child health services in Sweden. J Clin Nurs. 2021;30(5/6):819-829. https://doi.org/10.1111/jocn.15622

Article Title	Can a dairy-rich diet be effective in long-term weight control of young children	
Authors	Kelishadi R, Zemel MB, Hashemipour M, Hosseini M, Mohammadifard N, Poursafa P.	
		Page Number

		(Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	NR	
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	A longitudinal study on the effect of dairy intake in early childhood on the acquisition of body fat throughout childhood showed that suboptimal dairy intake during preschool was associated with greater gains in body fat throughout childhood. The objective of the current study was to determine the short- and long-term results of a randomized controlled trial of a dairy-rich diet on generalized and abdominal obesity, and the components of metabolic syndrome among obese pre-pubescent children.	Pg 602 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	All participants attended family-centered education sessions on healthy lifestyle (no materials provided)	Pg. 603 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**All Groups:** All participants attended 6 consecutive monthly family-centered education sessions about healthy lifestyle (healthy nutrition and increasing physical activity) conducted by a pediatrician and nutritionist. **Dairy Rich Group (DR):** A dairy rich diet of >800 mg of calcium per day with no change in energy or macronutrient intake was recommended to the children of this group. Were recommended to obtain most of their calcium from low-fat and regular milk, cheese and yogurt, as well as liquid and sold curt. **Energy-Restricted Group (ER)**: Participated in similar nutrition education session as	Pg. 603 (1)

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	the DR group. Were placed on a caloric restriction regimen with an energy content restricted to the caloric requirement for height. To ensure equal treatment intensity between the 2 intervention groups, the same team provided diet recommendations through educational sessions of the same number, duration and interval. **Control:** Received no dietary recommendations aside from what was discussed during the healthy lifestyle education sessions.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	The healthy lifestyle education sessions were provided by a pediatrician and nutritionist It is not clear who provided the DR and ER groups education, or whether this was done in conjunction with the healthy lifestyle education sessions.	Pg. 603 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	It is not stated whether the healthy lifestyle education sessions were provided as a group or individually. It is assumed that the groups were done in person, but not explicitly stated	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	The participants had been referred to the Obesity Research Clinic, Preventive Pediatric Cardiology Department, Isfahan Cardiovascular Research Centre (ICRC), a collaborating centre of the WHO in Isfahan, Iran.	Pg. 602 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their	All participants attended 6 consecutive monthly education sessions. Not stated duration of the sessions, and whether the intervention groups education sessions	Pg. 603 (1)

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duration, intensity or dose.	ran in conjunction with the general education sessions.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	None reported	
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	Intervention adherence was monitored by retention and attrition rates. No strategies regarding fidelity were discussed.	Pg. 604 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	120 children with obesity were randomized, and of these 95 participants completed the 36-month follow-up period (75%).	Pg 604 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Kelishadi R, Zemel MB, Hashemipour M, Hosseini M, Mohammadifard N, Poursafa P. Can a dairy-rich diet be effective in long-term weight control of young children?. J Am Coll Nutr 2009;28(5):601–10.

Article Title	Efficacy of family-based weight control program for preschool children in primary care	
Authors	Quattrin T, Roemmich JN, Paluch R, Yu J, Epstein LH, Ecker MA.	
		Page Number

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		(Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Buffalo Healthy Tots	Pg 684 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Goal was to test the efficacy of an innovative family-based intervention program designed for treating overweight/obese children (aged 2-5 years) and an overweight parent in the primary care setting. Hypothesized that children and parents randomized to a family-based behavioural intervention would have greater reductions in BMI percentile and BMI as the contributions of parental eating habits and obesity status is well linked to their child's weight status.	661 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**Control & Intervention** – standardized letter updating parents on the progress of their child, not provided **Intervention:** Icon diary used for documenting physical activity and sedentary behaviours. This was not provided	661 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**Information Control:** ten 60-minute sessions over 6 months in which a consistent group leader delivery dietary, physical and sedentary activities education. Children were engaged in active games. Parents received 8 phone calls between meetings by an assigned coach. Were given a child weight loss goal of 0.5 to 1lbs/week. Follow-up regarding progress was done with a standardized letter (*not provided)* at 3 months and at a well-child visit at 6 months. **Intervention:** Same as the information control. The group leader also emphasized parenting and behavioural strategies to promote child and parent-behaviour change. Parents also attended a 1:1 meeting at the time of group sessions with an assigned coach who assisted parents in shaping behaviour goals. Parents were instructed to record child and parent food intake, as well as physical and sedentary	661 (1)

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	behaviours using an icon-based diary (*not provided)*, this allowed for shaping of goals. Were asked to weight themselves and their child twice per week with a weight loss goal of 1-2 lbs/week.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**Information Control/Intervention:** trained group leader (*unclear role & training received*) as well as an assigned "coach" who provided mid-way point phone check ins (*unclear training of "coach"*) Pediatricians blind to the group assignments reviewed the child's progress at 3 and 6 month interval. Trained research assistant measured child's weight at every session	661 (1) 662 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Control & Intervention** – Group sessions, presumed to be in-person with individual phone calls by coach in between meetings. **Intervention -** Same as control with the addition of a 1:1 meeting at each group session, presumed to be in person	661-662 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Primary care setting → unclear location	660 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Control –** 53 randomized to control. Had ten 60-minute sessions over 6 months (4 weekly, 2 bimonthly and 4 monthly). Received 8 phone calls between meetings by their assigned "coach". **Intervention -** 52 randomized to the intervention. Also had ten 60-minute sessons over 6 months (4 weekly, 2 bimonthly and 4 monthly). Received 8 phone calls between meetings by their assigned "coach". Also met 1:1 with their coach either before or after each of the meetings	661,663 (1)
Tailoring		

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If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**Intervention –** Parents attended 1:1 meetings with an assigned coach who assisted parents in shaping behavioural goals after having reviewed food/activity diaries.	662 (1)
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Control & Intervention:** Group leaders were closely supervised by investigators. No other strategies were discussed regarding maintaining fidelity.	662 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	None reported	

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Quattrin T, Roemmich JN, Paluch R, Yu J, Epstein LH, Ecker MA. Efficacy of family-based weight control program for preschool children in primary care. Pediatrics 2012;130 (4):660–6.

Article Title	A pilot randomized controlled trial of a clinic and home-based behavioural intervention to decrease obesity in preschoolers	
Authors	Stark LJ, Spear S, Boles R, Kuhl E, Ratcliff M, Scharf C, et al. (2011)	
		Page Number (Location in Text)

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Brief Name		
Provide the name or a phrase that describes the intervention.	LAUNCH – Learning about Activity and Understanding Nutrition for Child Health	134 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	Developed a 6-month behavioural intervention for preschool obesity based on social cognitive theory and modeled on successful behavioural family-based intervention and modeled on successful behavioural family-based interventions for school-aged children. The program will specifically target preschool years, and was found to have a decrease in BMI z-score which was maintained at 12 months post-intervention. LAUNCH was designed to produce small decreases or stabilize the rate of children's weight gain, consistent with the current recommendations for treatment of preschool obesity.	134, 135 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	*Standard of Care:* Pediatrician used a scripted manual (not provided) to delivery dietary and physical activity recommendations outlined by the AAP (article with recommendations referenced). Each family was given a 1-page healthy food and activity brochure created by the Collaboration for Health Ohio (http://www.healthyohioprogram.org/healthylife/nutri2/nutrikids2/ounce.aspx). *LAUNCH*: Pedometer and diet diary were used as feedback tools. Parents were given a 14-day supply of vegetables at each of the six clinic visits. A written manual (not provided) was used by the clinical psychologist to conduct the parent group.	135 (1) (2)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	*Standard of Care*: Dietary and physical activity recommendations (outlined by AAP) were provided by a pediatrician using a scripted manual. Each family met with the pediatrician for a one-time 45-minute visit to review the growth chart and BMI. Each family was given a 1-page handout on healthy food and activity. *LAUNCH:* 6-month intervention consisting of 2 phases. Phase 1 (intensive phase) was 12 weekly sessions alternating between group-based clinic sessions (parents and	

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	child attending concurrent groups), and individual home visits. Phase 2 (maintenance) was 12 weeks of every other week sessions, alternating between group sessions in clinic and home sessions. Parent-group sessions addressed dietary education, physical activity and parenting skills. Separate sessions were done for snacks/beverages, breakfast/lunch and dinner. Sessions were also done on decreasing screen time, increasing physical activity. Education regarding child behaviour management skills was also taught. Parents were given a 14-day supply of vegetables at each of the six clinic visits. Children attended a group session concurrently – nutrition education done through various activities as well as completed 15 minutes of moderate to vigorous activity. In-home sessions supported generalization of clinic-taught skills to the home environment. Phase 2 (maintenance) focused on helping families continue to make/maintain changes in eating and physical activity.	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	*Control:* Pediatricians *LAUNCH:* The parent group during the intensive phase was conducted by a licensed clinical psychologist following a written manual. Child groups were conducted by a pediatric psychology postdoctoral fellow and a research coordinator. In-home sessions were conducted by psychology postdoctoral fellows. No mentioned of specific training for implementation of the intervention. Anthropometric measurements were done by trained personnel from the research centre → did not mention what training required.	135-136 (1)
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**Control**: Delivered via a clinic appointment in person **LAUNCH:** Group sessions for both parents and children were done in person. Home visits were individual and done in person. Unclear if the maintenance phase (phase	

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	2) were done in-person or whether done as a group/individual session.	
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	**Control:** Pediatrician office **LAUNCH:** Group sessions were done in a clinic location. Home sessions done in participant's houses. Did not specify where phase 2 occurred.	
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**Control:** One 45 minutes session **LAUNCH:** Phase 1 consisted for 12 weekly sessions alternating between group sessions and home visits. Group sessions were 90 minutes in duration and covered 1 topic relating to diet and physical activity. In-home sessions lasted 60-90 minutes.	
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	**LAUNCH:** Unclear whether home visits were personalized to address clinic-taught skills	
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	**Control –** The pediatricians followed a scripted manual **LAUNCH** – Parent group was conducted by a licensed clinical psychologist following a written manual. Inter-rater reliability was conducted by an independent coder for home food environment assessments. No other outcomes were reported to have this. No other information regarding planned fidelity	
Actual: If intervention adherence or fidelity was		

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assessed, describe the extent to which the intervention was delivered as planned.	Not reported	

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark LJ, Spear S, Boles R, Kuhl E, Ratcliff M, Scharf C, et al. A pilot randomized controlled trial of a clinic and home-based behavioral intervention to decrease obesity in preschoolers. Obesity (Silver Spring) 2011;19(1):134–41.
2. Spear BA, Barlow SE, Ervin C et al. Recommendations for treatment of child and adolescent overweight and obesity. Pediatrics 2007;120 Suppl 4:S254–S288.

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Article Title	A Pilot Randomized Controlled Trial of a Behavioral Family-Based Intervention With and Without Home Visits to Decrease Obesity in Preschoolers	
Authors	Stark LJ, Clifford LM, Towner EK, Filigno SS, Zion C, Bolling C, et al. (2014)	

		Page Number (Location in Text)
Brief Name		
Provide the name or a phrase that describes the intervention.	Learning about Activity and Understanding Nutrition for Child Health (LAUNCH)	Pg 1002 (1)
Why		
Describe any rationale, theory or goal of the elements essential to the intervention	To address the unique aspects of treating obesity in preschoolers (food neophobia or unwillingness to try new foods, tantrumming peaks in this age group). Changing the diet of preschoolers would necessitate changing parents' interaction with their child and their home environment. Developed and piloted a behavioural intervention for preschool obesity modelled on successful behaviour family-based intervention for school-age children, while addressing the unique challenges of preschoolers. Sought to examine a version of LAUNCH that conforms to Expert Committee guidelines for Stage 3 intervention on the number of sessions, structure, and support compared with a condition similar to Stage 1 intervention.	Pg. 1002 (1)
What		
Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	**LAUNCH-Home Visits (HV) & Clinic:** Required a diet diary, vegetables. LAUNCH-clinic required a home clean-out box. **Control:** A 1-page healthy food and activity brochure created by the Collaboration for Healthy Ohio (https://www.nationwidechildrens.org/specialties/center-for-healthy-weight-and-nutrition/services-we-offer/ounce-of-prevention)	Pg. 1003, 1005 (1)
Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	**LAUNCH-HV**: Manualized intervention consisting of two phases (Intensive & Maintenance), alternating between group clinic and individual home sessions. *Phase 1:* Months 1-3 targeted lifestyle behaviour modifications and improving parenting skills (supplementary material available online for treatment flow and session topics). Weeks 2-7 included education on diet, weeks 8-12 education on physical activity and parenting skills at all visits. Parents kept a 7-day diet diary for themselves and their child. They were also provided with vegetables at each session	Pg. 1003-1005 (1)

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	for daily taste tests between sessions. Children participated in a concurrent manualized group-based intervention paralleling the topics covered in the parent sessions focusing on education about healthy education, opportunities to try new foods and engage in physical activity. Home sessions followed a manualized protocol to support generalization of clinic-taught skills to the home environment through instruction, therapist modeling and parent rehearsal of dietary changes, physical activity and behaviour techniques. "Home clean-outs" were conducted following each clinic session on diet. *Phase 2:* Months 4-6 focused on helping families maintain treatment gains by engaging parents in long-term planning, problem-solving and use of parenting skills to promote maintenance of diet and activity changes. **LAUNCH-C:** Intervention content was identical to LAUNCH-HV clinic sessions. Instead of home-visits, parents were provided a "home clean-out" box to use on their own to eliminate high-calorie low-nutrient foods from the home. **Standard of Care:** Manualized intervention designed to deliver dietary and physical activity recommendations outlined by the AAP (2). Pediatrician explained BMI, BMI percentile and reviewed the child's growth chart. Pediatrician recommended screen time ≤ 2 hours; active play ≥ 60 mins daily, eliminating soda and ≤ 4 oz daily; fruits and vegetables ≥ 5 servings daily, limiting eating out; and appropriate portion size for preschoolers. They were also given a 1-page healthy food and activity brochure (3).	
Who Provided		
For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	**LAUNCH-HV:** Parent-group clinic sessions were conducted by a licensed clinical psychologist or 2nd year psychology postdoctoral fellow. The kid groups were led by a pediatric psychology postdoctoral fellow and a research coordinator. Home sessions were conducted by a pediatric psychology fellow.	Pg. 1002-1003 (1)

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	LAUNCH-C: Same as the LAUNCH-HV **Standard of Care:** Pediatrician	
How		
Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	**LAUNCH-HV**: The education sessions were provided in the clinic as a group. The home-visits were individual. **LAUNCH-C:** Education sessions were in-person group sessions. **Standard of Care:** Individual in-person sessions.	Pg. 1002-1003 (1)
Where		
Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	Conducted at the Cincinnati Children's Hospital Medical Centre as well as home visits (LAUNCH-HV)	Pg. 1003 (1)
When and How Much		
Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	**LAUNCH-HV:** 18-sessions, 6-month intervention consisting of two phases. Phase 1 was 3 months in duration consisting of 12 weekly sessions alternating between group-based clinic sessions and individual home visits. The parent group sessions were 90 minutes in duration. Home visits lasted 6-90 minutes. Phase 2 consisted of 12 weeks of every-other-week sessions alternating between group clinic sessions and individual home visits. **LAUNCH-C:** Sessions were conducted every other week during months 1-3 and monthly during months 4-6 of the intervention, for 10 treatment sessions. **Standard of Care:** Pediatrician met with each family individually for one 45 minutes visit.	Pg. 1003-1005 (1)
Tailoring		
If the intervention was planned to be personalised, titrated or adapted, then describe what, why,	**LAUNCH-HV/C:** The maintenance phase focusing on helping to maintain treatment gains focused on problem-solving around individual barriers, and use of parenting	Pg. 1004 (1)

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when, and how.	skills.	
Modifications		
If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	None reported	
How Well		
Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	No planned fidelity assessments were discussed. Intervention adherence was monitored via attrition	Pg. 1006 (1)
Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	15 families were randomized to LAUNCH-HV, 14 to LAUNCH-C, and 13 to standard of care. Of those, 3 families in LAUNCH-HV, 1 in LAUNCH-C and 1 in standard of care, dropped out before completing all the treatment sessions. The 1 in standard of care dropped out after completing the session.	Pg. 1006 (1)

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use '?' if information about the element is not reported/not sufficiently reported.

References:

1. Stark LJ, Clifford LM, Towner EK, Filigno SS, Zion C, Bolling C, et al. A pilot randomized controlled trial of a behavioral family-based intervention with and without home visits to decrease obesity in preschoolers. J Pediatr Psychol 2014;39(9):1001–12.

2. Spear, B. A., Barlow, S. E., Ervin, C., Ludwig, D. S.,Saelens, B. E., Schetzina, K. E., & Taveras, E. M.(2007). Recommendations for treatment of child and adolescent overweight and obesity. Pediatrics, 120(Suppl. 4), S254–S288. doi:10.1542/peds.2007-2329F

3. Toolkit (2007). Ounce of prevention is worth a pound: Preventing childhood obesity starting from birth. Retrieved December 26, 2021. https://www.nationwidechildrens.org/specialties/center-for-healthy-weight-and-nutrition/services-we-offer/ounce-of-prevention

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RITES TOOL

Explanatory Design: determines whether an intervention produces the expected result under ideal research circumstances and are intended to provide evidence on the *efficacy* of an intervention.

Pragmatic Design: determines the degree of beneficial effect under "real world" clinical conditions and are intended to provide evidence on the *effectiveness* of an intervention.

Criteria		Descriptor for trial with maximum *effectiveness* (pragmatic)	Descriptor for trial with maximum *efficacy* (explanatory)
Participants	**Characteristics/Description***	The participants are representative of the population who would receive the experimental intervention if it was part of usual care. ** They are of similar age, severity of illness and comorbidities to those patients who would be candidates for the intervention in usual care setting. They reflect diversity along parameters that could impact adherence.	The participants are a homogenous population and are markedly different from those seen in usual care. May have been deliberately selected to comply with treatment, respond to treatment or demonstrate the efficacy of the experimental intervention. There may be other exclusions that would not be seen in usual care ** (e.g. exclusion of participants with comorbidities).
	Eligibility	How similar are the participants to those who would receive the intervention if it was part of usual care?	There are many exclusions (e.g. those who don't comply, respond to treatment, or are not at high risk for the primary outcome, etc.), or the trial uses many selection tests not used in usual care.
	Recruitment	Recruitment is through usual appointments or clinics	Recruitment is through targeted information letters, advertising in newspapers, radio plus incentives and other routes that would not be used in usual care.
Trial Characteristics	**Setting**	The overall setting of the trial is similar to usual care – and includes diverse sub-settings common in usual care for this intervention (e.g. primary care, specialized care)	The setting is selected to maximize the ability to carry out the trial and identify an intervention effect if there is one. The setting may be more specialized than the setting in which the intervention would be delivered in usual care (e.g. a single center, specialized clinics or specialized academic centres or only providers with high levels of experience).

	Organization	The resources, provider expertise and the organization of care delivery in the intervention arm are identical to usual care	The trial increases staff levels, gives additional training, requires more than the usual experience or certification, or certification and increase resources.
	Comparison intervention	The control typically is applied by the full range of practitioners and in the full range of clinical settings, with only ordinary attention to the training, experience and performance of the practitioners.	The practitioner expertise is standardized in the control to maximize the chances of detecting whatever comparative benefits the experimental intervention might have
Flexibility	**Delivery**	Flexibility in the intervention is identical to that in usual care	Intervention delivery is less flexible than usual care. There is a strict protocol, monitoring and measures to improve compliance, with specific advice on allowed co-interventions and complications.
	Adherence	There is no more than usual encouragement to adhere to the intervention	Exclusion based on adherence and measures to improve adherence if found wanting.
	Practitioner adherence to study protocol	There is unobtrusive (or no) measure of practitioner adherence and no special strategies to maintain or improve it	There is close monitoring of how well the participating clinicians and centres are adhering to even the minute details in the trial protocol and "manual of procedures".
	Flexibility of the control	"Usual practice" or the best available alternative management strategy, offering practitioners considerable leeway in deciding how to apply it	Restrictive flexibility of the comparison intervention
Clinical relevance of experimental and comparison interventions	**Clinical Relevance**	Both the experimental and comparison intervention have the potential to be "best practice"	One or both of the experimental and comparison intervention is not a clinically relevant treatment (e.g. placebo, sub-clinical doses)
	Primary outcome	The primary outcome is of obvious importance to participants.	The primary outcome uses a surrogate, physiological outcome, central adjudication or uses assessment expertise that is not available in usual care, or the outcome is measured at an earlier time than in usual care.
	Primary analysis	The most pragmatic approach is to use intention to treat with all available data	The most explanatory analysis is one that excludes ineligible post-randomization participants and includes only those following the treatment protocol.

* Participants included in the study should be judged according to whether they are representative of the general population of those with the condition of interest who would receive usual care in the geographic area in which the trial is carried out

** Usual care describes the type of care routinely received by patients in the geographic area in which the trial is carried out. The care provided can vary by patient characteristics (eg. age, sex, education), the setting (inpatient/outpatient, primary care/specialized care) in which the patient is seen, individual providers and insurance plans.

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5-point Likert Scale:

1 = Strong Emphasis on Efficacy
2 = Rather strong emphasis on efficacy
3 = Balanced emphasis on both efficacy and effectiveness
4 = Rather strong emphasis on effectiveness
5 = Strong emphasis on effectiveness

Study Title: A Parent Treatment Program for Preschoolers with Obesity: A Randomized Control Trial

Authors: Ek A., Chamberlain, K., Sorjonen, K., Hammar, U., Malek, M., Sandvik, P., Somaraki, M., Nyman, J., Lindberg L., Nordin, K., Ejderhamn, J., Fisher, P., Chamberlain, P., Marcus, C., Nowicka, P.

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☒	☐	☐	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: Efficacy of a Community- Versus Primary Care-Centered Program for Childhood Obesity: TX CORD RCT

Authors: Butte, N., Hoelscher, D., Barlow, S., Pont, S., Durand, C., Vandewater, E., Liu, Y., Adolph, A., Pérez, A., Wilson, T., Gonzalez, A., Puyau, M., Sharma, S., Byrd-Williams, C., Oluyomi, A., Huang, T., Finkelstein, E., Sacher, P., Kelder, S.,

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☐	☒	☐	☐
Flexibility of Intervention	☐	☐	☐	☒	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

For peer review only

Study Title: Two-year follow-up of a primary care-based intervention to prevent and manage childhood obesity: the High Five for Kids study

Authors: Sheryl L. Rifas-Shiman, Elsie M. Taveras, Steven L. Gortmaker, Katherine H. Hohman, Christine M. Horan, Ken P. Kleinman, Kathleen Mitchell, Sarah Price, Lisa A. Prosser and Matthew W. Gillman

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☒	☐	☐
Trial Setting	☐	☐	☒	☐	☐
Flexibility of Intervention	☐	☒	☐	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: Clinic and Home-Based Behavioral Intervention for Obesity in Preschoolers: A Randomized Control Trial

Authors: Stark, L., Filigno, S., Bolling, C., Ratcliff, M., Kichler, J., Robson, S., Simon, S., McCullough, M., Clifford, L., Stough, C., Zion, C., Ittenbach, R.

Reviewer: SN, CD

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: A pilot randomized controlled trial of a behavioral family-based intervention with and without home visits to decrease obesity in preschoolers (2014)

Authors: Stark, L., Clifford, L., Towner, E., Filigno, S., Zion, C., Bolling, C., & Rausch, J.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☒	☐	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☒	☐	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

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Study Title: A pilot randomized controlled trial of a clinic and home-based behavioural intervention to decrease obesity in preschoolers (Stark 2011)

Authors: Stark, L., Spear, S., Boles, R., Kuhl, E., Ratcliff, M., Scharf, C., Bolling, C. & Rausch, J.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☒	☐	☐	☐
Flexibility of Intervention	☐	☒	☐	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

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Study Title: Efficacy of family-based weight control program for preschool children in primary care (Quattrin 2012)

Author: Quattrin, T., Roemmich, J., Paluch, R., Yu, J., Epstein, L., & Ecker, M.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☒	☐	☐
Trial Setting	☐	☐	☒	☐	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☐	☒

For peer review only

Study Title: Can a dairy-rich diet be effective in long-term weight control of young children

Author: Kelishadi, R., Zemel, M., Hashemipour, M., Hosseini, M., Mohammadifard, N. & Poursafa, P.

Reviewer: SN, RG

	Strong Emphasis on Efficacy 1	Rather strong emphasis on efficacy 2	Balanced emphasis on both efficacy and effectiveness 3	Rather strong emphasis on effectiveness 4	Strong emphasis on effectiveness 5
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☐	☐	☒	☐
Flexibility of Intervention	☐	☐	☒	☐	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

For peer review only

Study Title: A child-centered health dialogue for the prevention of obesity in child health services in Sweden – A randomized controlled trial including an economic evaluation

Authors: Derwig, M., Tiberg, I., Björk, J., Welander Tärneberg, A., & Kristensson Hallström, I.

Reviewer: SN, RG

	Strong Emphasis on Efficacy ❶	Rather strong emphasis on efficacy ❷	Balanced emphasis on both efficacy and effectiveness ❸	Rather strong emphasis on effectiveness ❹	Strong emphasis on effectiveness ❺
Participant Characteristics	☐	☐	☐	☒	☐
Trial Setting	☐	☐	☐	☒	☐
Flexibility of Intervention	☐	☐	☐	☒	☐
Clinical relevance of experimental and comparison intervention(s)	☐	☐	☐	☒	☐

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P.1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P.2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P.4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P.5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P.5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P.6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P.6-7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P.7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P.6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P.6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P.7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P.8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P.8-9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P.8-9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P.8-9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P.8-9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P.8-9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P.7-9
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P.7-9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P.9 Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	P.9 Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P.11 Figure 7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P.9-11 Figures 2-6
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P.11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P.9-11 Figures 2-6
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P.9-11 Figures 2-6
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P.11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P.13-15
	23b	Discuss any limitations of the evidence included in the review.	P.14-15
	23c	Discuss any limitations of the review processes used.	P.14-15
	23d	Discuss implications of the results for practice, policy, and future research.	P.15-16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P.3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P.3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P.16
Competing interests	26	Declare any competing interests of review authors.	P.16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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